Reinforcement Learning for Lean Proofs with a Small Language Model

Midterm report: formulation, experiments, and findings

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Reading dates. Historical sections retain their original reporting cutoff. Section 8 adds the prospective study designed on 16 October 2026. Appendix A describes every DEV and TEST theorem from the 17 October review pack. Section 9 adds the EXP-028 and main-run results of 18 and 19 October 2026. Section 9.6 adds a read-only analysis of those results from 20 October 2026, and Section 11 records two follow-up designs that did not run. Section 10 records the post-hoc analyses of 20 and 21 October 2026. Current execution status remains in the README; dated protocols and evidence remain in `docs/EXPERIMENTS.md`.

Abstract

This report asks whether reinforcement learning (RL) can teach a small language model, `Qwen2.5-0.5B-Instruct`, to write Lean 4 proofs of elementary number theory. We formulate proof writing as a Markov decision process (MDP) with a reward of 1 only for a complete proof that Lean accepts. Most experiments failed for one main reason: when Lean accepts no attempt, the RL update is exactly zero. RL helped only on theorems that the model could already prove some of the time. We also found ways to obtain the reward without proof skill, and several engineering faults.

Main findings

1. RL needs some accepted attempts. With no accepted attempt, the gradient is zero (Section 2.2).
2. The first barrier was the Lean language, not the mathematics (Section 4.1).
3. A reward that does not check correctness can increase while accuracy decreases (Section 4.3).
4. RL helped twice, both times on theorems that the model could already prove some of the time (Sections 4.4 and 4.6).
5. A Lean acceptance does not always show skill: the checker had faults, and some statements are trivial (Section 5).
6. The plan recorded on 13 October uses a corrected environment, a curriculum ordered by proof structure, and a cleaned held-out panel (Section 7).
7. In the runs of 18 and 19 October, filtered and curriculum-ordered RL again raised acceptance only on their training theorems, in one run each. No arm showed a detected held-out gain over S (Section 9).
8. A read-only analysis of these runs finds no sign of a fault in the RL update: on the 62 `RL_filtered` theorems, acceptance rises from 0.306 to 0.597. The reward is rarely nonzero: R's first visit finds a proof on 46 of 506 TRAIN theorems. RL concentrates on `simp_all` actions with a lemma list. On the selection set, 326 of `RL_filtered`'s 328 first actions start this way, and Lean rejects 224 of them at the first tactic. No run tests this account (Section 9.6).

1 The problem

We want a small language model that writes proofs that the proof assistant Lean 4 can check. The topic is school and early-university number theory: parity, divisibility, gcd and lcm, primes, and congruences. The midterm goal was a model that produces some complete proofs that Lean accepts. The goal for the end of the semester is proof strategy on competition-style theorems.

Motivation.

1. **Efficiency.** We want the most proofs for a fixed compute budget. We study the *curriculum* (which theorems, in which order) and the *distribution of reward* over the actions of a proof.
2. **Checked learning.** A model that learns without a checker can learn false statements. Lean should check the model while it learns, not only at the end.
3. **One model for informal and formal reasoning.** The model writes a short informal argument and then its Lean tactic. A separate translator would add errors and compute.
4. **No human graders.** Lean gives an exact reward. This setting is called RL with verifiable rewards (RLVR). The reward is *sparse*: a signal comes only from a complete proof.

What Lean does. A Lean theorem has a statement and a proof. We write the proof as a list of *tactics*. Each tactic changes the *goals*, the facts that remain to be proved. When no goals remain, the proof is complete. Table 1 shows a small example that we wrote to explain the idea, and Figure 1 shows when Lean accepts a proof.

Table 1: A tactic proof that $a \mid b$ and $b \mid c$ imply $a \mid c$ (here $a \mid b$ means $b = ak$). This example explains the idea; it is not a measured proof.

Action	Goal before the action	Tactic (the action)	Effect
1	$a \mid c$, given $a \mid b$, $b \mid c$	`obtain ⟨k, hk⟩ := h1`	Adds $b = ak$
2	$a \mid c$, given $b = ak$, $b \mid c$	`obtain ⟨l, hl⟩ := h2`	Adds $c = bl$
3	$a \mid c$, given $b = ak$, $c = bl$	`exact ⟨k * l, by rw [hl, hk, Nat.mul_assoc]⟩`	$c = (ak)l = a(kl)$. No goals remain

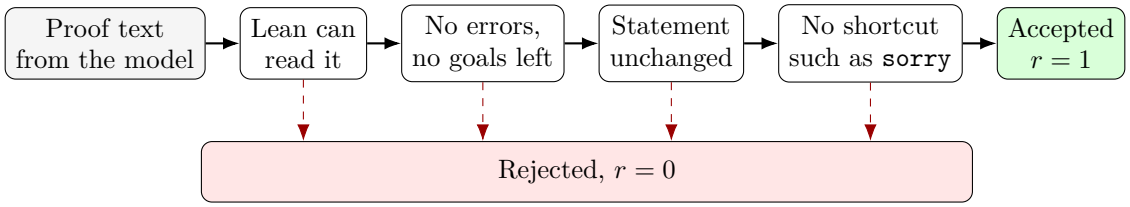


Figure 1: When Lean accepts a proof. Every check must pass. One wrong character or one invented lemma name causes a rejection.

Setup. Model: `Qwen/Qwen2.5-0.5B-Instruct` [1], with 494,032,768 parameters. Checker: Lean `v4.10.0-rc1` with one fixed mathlib version. Compute: one rented NVIDIA L40S GPU, and a shared CPU server.

2 Formulation as an MDP

An MDP describes an agent that acts step by step [3]. At step t, the agent observes a state s_t, selects an action a_t, receives a reward r_t, and moves to a state s_{t+1}. Here the agent (the *policy* π_θ) is the language model, and the environment is Lean. We used two formulations, which differ only in the size of one action (Figure 2).

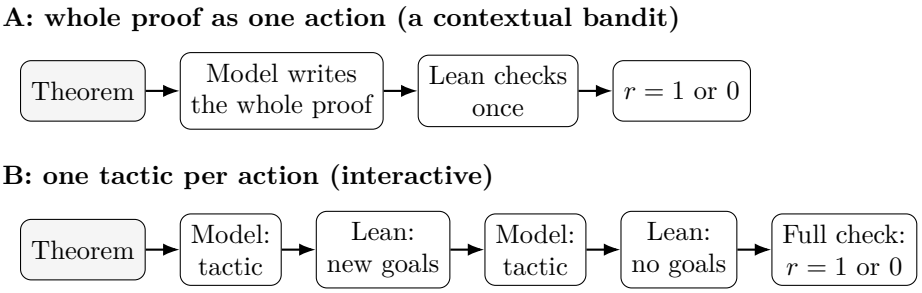


Figure 2: The two formulations. In A, the model receives no feedback while it writes (Experiments 4 to 6, 20 to 22). In B, Lean runs each tactic first (Experiments 19 and 23).

Table 2 defines Formulation B, and Figure 3 shows it as a graph.

Table 2: The interactive MDP (environment version 1).

Part	Definition
State s_t	The current Lean goals. At $t = 0$, the theorem. In an *assisted* attempt, part of a known proof is applied first
Action a_t	A short tactic block (at most 64 tokens in the main configuration of this version)
Transition	Lean runs the block. If it succeeds, the new goals are s_{t+1}
End of episode	No goals remain (then Lean checks the full proof), Lean reports an error, or the action budget is used (4 actions)
Reward	1 at the end if Lean accepts the full proof, otherwise 0
Discount	$\gamma = 1$

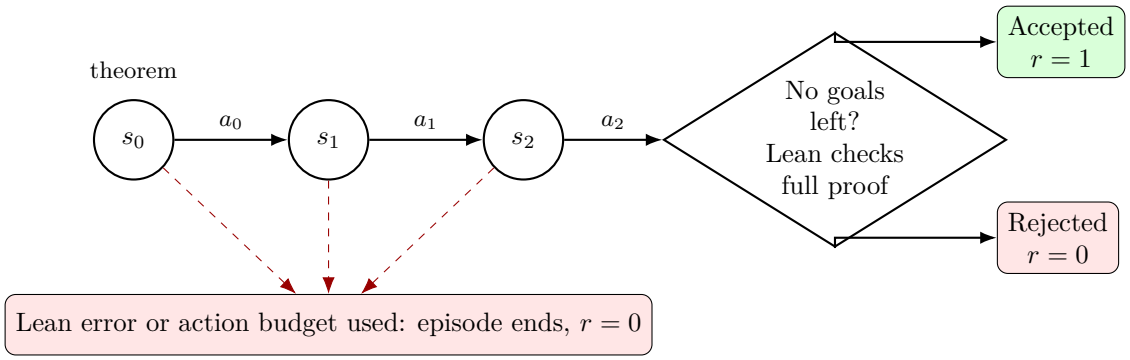


Figure 3: Formulation B for the proof of Table 1. Solid arrows: the tactic succeeds. Dashed arrows: an error ends the episode with reward 0.

Version 1 has three defects. The model does not see the remaining actions, the theorem, or the proof so far. With an action limit that the model cannot see, the process is not exactly Markov. One error ends the episode. Section 7 corrects these defects.

2.1 Learning algorithm

We maximize the expected return $J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\big[\sum_t \gamma^t r_t\big]$. The only reward is at the end, so the return of an episode τ is $R(\tau) \in \{0, 1\}$. For one theorem, we sample G episodes (one episode is one attempt) and use the REINFORCE estimator [4]

$$\nabla_\theta J(\theta) \approx \frac{1}{G}\sum_{i=1}^{G} A_i \, \nabla_\theta \log \pi_\theta(\tau_i). \tag{1}$$

The *advantage* A_i compares attempt i with the other attempts. The interactive runs use the leave-one-out baseline (RLOO) [5], and the whole-proof runs use GRPO [6, 7]:

$$A_i^{\text{RLOO}} = R_i - \frac{1}{G-1}\sum_{j\neq i} R_j, \qquad A_i^{\text{GRPO}} = \frac{R_i - \bar{R}}{s_R + \varepsilon}, \tag{2}$$

where $\bar{R}$ and s_R are the mean and standard deviation of the G rewards.

2.2 Main consequence: no accepted attempt, no learning

If all G attempts fail, every $R_i = 0$, so every $A_i = 0$, and the update in Equation (1) is exactly zero. The same is true when Lean accepts all attempts. The model learns only from *mixed groups*, which contain at least one accepted and one rejected attempt (Figure 4).

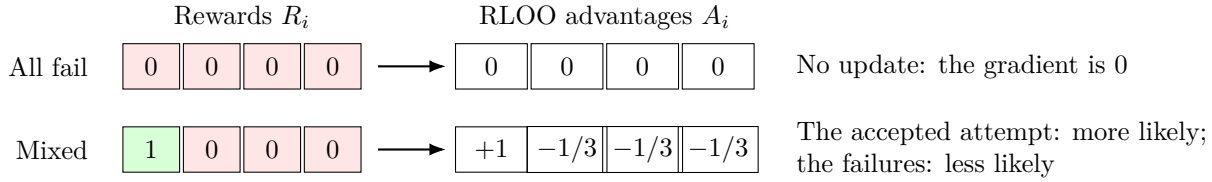


Figure 4: A group needs both an accepted and a rejected attempt ($G = 4$).

If Lean accepts each attempt with probability p, a group of G independent attempts is mixed with probability

$$P_{\text{mix}}(p, G) = 1 - p^G - (1-p)^G. \tag{3}$$

Figure 5 shows that this probability is small when p is small. This equation explains most of our failures.

2.3 A second reward: RLP

Some experiments used the RLP reward [8] instead of Lean. For a question x, reasoning c, and known answer y, the reward is $\log p_\theta(y \mid x, c) - \log p_{\bar{\theta}}(y \mid x)$, where $p_{\bar{\theta}}$ is a slowly updated copy of the model. The reward is high when the reasoning makes the correct answer more probable. It does not check whether the reasoning is correct (Figure 6). We call it a *heuristic* reward.

Measures. *Accepted attempts* counts sampled proofs that Lean accepts, for example 26/48. *Theorem coverage* counts theorems with at least one accepted attempt. One standard tactic proves 28 of the 420 validation statements without a model, so the main validation set is the other 392 (*non-trivial*) statements.

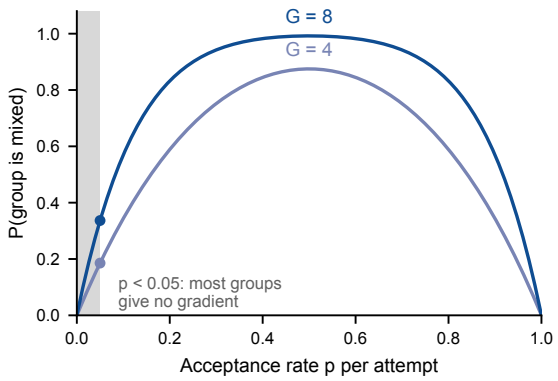


Figure 5: Probability that a group is mixed (Equation 3). At $p = 0.05$ it is 0.185 for $G = 4$ and 0.337 for $G = 8$.

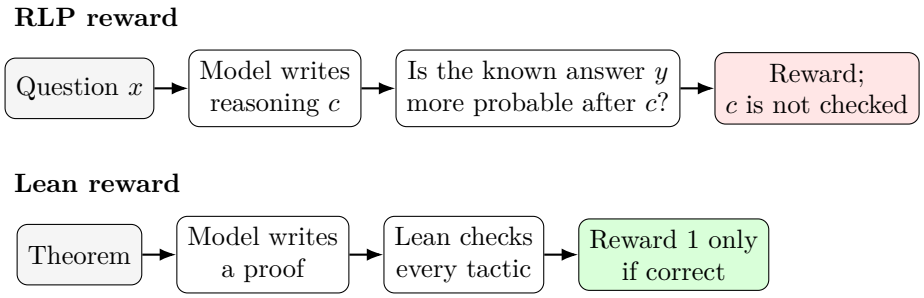


Figure 6: The two rewards. RLP rewards reasoning that makes a known answer more probable, correct or not. Lean rewards only a correct proof.

3 Data, training stages, and curricula

Table 3: Data sources and their use.

Source	Content	Use
LeanDojo Benchmark 4 [2]	Theorems and tactic proofs from mathlib; number-theory subset of 706 training, 17 validation, 15 test theorems	Supervised training; short proofs for interactive RL
Lean Workbook [9]	Competition-style theorems as Lean statements	Pool of 4,306 number-theory statements that Lean checks
Number-theory SFT set	5,812 examples: Lean-accepted proofs, mathlib syntax examples, proof variants	Second supervised stage
Hand-written Lean theorems	Logic, arithmetic, divisibility, induction	Experiments 5, 6; DEV of Experiment 19
Hand-written arithmetic questions	448 questions on gcd, inverses, primes, congruences	RLP (Experiment 9)
Open textbook (Judson)	Chapters of *Abstract Algebra: Theory and Applications*	RLP (Experiment 7)
Curriculum templates	192 training theorems in 5 stages and 152 held-out theorems, all checked by Lean	Final run (Section 7)

Table 3 lists the data. Figure 7 shows how we keep test data out of training. Renamed copies of one theorem always go to the same split, and we never train on the DEV set.

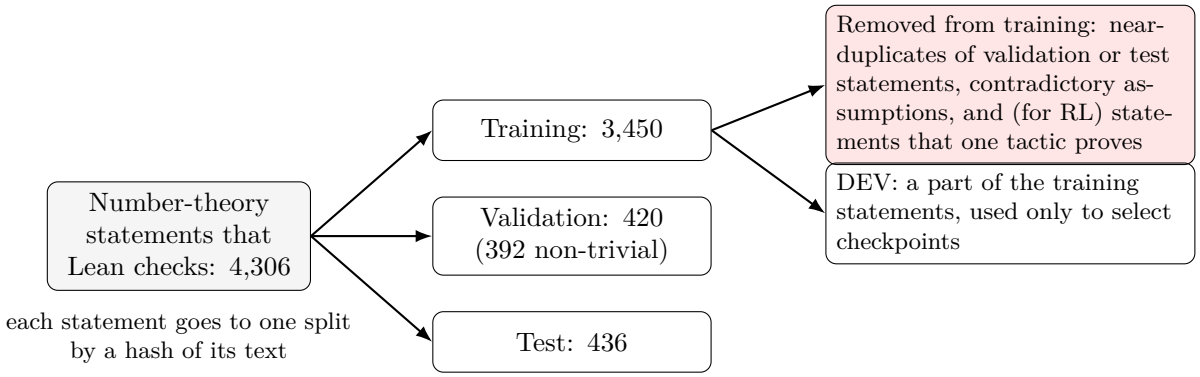


Figure 7: Data splits. Two statements are near-duplicates when their token sets overlap by at least 85%.

Training stages. All models come from one chain of checkpoints (Figure 8). SFT v2 is one epoch over Lean proofs from the LeanDojo training files (4,526 optimizer updates of 64 examples), without any file that contains a validation or test theorem. Common SFT is a second supervised stage on the number-theory SFT set. Supervised fine-tuning (SFT) copies existing proofs. RL learns from the model's own attempts, so Lean must accept some attempts.

Curricula. RL needs theorems that the model can prove some of the time. A *curriculum* starts with easier versions of a theorem. We made theorems easier to prove in three ways: we supplied knowledge

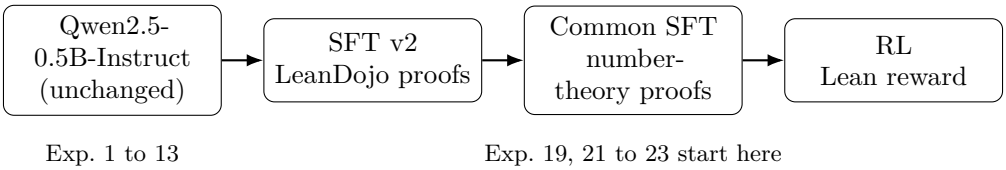


Figure 8: The training chain.

in the prompt, we supplied part of the proof (a *reverse curriculum* [10], Figure 9), or we asked for one action instead of a whole proof. Most known proofs are short: 85% of the linear tactic proofs in our LeanDojo pool have one or two tactics, and only 24 have five or more. Therefore a reverse curriculum on these proofs has few levels. To test whether *order* matters, an *ordered* arm and a *shuffled* arm receive the same theorems the same number of times. Table 4 lists the curriculum of each experiment.

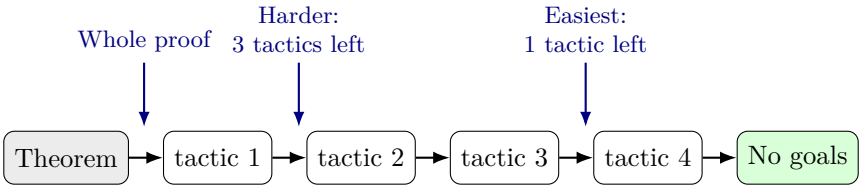


Figure 9: A reverse curriculum on one known proof. Each blue arrow is a start point; the earlier tactics are given. Training moves the start point toward the theorem.

Table 4: Curricula by experiment.

Exp.	Training data	Order and support	Comparison
4	LeanDojo number-theory theorems	Grouped by topic	Random order
5	Hand-written Lean theorems	Logic, arithmetic, divisibility, induction; support decreases	Random order
6	Theorems with part of the proof given	Missing expression, then tactic, then body	Random order
9	448 arithmetic questions	Prerequisites first, with replay	None
13	72 questions in 5 versions	Support decreases from full information to none	Shuffled order
20	Number-theory proofs	Short proof endings, longer endings, whole proofs	SFT parent
19	3 practice proof endings	Fixed mixture	SFT parent
23	22 Lean Workbook theorems	Structure, lemma use, combined steps, whole proof	Shuffled order
Final	S0 to S4 templates	Identity, one lemma, two steps, chain of subgoals, 3 to 6 steps	Shuffled; adaptive

4 Experiments

Figure 10 shows all experiments in order, with the lesson that led to each next experiment. A cause is *tested* when an experiment could have shown it to be false; otherwise it is a *candidate*.

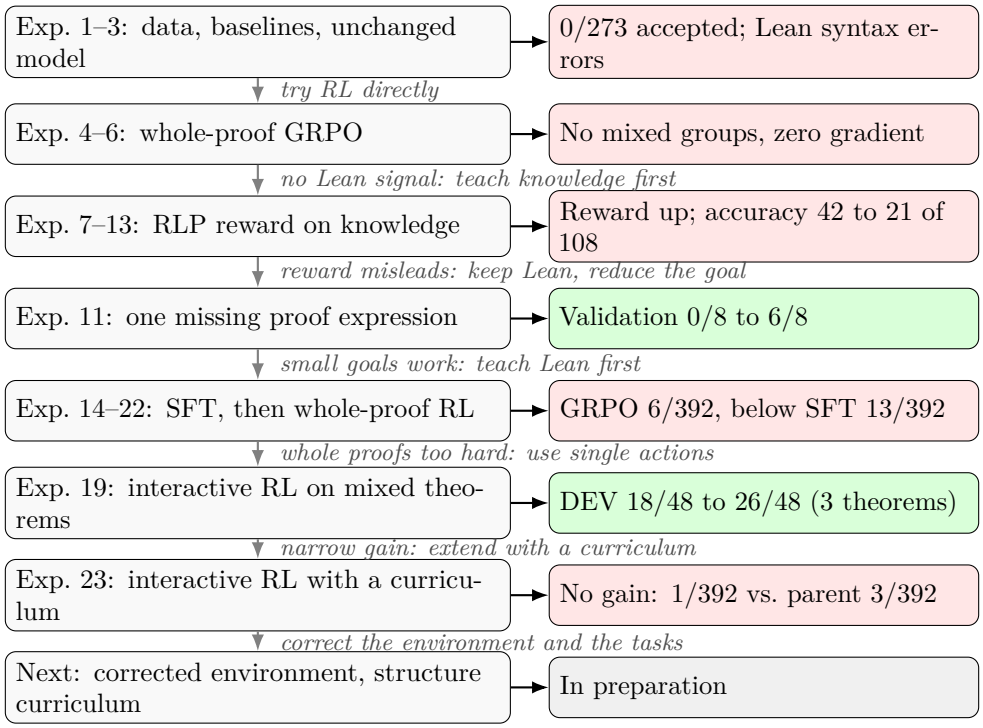


Figure 10: The sequence of experiments. Red: failure. Green: RL gain. Grey: in preparation.

4.1 Data, baseline, and the unchanged model (Experiments 1 to 3)

Experiment 1 selected a number-theory subset of LeanDojo. Its authors report that the random split overestimates performance [2], so we later built stricter splits (Figure 7). Experiment 2 tested the base model on school mathematics: 9 of 24 GSM8K and 3 of 20 MATH number-theory problems. A small fine-tune lowered the held-out loss, but the scores became 8 of 24 and 3 of 20. A lower loss did not mean better answers.

Experiment 3 asked the unchanged model to prove 21 statements. Lean accepted 0 of 63 local attempts and 0 of 273 GPU attempts. Figure 11 shows that no attempt reached the mathematics. A pattern match found old Lean 3 syntax in 33 of the 63 attempts. In 34 of the 63, the model wrote a new theorem when we asked only for a proof. The first barrier is the language (*tested*: the classification reads Lean's error messages).

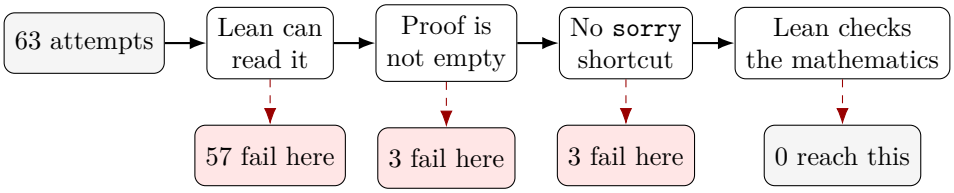


Figure 11: Where the 63 attempts of Experiment 3 failed.

4.2 Whole-proof RL from the unchanged model (Experiments 4 to 6)

We ran GRPO in Formulation A with $G = 4$. In Experiment 4, Lean accepted no proof, no group was mixed, and the gradient was zero for all 32 optimizer updates. Validation stayed at 0 of 8 theorems. In Experiment 5, easier hand-written theorems gave 0 of 256 accepted proofs, so we did not train. In

Experiment 6, with one expression missing from a given proof, Lean accepted 2 of 64 attempts, but only one theorem gave mixed groups, and held-out attempts stayed at 0 of 16.

Why it failed (*tested*): every group had reward 0, so every update was 0 (Section 2.2). Therefore Experiment 4 gives no information about curriculum order.

4.3 RL with a reward that does not check correctness (Experiments 7 to 10, 12, 13)

Lean gave no signal, so we tried to teach knowledge first with the RLP reward.

- Experiment 7 (textbook text): held-out accuracy was 11/48 before and after.
- Experiment 8: only 8,557,971 of 494,032,768 weights had changed, because low-precision (BF16) rounding discards small updates. With full precision, accuracy stayed at 11/48.
- Experiment 9 (448 arithmetic questions): the reward increased, but held-out accuracy decreased from 42/108 to 21/108 (Figure 12). Answers without a readable number increased from 17 to 69, and answers at the length limit from 15 to 36.
- Experiment 10: an early checkpoint improved at recognizing concepts (7 to 14 of 32), not at using them. Two AI graders disagreed on 44 of 321 answers.
- Experiment 12: all 612 numeric answers in the data were correct, but 160 of 448 rules omitted a condition (for example, $\gcd(a, n) = 1$ for a modular inverse). A constant answer solved every proof-by-contradiction item.
- Experiment 13 (decreasing support): 0 of 18 answers were fully correct at all five checkpoints.

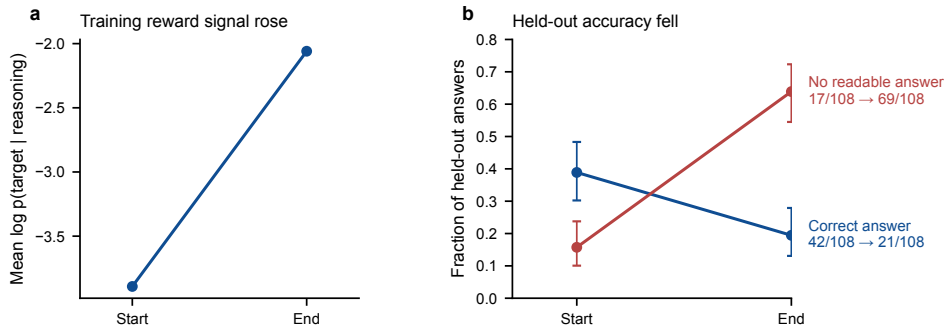


Figure 12: Experiment 9. (a) The quantity that the RLP reward pays for increased. (b) Held-out answers, $n = 108$. Error bars: 95% Wilson score intervals.

Why it failed: the reward pays for reasoning that makes the answer more probable, not for correct reasoning. This is *reward hacking*. In Experiment 9, 17 incorrect answers received a positive advantage. A deliberately false line of reasoning increased the reward in both inspected examples. A *candidate* reason for the accuracy decrease is that training supplied the start of the answer format, but the test did not.

4.4 A smaller goal (Experiment 11)

We reduced the proof to one missing expression with the mathematics supplied (Figure 13), and ran GRPO with $G = 8$. Validation accuracy increased from 0/8 to 6/8, and held-out accuracy from 0/4 to 2/4. It worked because some groups were mixed. The sets are small, and Lean checked only the expression: only 2 of 8 explanations had no mathematical error.

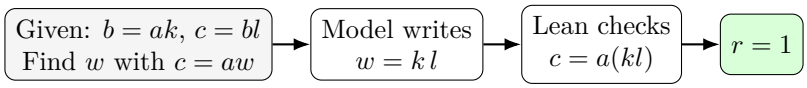


Figure 13: The one-expression goal of Experiment 11. Lean checks the expression, not the explanation.

4.5 Supervised training, then whole-proof RL (Experiments 14 to 22)

After SFT v2, the model proved 0 of the 392 non-trivial validation statements (2 of all 420). On 200 training statements it proved 0 in 1,600 attempts. The most frequent first errors were an invented lemma name (582), a failed rewrite (329), and the length limit (253). The barrier moved from syntax to library knowledge.

Experiment 20 added a common SFT stage, then alternated SFT with whole-proof GRPO in one branch and with RLP in another (Figure 14). Common SFT increased the count from 0 to 13. The GRPO branch ended at 6, below its parent, and the RLP branch at 15. The GRPO model wrote the single tactic `simp_all` in 1,637 of 3,360 attempts; the parent did so in 62.

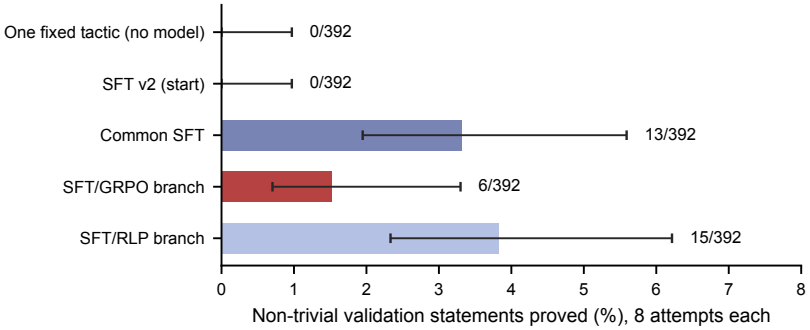


Figure 14: Experiment 20: non-trivial validation statements with an accepted proof in 8 attempts, $n = 392$. Error bars: 95% Wilson score intervals.

Experiment 21 used rejection-sampling fine-tuning (RFT: train on sampled proofs that Lean accepts [11]), then GRPO. On DEV (64 attempts per model), no model had an accepted proof. The first errors of the final model were proof construction (43), unknown names or types (17), syntax (3), and timeout (1). A parser fault rejected every action in a code block, so its interactive pilot made no update. Experiment 22 prepared PPO, but the GPU run failed before RL started.

Why it failed (*candidates*): the low acceptance rate gives few mixed groups; the GRPO model concentrated on `simp_all`, which closes only easy goals. The branches also had extra SFT stages and different data, so the comparison does not isolate RL.

4.6 Interactive RL: the clear gain (Experiment 19)

We used Formulation B from common SFT, with REINFORCE, RLOO, $G = 4$, and at most 4 actions. We trained only on three practice theorems that the model proved some of the time. The selection rule, fixed before the run, required more accepted DEV attempts than the parent over two seeds and no lost DEV theorem. The rule selected the RL model (Figure 15). It proved 7 of 32 original theorems without help (parent: 4 of 32), and all 7 proofs passed a new Lean check. An earlier run of this type lost one DEV theorem, so the rule kept its parent. Not every theorem improved: `Nat.not_even_one` decreased from 4/4 to 2/4.

Limits: DEV has only 3 theorems, there is one training seed, and the 32 original theorems are from the training domain. RL made the model more reliable on proofs that it could already find; it did not

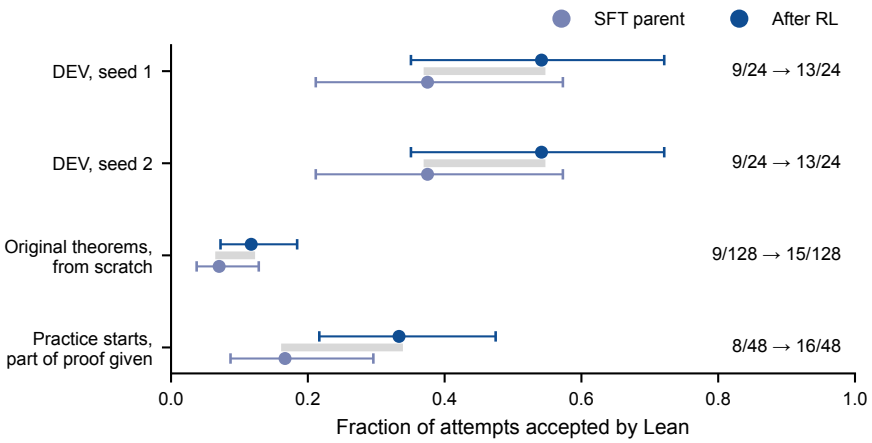


Figure 15: Experiment 19: SFT parent and RL model on the same theorems and seeds. Error bars: 95% Wilson score intervals, which treat correlated attempts as independent and so understate the uncertainty.

show that RL teaches new proofs.

4.7 Interactive RL with a curriculum (Experiment 23)

On 22 broader theorems, a short warm-start first increased accepted whole proofs on 16 training theorems from 0 to 13. Then an *ordered* arm (answer given, one lemma, combined steps, whole proof) and a *shuffled* arm trained on the same theorems. Neither arm improved on the parent. On broader DEV, all three models had 0 of 64. On easy retention, the parent had 24/48, ordered 20/48, and shuffled 21/48. On non-trivial validation, the parent proved 3 of 392 and ordered RL 1 of 392; one of the parent's accepted statements has the empty set as an answer. The ordered model made more valid first moves than the parent (88 against 40), but it completed no proof in its first move (Figure 16).

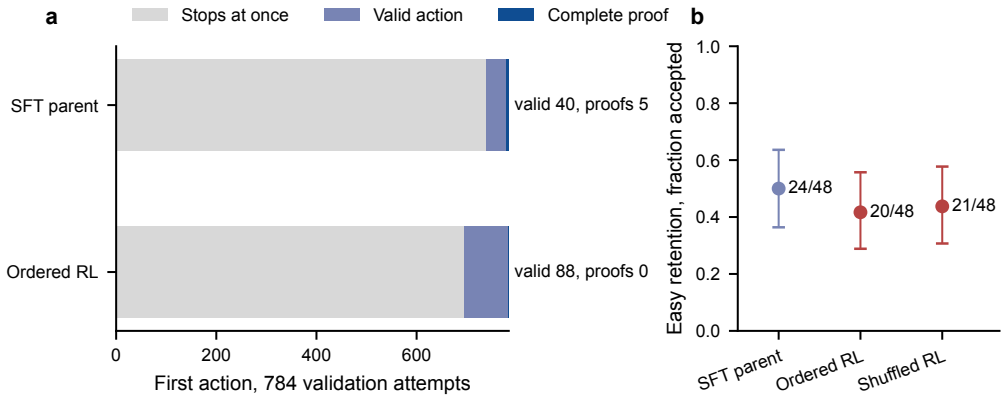


Figure 16: Experiment 23. (a) First action on the same 784 validation attempts. (b) Easy retention, $n = 48$. Error bars: 95% Wilson score intervals.

Why it failed (four *candidates*, not yet tested):

1. A valid action is not progress. The model learned valid actions, not complete proofs.

2. The curriculum had gaps: the lemma stage covered 9 of 22 theorems, the combined stage 11 of 22, and the early stages contained the answer.
3. For one Lean error type, the interface lost the message and gave reward 0, while ordinary errors received a penalty. A failure with a lost message therefore scored better (186 actions).
4. The environment did not show the theorem or the remaining actions, and one error ended the attempt.

One common explanation is not supported: most repeated-state failures occurred without `simp_all`.

5 Ways to obtain the reward without proof skill

We searched for every way to obtain the Lean reward without real proof skill and found six types (Figure 17).

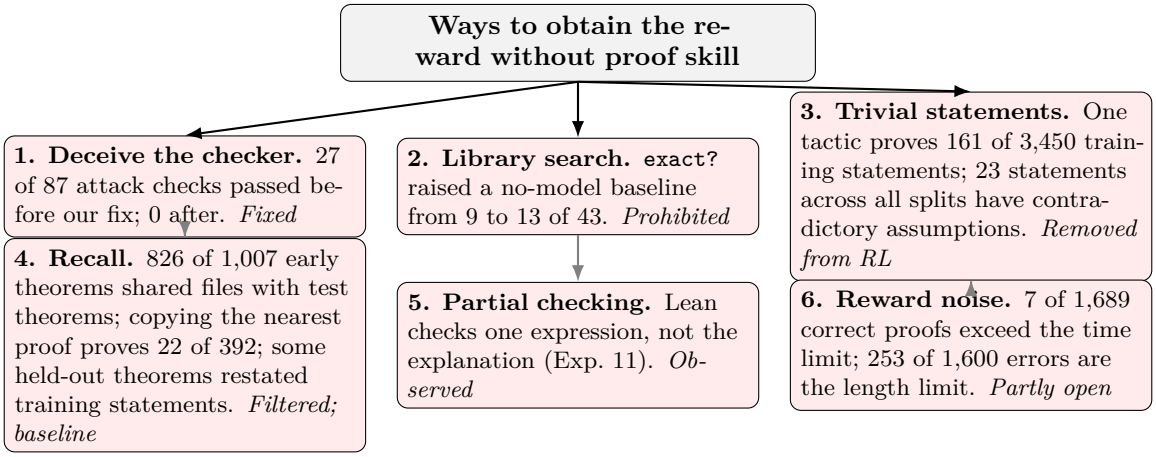


Figure 17: Six types of reward without real proof skill, with measurements and status.

The checker attacks included false "no axioms" messages, a replaced axiom report, and a command after the proof. A scan of 10,869 accepted proofs found no attack in real runs. Some statements are also formalized incorrectly; for example, a divisibility statement about $(5+\sqrt{35})^{2n-1}$ is about real numbers, where it is trivially true. Before we report an RL gain, we therefore separate trivial, contradictory, near-duplicate, and new statements, and report the new statements.

6 Engineering faults

Some failures were engineering faults, not scientific results (Table 5).

Table 5: Engineering faults that we found.

Fault	Effect	Status
BF16 optimizer state	Small updates rounded away	Fixed (full precision)
Checker attacks	27 of 87 attack checks accepted	Fixed (0 of 87)
Batch checking	One broken proof caused the rejection of the next proof; 15 of 40 correct proofs rejected in one test	Fixed
Action parser	Every action in a code block rejected before Lean	Fixed
Lost Lean error messages	One error type gave reward 0 instead of a penalty (186 actions)	Patch not yet verified
Proof export	Indentation of multi-line proofs changed, so exports failed	Fixed
Full-file compilation	DEV files exceeded the memory limit without a proof	Open
Prohibited-word filter	A string trick hid a prohibited command, and Lean accepted it	Fixed (version 2.2)
Run failures	Full disk at checkpoint save; missing PPO method; launch-script name conflict; wrong prompt-length count	Fixed

7 Plan recorded on 13 October

Corrected environment. Version 2.1 of the environment corrects the defects of version 1 (Figure 18). The model sees the theorem, the accepted actions, the current goals, the last Lean message, the tactics that already failed, and the remaining actions. An action contains up to four tactics, and Lean keeps all tactics before the first error, as in DeepSeek-Prover-V1.5 [11]. The reward is 1 for a proof and 0 otherwise, with $\gamma = 0.95$, so shorter proofs receive more reward. Version 2.2 corrected further faults. Integration tests with real Lean passed: 8 of 8 for version 2.1, and 86 of 86 for the version 2.2 trainer.

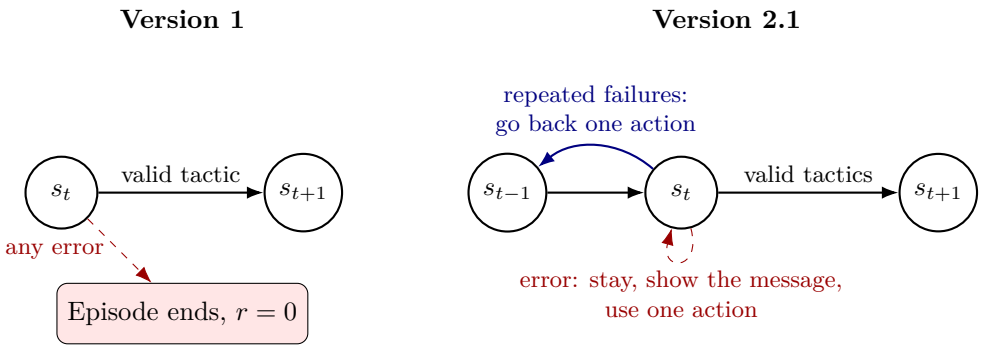


Figure 18: Treatment of a failed action in version 1 and version 2.1.

Curriculum ordered by proof structure. The final curriculum has five stages, ordered by the type of reasoning that a proof needs (Figure 19). All theorems come from templates in families of 8, and Lean accepts the reference proof of every theorem. A family that one fixed tactic mostly proves (4 or more of 8) moves to S0. We defined this relabel rule after we saw the first probe results, so it was not chosen blind. Within the training set, exact duplicates were removed, which leaves 192 training theorems.

Cleaning the held-out set. A semantic check compared held-out theorems with the statements behind the SFT data. In the first panel, it found 26 held-out–training pairs with the same content and

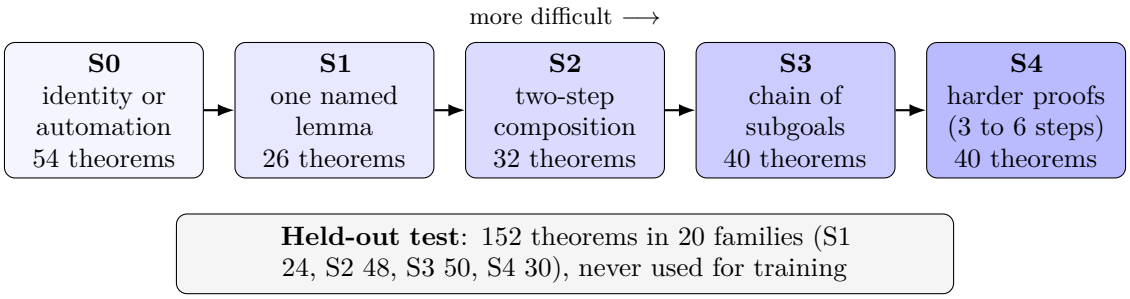


Figure 19: The final structure curriculum: 192 training theorems and 152 held-out theorems. Training excludes 670 names: every held-out, evaluation, reserve, demo, and DEV name.

3 near-transfers; for example, `ho_coprimechain_01` restates `lean_workbook_22701`. We removed 35 theorems, which left 61 clean theorems. A second set of new families went through the same check, which removed 7 contaminated held-out families and 2 training families. We then dropped the 5 remaining held-out theorems rated "related". The final held-out set has 152 theorems: the 61 original clean theorems and 91 new ones. On the 61 original clean theorems, the SFT parent (environment version 1) proved 6, with 8 of 488 attempts accepted.

Can RL start on these stages? Before training, we ran the SFT parent 8 times on each S0 to S2 theorem of the first template set, with the version 1 prompt (Figure 20). With the relabelled stages, S0 had 187 of 448 attempts accepted (44 of 56 theorems mixed), S1 26 of 256 (14 of 32 mixed), and S2 10 of 256 (8 of 32 mixed). Every stage has mixed groups, so RL has a signal, but the signal is weak in S2. Under the original labels, one family that `omega` proves alone gave 32 of the 42 S2 accepted attempts, and the two real two-step families gave 8 of 64 and 0 of 64. With the version 2.1 prompt, 186 of 386 first actions of the parent failed on format. In 145 of these 186, the code block had a label other than `lean`, for example `liquorice`; no run has tested whether the label causes the failure. Therefore a short warm-start in the new format comes before RL. Its data has 1,969 training rows on 468 theorems, each traced again through the version 2.1 environment.

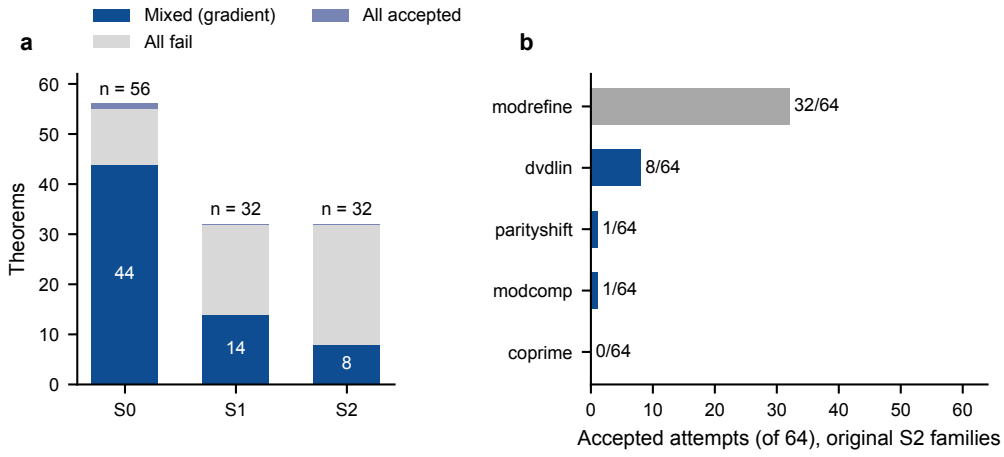


Figure 20: Readiness test of the SFT parent on the first template set, version 1 prompt, 8 attempts per theorem, one seed. (a) Theorems per relabelled stage by outcome. (b) Accepted attempts per family of the original S2 stage, $n = 64$; grey: the family that `omega` proves alone, later moved to S0.

The frozen plan for the final run. We recorded this protocol before the launch, when no result was available. The subsequent outcome belongs to the EXP-023 completion handoff in `docs/EXPERIMENTS.md`; the protocol below remains historical. The run tests two claims: RL improves the model over its SFT model W on held-out theorems, and a curriculum makes RL more efficient (Figure 21).

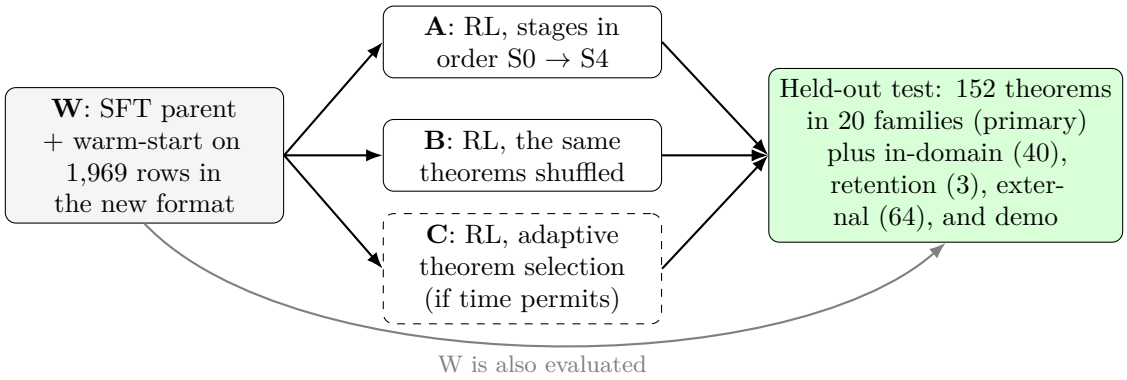


Figure 21: The frozen comparison. A and B against W test whether RL helps. A against B tests whether order helps. C tests whether theorems near a 50% acceptance rate give more mixed groups for the same compute.

- **Settings.** Environment version 2.2, $\gamma = 0.95$, $G = 8$, at most 6 actions, REINFORCE with the leave-one-out baseline on discounted returns, learning rate 10^{-6}, seed 20261013 for every arm, at most 150 groups per arm.
- **Admission.** W makes 8 attempts on each training theorem. A theorem enters training only if Lean accepts 1 to 6 of the 8 attempts, so every admitted theorem can give mixed groups. A stage with too few admitted theorems is dropped from both arms.
- **Primary measure.** The mean fraction of the 8 attempts that Lean accepts, per held-out theorem. The primary test permutes whole families, because theorems in one family are correlated; Holm correction controls the two comparisons of each claim.
- **Efficiency.** For each arm, the gain in proved theorems over W per million generated tokens, together with Lean calls, updates, and the share of mixed groups.
- **Validity gates.** A against B counts as confirmatory only if both arms complete the same number of groups, with updates within 10% and censored groups within 2. The efficiency claim also needs generated tokens within 10%. If training acceptance does not increase during RL, a null result is uninformative. No checkpoint, setting, or theorem is chosen from evaluation results.
- **Sample size.** In simulation, the McNemar test gave a false-positive rate of up to 0.14 against the intended 0.05; the family test stays at 0.05 or less. With the final 152 theorems, the family test detects 8 to 19 percentage points in the fraction of theorems proved, and 11 to 15 points at a baseline like the parent's. These values come from an assumed model, not from data.
- **Compute.** One L4 GPU for 12 hours, with a reservation of USD 37.01 and one attempt. A CPU fallback runs as a labelled deviation if the attempt fails.
- **Demo.** `demo_coprime` and one held-out theorem per level (a common divisor divides the gcd; odd m and n give an even $mn + 1$; ab or $a + b$ is even), run on A if A proves at least as many held-out theorems as W, otherwise on W. Six earlier demo theorems matched statements in the training or SFT data and were withdrawn.

7.1 Familiar-family reliability improves without an established transfer gain

Results recounted on 18 October 2026. The completed EXP-023 final run gives more accepted attempts on familiar number-theory constructions after RL (Table 6). These evaluations start from whole goals without supplied proof lines. They compare fixed final checkpoints with the SFT model W, using the same attempts per statement within each panel. Later data amendments leave a smaller held-out panel than the historical plan above.

Model	Familiar attempts	Familiar theorems	Held-out attempts	Held-out theorems
W: SFT model	148/320	31/40	258/1,136	59/142
A: ordered RL	208/320	32/40	283/1,136	53/142
B: shuffled RL	213/320	34/40	302/1,136	54/142
C: adaptive RL	242/320	35/40	249/1,136	46/142

Table 6: Recorded Lean-accepted complete proofs in the final curriculum run. Each statement has eight attempts. A theorem counts as proved when Lean accepts at least one attempt. Familiar constructions and held-out constructions are separate populations; their denominators must not be pooled.

More accepted attempts need not mean broader proof coverage. Every RL final proves fewer distinct held-out theorems than W in these attempts. The family-level tests find no overall held-out benefit after Holm correction; each adjusted p-value equals 1. The familiar-family tests support higher sampled reliability, but this is a secondary panel from a single training seed. It does not establish broad acquisition of number-theory proofs.

What the comparison does not explain. The training record contains no group from the hardest stage S4. On held-out S4 statements, W accepts 14/240 attempts; A accepts 1/240, B accepts 7/240, and C accepts 0/240. Missing practice is a candidate explanation, but no causal ablation tests it. Actual generated training tokens differ across arms, so the registered token gate rejects the efficiency claim. Earlier supervised exposure required benchmark removals; remaining semantic and pretraining familiarity are not certified absent. These limits prevent an explanation that attributes the transfer result to a specific missing skill or optimizer property.

Evidence boundary. The recount reads saved strict-replay verdicts; it does not run fresh Lean replay or model inference. The raw inputs and counted denominators are recorded in `nt_wednesday_evidence_20261018_v1/recount.json`, with their source paths and planted counter controls. The original final-run results and amendments remain in `docs/EXPERIMENTS.md`. This REINFORCE comparison does not measure PPO or the later candidate-selection policy.

8 Two major experiments: generalization and curriculum

Revised framing, 16 October 2026. The project has two major experiments. The generalization experiment compares the unchanged Qwen model, project SFT, and project SFT followed by RLVR. The curriculum experiment asks whether ordering RL practice improves familiar benchmark performance, generalization, or both. They share a benchmark and model lineage, but answer different questions. This revision supersedes the earlier curriculum-only framing. Both experiments remain prospective.

Table 7: The two major experiments. Benchmark preparation and calibration support these comparisons.

Experiment	Comparison	Question
Generalization, EXP-026	Unchanged Qwen; SFT; SFT + RLVR	What does each training stage change on familiar and shifted theorems?
Curriculum, EXP-027	Shuffled RL; prerequisite-first RL, from the same SFT checkpoint	Does curriculum improve familiar performance, generalization, or both?

8.1 Experiment 1: generalization across training stages

The first experiment asks what SFT and RLVR add to the model's proof capability. Its conditions are B, the unchanged model; S, B after project SFT; and R, the exact S checkpoint after RLVR. Use one frozen, non-curricular RL recipe for R. S against B measures the effect of project SFT. R against S measures the effect of adding this RLVR stage. Evaluate every condition on the same frozen benchmark panels.

Our selected model is `Qwen2.5-0.5B-Instruct`. *Without SFT* means without additional project SFT. The official model card identifies it as instruction-tuned with upstream post-training. We therefore cannot describe B as a model that never received supervised instruction tuning. A raw Qwen Base checkpoint would define a different starting model. Do not substitute unrelated historical checkpoints for the B-to-S-to-R lineage.

DeepSeek-Prover-V1.5 evaluates capabilities across training stages [11]. That supports the comparison structure, not a prediction of our outcome. The stages receive different amounts of training. R against S therefore does not establish superiority over equal-cost continued SFT. Record training compute separately from the matched inference budget.

Generalization needs a specified change from training to evaluation. Lake and Baroni distinguish nearby generalization from systematic composition in SCAN [13]. This motivates separate panels, without making composition the only research question. Table 8 specifies the intended shifts. Familiar-family performance uses held-out theorems, not training accuracy.

Table 8: Shared evaluation panels. Claims depend on the relation between each panel and known training exposure.

Panel	Change from training	Permitted conclusion
Familiar-family instances	New statements from represented constructions	Reliability within familiar constructions
Held-out families or topics	A reviewed construction or method absent from known training	Transfer under this specified shift
Unseen skill combinations	Components occur separately in training; their combination does not	Transfer to this combination, after exposure review
Longer reference dependencies	Longer chains of represented intermediate claims	Reference-structure extrapolation, not minimum proof depth

Freeze primary panels and comparisons before sampling. Report B, S and R on each panel; do not hide different shifts inside one aggregate. A compatible established benchmark can provide an additional fixed panel after explicit adoption and exposure review. Its name alone does not establish generalization. Its relation to training determines what the result means.

EXP-025 supplies candidate inputs, not a certified structural benchmark. Manas reviews intended meaning, semantic families, component skills, reference strategies and known SFT/RL exposure. Reference dependency depth and tactic length remain separate from minimum mathematical difficulty. Keep current TRAIN, DEV and TEST membership unchanged. Any required expansion needs a separately versioned release. If a panel cannot be supported, report that limitation and retain an explicitly labelled pilot.

8.2 Experiment 2: curriculum effects on performance and generalization

The second experiment holds the SFT checkpoint S and RLVR implementation fixed. Its conditions are R, shuffled RL practice, and C, prerequisite-first RL practice. Both start from the exact S checkpoint. The first intervention changes RL order while keeping SFT unchanged. This measures the total effect of the schedule, without assuming a transfer benefit.

R and C receive permutations of the same TRAIN task-ID multiset, including identical repeat counts. Reviewed skill prerequisites determine ordered stages; freeze within-stage tie-breaking and permutation seeds. Keep frozen-S acceptance as a separate empirical difficulty measure. If calibration determines TRAIN admission, apply one common rule to both arms. Retain exclusions and never filter TEST by its acceptance rate. Use whole theorems without supplied proof prefixes.

R can serve both experiments when every resolved setting and seed matches. Reusing the run avoids duplicate training, but does not add an independent replication. Adaptive sampling, changed theorem allocation and supplied-prefix assistance remain distinct later curriculum interventions. DeepSeek-Prover-V2 incorporates subgoal-derived problems into a curriculum [12]. Our first comparison fixes the theorem multiset and tests order.

Measure C minus R on familiar benchmark theorems and separately on each generalization panel. Table 9 gives the possible interpretations. A familiar-panel gain does not establish transfer. No detected gain does not establish absence of an effect. Report uncertainty and assessment completion alongside every contrast.

Table 9: Interpretation of curriculum results, conditional on a valid comparison and the stated uncertainty analysis.

Familiar-panel result	Generalization-panel result	Interpretation
Gain detected	Gain detected	Evidence for benefits on both named panels
Gain detected	No gain detected	Familiar performance improves; transfer benefit remains unestablished
No gain detected	Gain detected	Evidence for transfer without a detected familiar-panel benefit
No gain detected	No gain detected	No measured curriculum benefit under this protocol

Efficiency is a secondary outcome of the curriculum experiment. Plot prescheduled DEV measurements against actual generated tokens, alongside physical Lean calls and GPU time. Equal group counts do not imply equal cost. Freeze any target score before training and keep an unreached target as unreached. The earlier EXP-023 null order result and failed efficiency gate remain unchanged.

8.3 Common controls and interpretation limits

At evaluation, all conditions receive the same interface, statements, library access, feedback and per-attempt limits. An interface warm-up that updates B's weights is additional training and must not be hidden. Keep evaluation proofs outside SFT, RL, demonstrations and retrieval. Choose checkpoints

and settings on DEV; evaluate TEST after those choices freeze. SFT validation remains a TRAIN-only monitor.

Report complete-proof accepted-attempt fraction and proved theorems within the same fixed attempt budget. Acceptance requires an unchanged statement and complete replay under the pinned verifier. Every planned theorem–attempt pair stays in the operational denominator. Retain rejection, timeout, infrastructure failure, truncation and missing output as separate categories. Material infrastructure failures make the scientific comparison inconclusive. Report completion coverage and resources for failed attempts as well as accepted attempts.

Use paired RL seeds for R against C and retain semantic-family dependence in uncertainty calculations. Freeze primary contrasts and any multiplicity correction before evaluation. Repeated generations of one model are not independent training replications. Few seed pairs or families support descriptive conclusions.

For the curriculum contrast, freeze the optimizer, reward, discount, interaction protocol and hard resource caps. Use no extra shaping or exploration bonus. Compare planned, consumed and uncensored theorem multisets; a cap stop can prevent equal realized exposure. Preserve incomplete runs without raising caps or adding selected retries. Topic clustering and recency remain alternatives to a prerequisite-learning mechanism. Mixed-reward groups do not by themselves explain generalization.

The RLVR condition must name its optimizer. REINFORCE and GRPO can be method arms within generalization, rather than additional major experiments. Their current interactive and whole-proof paths change feedback as well as optimization. Begin with one fixed RLVR recipe; an optimizer comparison requires matched interaction and budget. This avoids a full optimizer-by-curriculum grid in the first study.

8.4 Ownership and shared handoff

Manas owns the curriculum experiment: its design, prerequisite schedules, C training and R/C analysis. Abhinav owns the generalization experiment: its design, B/S/R training and evaluation. Anwesha owns Lean interaction and replay. Abhinav binds B, derives S with the frozen SFT recipe, and derives R from that exact S with the frozen RLVR recipe. Historical checkpoints need verified lineage before reuse. He delivers a B/S/R table for generalization, with separate benchmark-panel columns.

For curriculum, Manas derives C from the same S and compares it with R. Abhinav supplies the immutable S checkpoint, R schedule, resolved settings and R outcomes. The existing `TaskSchedule.frozen` and trainer `--task-order` accept explicit theorem repetitions. The study manifest must check whole-theorem membership and equal theorem multiplicities across schedules. The scheduler does not enforce those cross-arm conditions. Manas delivers the R/C table with familiar performance, generalization and measured cost.

Use optimizer-disabled TRAIN/DEV calibration before fixing training settings. The existing `--assessment-only` interface uses `--assessment-split validation` for DEV and `test` for TEST. Preserve configs, source identities, raw attempts, consumed schedules, failures and checkpoint provenance. Resolve panel counts, model revisions, learning settings, seeds, resource caps and launch costs before execution. The guides are in EXP-026 and EXP-027 of `docs/EXPERIMENTS.md`. These arm definitions do not themselves authorize training or new source admission.

8.5 Execution plan for the number theory study

The immediate study asks whether RL improves independent number theory proofs and whether prerequisite order changes that improvement. The primary sequence is B, one fixed SFT checkpoint S, then matched shuffled R and curriculum C branches. Use the interactive REINFORCE implementation for both branches, with whole-theorem starts and the same theorem repetitions. The separate small

PPO comparison tests optimizer feasibility under its own interface. It cannot supply R for this curriculum study.

Data before theorem difficulty. Broad prerequisite coverage takes priority over selecting theorems the model already proves. Keep the checked interactive action corpus separate from the checked complete-proof corpus. Action SFT targets responses to live states, including format and Lean-error recovery. Complete-proof SFT targets complete reference constructions under a different prompt contract. Do not concatenate these formats or change their mixture without registering the training recipe. Repeated SFT examples do not count as additional mathematical statements. Keep one RL theorem per admitted statement, started from the whole theorem, with an alias ledger that preserves its source proofs.

The current consumer release is `nt_curriculum_data_20261018_v1/validated_v2/`. It supersedes the initial common release for new training. Fresh strict replay accepts all 580 source proofs under the pinned Lean and mathlib pair. All 536 source theorems reconstruct through the selected interaction environment. Full-proof acceptance does not certify the extracted SFT examples. Their separate live checks reject examples that start after closure, continue after closure, or fail the required transition checks.

This release retains 1,880 TRAIN SFT examples. The separate SFT validation retains 125 SFT examples from 32 TRAIN-only theorem names. The primary RL bank contains 506 theorems with whole-theorem starts. Its alias ledger preserves 516 source proof variants. Every source record has a retained or quarantine disposition; no record remains pending. All retained action messages pass exact v2 rendering, parsing and tokenizer checks without truncation. The limits remain 2,048 tokens for context and 192 tokens for an action completion, including its chat terminator. The evidence is in `validated_v2/data_consumer_receipt.json` and `model_contract_v2/validated_v2_fit.json` on the remote artifact store.

Table 10: Source topic labels in the filtered TRAIN bank. The denominator is 506 theorems. These lexical labels do not certify mathematical skill coverage.

Source label	Theorems
Divisibility	217
Primes and factorization	180
Congruences	77
Bézout and coprimality	32

The semantic review reads all 536 canonical TRAIN statements and all 548 TRAIN proof variants. The primary policy excludes 19 valid foundational controls into a separate source pack. It excludes ten AI-reviewed redundancy cases without claiming certified equivalence. It excludes one theorem whose reference invokes a reserved DEV target lemma. These policy exclusions are distinct from rejected action transitions and contradictory assumptions. The source admission certificates remain bound to the actual proof identities and statements. Their scope is formal premise satisfiability, not informal source alignment or unknown pretraining exposure.

Both runtime schedules contain 1,012 groups with identical theorem multiplicities. The reviewed candidate order retains all 54 proposed pedagogical edges and their providers. No annotation has human confirmation. These assignments describe source operations and a proposed teaching order; they do not measure model mastery or establish prerequisite necessity. The current comparison can measure an order effect under these fixed assignments. It cannot isolate prerequisite learning from recency or topic clustering.

Training gaps and reserved transfer concepts require different actions. For a missing TRAIN-eligible prerequisite, add source-bound examples through the existing admission and replay path in a new data revision. Do not relax admission, copy evaluation solutions or move reserved concepts into TRAIN to fill the coverage table. The existing undergraduate syllabus reserves quadratic residues and arithmetic functions for topic transfer. Its capability labels and the informal school-level corpus do not certify formal skill coverage. The review therefore distinguishes observed source coverage, missing annotations, missing examples and deliberately held-out material. The fixed order comparison does not consume a curriculum-validation set or use adaptive stage promotion. A distinct curriculum-validation reservation remains required before an adaptive extension.

Bind S and the comparison. The historical SFT checkpoint is ineligible for the clean primary comparison. Its recorded training examples explicitly invoke `Nat.factorial_mul_factorial_dvd_factorial`, which proves a reserved DEV target up to product order. Removing the source examples cannot remove their prior effect on those weights. Train a fresh S from the pinned upstream B using the SFT examples of the `validated_v2` release, then derive both RL branches from that exact final checkpoint. The historical checkpoint remains available for a labelled familiar diagnostic. Direct-name and exact-statement scans do not establish complete semantic alias independence. Upstream pretraining exposure remains unknown.

The base and historical tokenizers produce identical token IDs for every retained message and theorem prompt. Both model files pass a bounded finite-forward check without generation or updates. These checks establish file and input compatibility; they do not establish proof capability or validate a future GPU training run. Future SFT workers record exact consumed source bytes, selected rows, configuration and tokenizer metadata before loading weights. The fresh S checkpoint does not yet exist.

Before training, use a bounded optimizer-disabled TRAIN/DEV calibration to inspect reward variation and failures. Do not select theorems from TEST or treat loss, valid local tactics or supplied-prefix acceptance as complete-proof capability. In the selected environment version, intermediate goal changes and Lean diagnostics receive zero reward. Terminal reward requires accepted complete replay; the legacy `goal_change_v1` setting does not add shaping in this version. Freeze prompts, action and context limits, discount, optimizer, schedule repetitions, evaluation attempts and hard resource caps together. Bind the completed TRAIN theorem replay and reconstruct the fixed DEV theorems in the actual consumer environment before loading the training model.

Delivery and failure interpretation. Complete semantic annotation, input binding and the costed execution package first. After exact-cost approval, run the S checks and matched RL branches early enough to finish evaluation before Tuesday close. Reserve Wednesday, 21 October 2026, for the counted comparison and its limitations. Use the fixed final checkpoints and preserve every planned theorem–attempt pair, failure and cap stop. Report familiar performance, reviewed transfer panels, completion coverage and physical resource use separately. If semantic review remains incomplete, report exploratory order results without a prerequisite mechanism claim. If rewards remain constant or an arm stops before matched exposure, report that limit without extending caps or selecting replacement theorems. The historical help, null and retention results remain available even if the new comparison is incomplete. The prepared optimizer pilot and this broader program each require their own concrete launch approval. The staged broad-bank package has a quoted reservation of USD 102.31. Its proposed ceiling is USD 110. Its maximum duration is 36 hours. The earlier fallback requires an admitted historical S checkpoint. The confirmed target exposure makes that fallback ineligible for the clean study. These planning envelopes do not authorize spend or guarantee completion.

9 Results of 18 and 19 October 2026

This section reports EXP-028 and the R and C branches of the main run. EXP-028 measures S, arms derived from S, and two SFT runs from B on a new held-out selection set, with S (E0) as the baseline. Table 11 lists the runs. Here B, S, R and C name the models of the main run (Section 8.5), not the EXP-023 arms of Table 6. The run outcome counts below were recounted from the saved run records, without new Lean or model runs. Costs and the selection-set construction counts come from the dated record. Sign-flip p-values below are exact. The original dated entries give Monte Carlo values, and the recount entry of 19 October gives the exact values. The dated records are in EXP-028 of `docs/EXPERIMENTS.md`.

Selection set and decision rule. The EXP-028 selection set comes from the existing corpus. Its construction excludes the TRAIN theorems and their aliases, every SFT source, and the DEV and TEST sets. Of 216 candidates, admission removes 14 that a single fixed tactic proves and 1 with known exposure of the parent. The selection set holds 164 of the 201 admitted theorems. Of the 164 theorems, 79 fall in the four TRAIN topics and 85 in other topics. The median reference proof has 29 lines. Each arm makes 2 attempts per theorem, and each comparison pairs the theorems that both arms assess. The registered rule promotes an arm only if its paired gain over E0 is at least 5 points. Its paired bootstrap 95% interval must also exclude 0. E3 is compared with E5, its SFT seed control, instead of E0. The bootstrap resamples theorems with 10,000 replicates. TEST remains unused.

What did not run. E4 was to test SFT from B on the 1,880 base SFT examples plus 64 SFT examples mined from step search on TRAIN. Its SFT stage failed after 4 s, because the mining script wrote rows without the `format` field that the trainer requires. The frozen lease program could not rerun the stage, so E4 needs a later approved lease. E1, step search without the automation fallback, was dropped before any paid run. Automation runs only after a failed policy attempt, so E1's model-only outcome equals E2's up to Lean timing.

Cost. EXP-028 used an estimated USD 52.28 of instance compute against USD 165.32 reserved. The main run, which trained S, R and C, used an estimated USD 33.77 against USD 102.31 reserved. Both figures use the quoted on-demand rate and exclude storage and transfer. They are estimates, not invoices. The EXP-028 total uses USD 6.23 for Lease 3; another dated entry gives USD 6.59 for the same lease.

9.1 Environment and learner of these runs

Every run in this section uses environment version 2.2 (Section 7). Table 12 defines it with the settings of these runs. Figure 22 shows how the environment processes one action. The settings come from the recorded run configurations. The EXP-030 record in `docs/EXPERIMENTS.md` names the keys in which its configuration differs from R's. The status names in Figure 22 are the action categories of Figure 26 and Section 9.6.

The learner uses the REINFORCE estimator of Equation (1) with a leave-one-out baseline on discounted returns. The loss of action t in attempt i is $-A_{i,t} \log \pi_\theta(a_{i,t} \mid s_{i,t})/G$, and the gradient norm is clipped at 1.0. For attempt i of a group of G attempts, the advantage of action t is

$$A_{i,t} = \sum_{k \geq t} \gamma^k r_{i,k} - \frac{1}{G-1} \sum_{j \neq i} \sum_{k \geq 0} \gamma^k r_{j,k}, \tag{4}$$

where $r_{i,k}$ is the reward after action k of attempt i. The reward comes only at the end, so every action of an attempt gets the same advantage. If all G returns of a group are equal, every advantage is 0, and

Table 11: Runs reported in this section. S is the SFT checkpoint of the main run. B is its pinned upstream model. The selection set has 164 theorems. E0 and E2 assess 163, because one group is censored.

Run	Start	What changed	Population	Attempts
S (E0)	S	Nothing: one candidate action per state, model-only; the baseline	Selection set (163 assessed)	2 per theorem
E2, step search	S	4 candidate actions per state, with an automation fallback	Selection set (163 assessed)	2 per theorem
E3, lemma-section SFT	B	SFT on the SFT examples with a lemma section; evaluated with retrieval	Selection set	2 per theorem
E5, SFT seed 2	B	SFT on the SFT examples without the lemma section, with a second seed	Selection set	2 per theorem
R (E7)	S	REINFORCE on all 506 TRAIN theorems in shuffled order	Selection set	2 per theorem
`RL_filtered` (E6)	S	REINFORCE on 62 TRAIN theorems with S acceptance in $(0, 0.75]$; 1,012 groups of 4	Selection set	2 per theorem
Kimina-Prover-Distill-0.6B, two draws	Released model	A different model; whole proofs with long reasoning; 8,096-token limit per attempt	Selection set	8 per theorem per draw
`pass_rates`	S	Nothing; measures S on TRAIN	506 TRAIN theorems	4 per theorem
R and C training	S	Same theorem multiset; R in shuffled order, C in prerequisite-first order run twice	506 TRAIN theorems	1,012 groups of 4
`dev_R`, `dev_C`	R, C	Nothing; final weights	8 DEV theorems	32 per model

the group gives no update (Section 2.2). With $\gamma < 1$, a group in which Lean accepts every attempt can still update, because proofs that close at different actions have different returns.

9.2 No arm shows a detected held-out gain over S

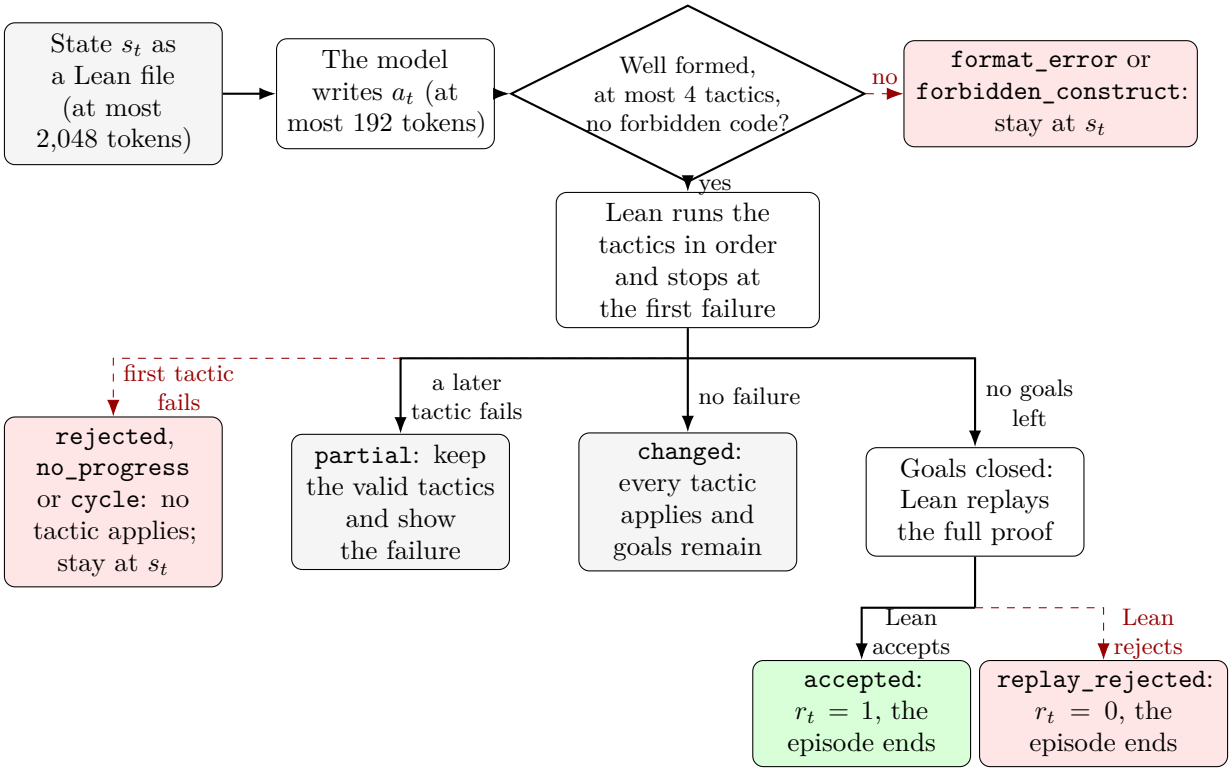
In E0, S proves 1 of 163 selection theorems and accepts 1 of 326 attempts (Figure 23a). E0 assesses 163 of the 164 theorems. On group 143, a root automation probe hit the 20 s Lean timeout, and the running code censored the whole group. A timeout is a failed automation outcome, not an infrastructure fault; later runs record it and continue. Comparisons with E0 therefore pair 163 theorems. E2 loses the same group to the same defect.

E2 proves 1 of 163 theorems, E3 1 of 164, E5 0 of 164, R (E7) 1 of 164 and `RL_filtered` (E6) 2 of 164. No arm meets the promotion rule, and no paired interval excludes 0. The largest gain is E6 against E0: +0.92 points, with a bootstrap 95% interval of $[-0.61, +3.07]$. E3 against its control E5 gains +0.30 points, with an interval of $[0.00, +0.91]$. E0, E2, E3 and E7 each prove 1 theorem. E2 and E3 prove the same theorem, and E0 and E7 each prove a different one. This pattern fits sampling noise at the floor.

Capability probe. The probe asks whether S's near-zero acceptance comes from S or from the difficulty of the selection set. It uses Kimina-Prover-Distill-0.6B, which its model card describes as a distillation of Kimina-Prover-72B [14]. The probe uses the model card's sampling settings and checks each proof through the same verifier path as S. In draw 1, Kimina proves 47 of 164 theorems within 8 attempts. In draw 2, which changes only the batching, it proves 40 of 164. An exact McNemar test on the 25 theorems proved in only one draw gives $p = 0.23$.

Table 12: The interactive MDP, environment version 2.2, with the settings of the runs in this section.

Part	Definition
Theorem	One statement, from TRAIN or a held-out set. No proof lines are supplied
State s_t	The theorem, the policy actions that Lean accepted, the current goals, the result of the last action, the tactics that already failed at the current goals, and the number of actions left
Observation	The state written as a Lean file, with the goals in a `/- tactic state -/` comment. The prompt holds at most 2,048 tokens. If the prompt is longer, the oldest policy actions are hidden. If it is still longer after an action, the episode ends
Action a_t	Lean tactic text of at most 192 generated tokens, bare or in one code fence labelled `lean` or `lean4`, with at most 4 top-level tactics and optional `-` comment lines. Text around a fence, or a second fence, is a format error
Transition	Lean runs the tactics in order and keeps the valid tactics before the first failure (Figure 22). A failure is a Lean error, unchanged goals, or a repeat of an earlier state on the current proof path
Harness revert	After 2 distinct tactics fail at one state, the harness removes the last accepted action and records it as failed at the earlier state. It reverts at most once per action
End of episode	Lean replays a closed proof; 6 actions are used; a Lean call exceeds the 20 s timeout; or the next prompt exceeds 2,048 tokens with every policy action hidden. A REPL crash, or a first prompt over the limit, censors the whole group
Reward	$r_t = 1$ if Lean accepts the replayed full proof after action a_t, otherwise 0. The reward is Lean-verified and has no heuristic term
Discount	$\gamma = 0.95$. A proof closed by a_t, with $t = 0, \ldots, 5$, has return γ^t: 1 for the first action and 0.774 for the sixth
Learner	Equation (4) with $G = 4$ attempts per group. AdamW with learning rate 10^{-6} and no weight decay; gradient norm clipped at 1.0. No KL, entropy or reference-model term
Evaluation	The same environment with no update, 2 attempts per selection theorem. A theorem counts as proved if Lean accepts any attempt



Each action uses one of the 6 actions of an episode. Outcomes other than a closed proof give the next state s_{t+1}; after a self-loop, s_{t+1} has the same goals and a new failure message. After 2 distinct tactics fail at one state, the harness removes the last accepted action, at most once per action. After 6 actions without a proof, a Lean timeout of 20 s, or a next prompt over 2,048 tokens with every policy action hidden, the episode ends with reward 0. A REPL crash censors the group.

Figure 22: How environment version 2.2 processes one action a_t. Solid arrows lead to progress or a proof; dashed arrows lead to a self-loop or a failed replay. The status names are the action categories of Figures 26 and 28.

Limits: the probe is not an arm matched to S. Kimina generates whole proofs with long reasoning, not S's action format. In draw 1, Kimina generates 6,444 tokens per attempt on average, with a limit of 8,096. In E0, S generates about 574 tokens per attempt. Kimina's 8 attempts therefore use far more generated tokens than S's 2 attempts. In draw 1, 853 of 1,312 attempts reach the token limit without a closed proof and count as failures. The pinned Lean and mathlib are probably older than Kimina's training environment, so some lemma names may not exist. The selection theorems have no informal problem text, so the problem field of Kimina's prompt is empty. These three effects bias Kimina's score down. A public prover may have trained on these statements, and this unknown exposure could bias the score up.

Figure 23a shows Kimina's unbiased pass@2 estimates from 8 attempts per theorem. The estimate is 11.35% in draw 1 and 11.67% in draw 2. S proves 0.61% of its 163 theorems within 2 attempts. The panel compares models at an equal number of attempts, not at equal generated tokens. A 0.6B prover with whole-proof generation and a larger generation budget therefore proves many of these theorems. The probe does not separate S's model and training data from its action format and generation budget.

Kimina's proved fraction falls with reference-proof length in both draws (Figure 23b). For references of at most 20 lines, it proves 28 of 62 theorems in draw 1 and 20 of 62 in draw 2. For 21 to 40 lines, it proves 13 of 42 and 12 of 42. For more than 40 lines, it proves 6 of 60 and 8 of 60. Reference length describes one recorded proof; it is not a measured minimum difficulty.

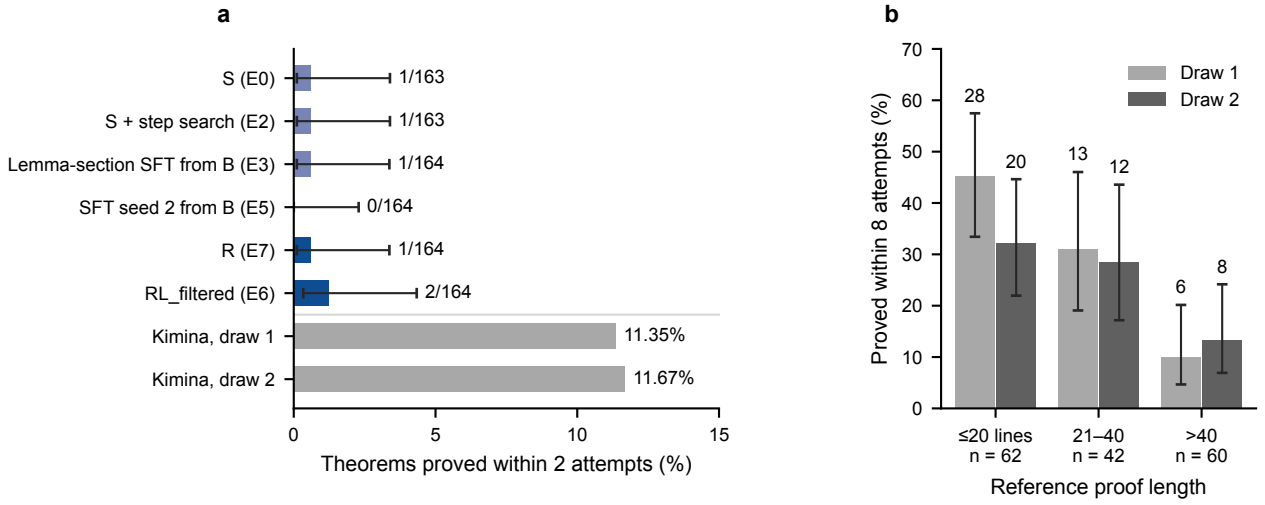


Figure 23: Held-out results on the EXP-028 selection set of 164 theorems. (a) Theorems proved within 2 attempts per theorem: E0 1/163, E2 1/163, E3 1/164, E5 0/164, E7 1/164 and E6 2/164. Kimina draws 1 and 2 appear as unbiased pass@2 estimates from 8 attempts per theorem, 11.35% and 11.67%, with no interval. Blue-grey bars are SFT models, dark blue bars are RL from S, and grey bars are Kimina. The panel holds attempts equal, not generated tokens. (b) Kimina theorems proved within 8 attempts, by reference-proof length, for draws 1 and 2: at most 20 lines, 28/62 and 20/62; 21 to 40 lines, 13/42 and 12/42; over 40 lines, 6/60 and 8/60. Error bars in both panels: 95% Wilson score intervals over theorems, which treat theorems as independent.

9.3 RL raises acceptance on its training theorems without measured transfer

`pass_rates` measures S with 4 attempts on each of the 506 TRAIN theorems. S accepts 84 of 2,024 attempts and proves 63 of 506 theorems. Of the 506 theorems, 443 have no accepted attempt. The registered filter keeps the 62 theorems with an S acceptance rate in $(0, 0.75]$. `RL_filtered` runs

REINFORCE from S on these 62 theorems, with the resource caps of R. It completes all 1,012 groups of 4 attempts, with 337 optimizer updates and no censored group.

Acceptance rises by training quarter: 404, 500, 576 and 602 of 1,012 attempts (Figure 24a). Each theorem is visited 16 or 17 times. The mean first-visit acceptance per theorem is 0.306. The mean last-visit acceptance is 0.597. Of the 62 theorems, 33 are higher at the last visit, 7 are lower and 22 are equal (Figure 24b). An exact paired sign-flip test gives $p < 0.001$ ($p = 2.4 \times 10^{-6}$). On the same 62 theorems, S's `pass_rates` acceptance is 0.323. The first visit is close to this value, so regression to the mean after selection does not explain the rise. R, trained on all 506 theorems, shows no detected rise. Its first visits accept 81 of 2,024 attempts and its second visits 83 of 2,024 ($p = 0.93$).

E6 evaluates the final `RL_filtered` weights on the selection set. It proves 2 of 164 theorems, each in both attempts; E0 proves 1 of 163. One of its theorems is the one that E2 and E3 prove, and the other is the one that E7 proves. Against R (E7), E6 gains +0.91 points, with an interval of $[0.00, +2.44]$. The training gain on 62 theorems therefore does not transfer measurably to the selection set at this set size.

Limits: the rise measures fit to the 62 trained theorems, and it may be memorization. S's SFT examples are extracted from reference proofs of TRAIN statements. The theorem exclusions that define the RL bank also apply to these SFT examples. S may therefore have seen actions from reference proofs of the trained theorems. Recall of SFT-seen proofs is a candidate explanation of the rise; no run here tests it. The data audit of 19 October counts them: 59 of the 62 trained theorems have at least one SFT example from their own reference proof. R and `RL_filtered` also differ in theorem set (506 against 62 theorems), visits per theorem (2 against 16 or 17) and order. No run separates these differences.

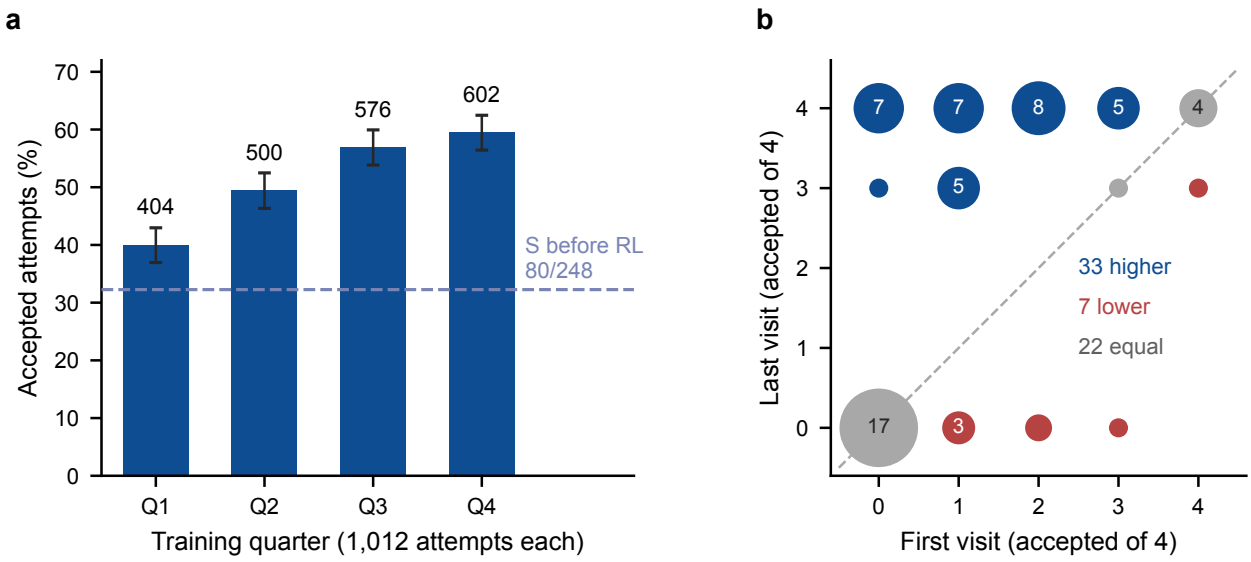


Figure 24: `RL_filtered` on its 62 TRAIN theorems. (a) Accepted attempts by training quarter: 404, 500, 576 and 602 of 1,012 attempts. The dashed line marks S before RL on the same 62 theorems: 80/248 attempts in `pass_rates`, with 4 attempts per theorem. Error bars: 95% Wilson score intervals, which treat correlated attempts as independent and so understate the uncertainty. (b) Accepted attempts of 4 per theorem at the first visit against the last visit, for 62 theorems. Bubble area shows the number of theorems, and the dashed line is $y = x$. Of the 62 theorems, 33 are higher at the last visit, 7 lower and 22 equal. Panel (b) has no error bars.

9.4 In one run per arm, curriculum order raises TRAIN acceptance and narrows the actions

R and C start from the same S and run 1,012 groups of 4 attempts on the same 506 TRAIN theorems. Their configurations differ only in identifiers, and their commands differ only in the task order. R uses a shuffled order, and C runs its prerequisite-first order twice. Both complete all groups with no censored group. C accepts 255 of 4,048 attempts, and R accepts 164 of 4,048. C proves 93 of the 506 theorems, and R proves 63. Per theorem, C minus R is positive for 73 theorems and negative for 32 (paired sign-flip $p = 0.0002$). Each arm is one training run. The sign-flip test treats theorems as independent, so it does not include run-to-run variation or dependence between theorem families.

When the gain appears. S's `pass_rates` run gives a measurement before RL on the same theorems. C's first visits accept 130 of 2,024 attempts, against 84 for S ($p = 0.0019$). R's first visits accept 81 of 2,024 ($p = 0.87$ against S). C's second epoch accepts 125 of 2,024 attempts ($p = 0.79$ against its first epoch). C's gain therefore appears within its first epoch and does not grow in the second. By quarter of C's first-epoch order, C's first visits accept 56, 35, 33 and 6 attempts (Figure 25a). On the same quarters, S accepts 42, 27, 12 and 3 attempts. The quarters hold 504, 504, 504 and 512 attempts.

Exploratory automation check. This check is exploratory, and its rule was not registered. It asks how much of the late-quarter gain consists of proofs that use only automation tactics. Under the strict rule, an accepted attempt is *automation-only* if every action is exactly one automation tactic name with no arguments. Under the argument rule, a tactic with arguments, for example `simp [h]`, also counts. In quarter 3, 3 of C's 33 accepted first-visit attempts are automation-only under the strict rule. Under the argument rule, 15 of the 33 are automation-only. S has 0 of 12 under both rules. In quarter 4, C has 0 of 6 under the strict rule and 4 of 6 under the argument rule. S has 0 of 3 in quarter 4. The dated count did not record its tactic set; the recount uses a set that reproduces the recorded counts.

Narrower actions. Here a quarter is one quarter of a run's training actions, in training order. C's actions narrow over training (Figure 25b and c). C's format-error rate falls from 4.7% of actions in the first quarter to 0.3% in the last. R's format-error rate is 18.1%, 6.7%, 18.6% and 47.9% by quarter. It therefore rises to 47.9% in the last quarter, after a dip in the second. C's mean action length falls from 56.3 tokens in the first quarter to 10.4 in the last. In C's last quarter, actions that are exactly `omega`, `simp` or `simp_all` make 4,597 of 5,845 actions. Under the strict rule, 50 of C's 255 accepted attempts are automation-only, against 1 of R's 164. Under the argument rule, the counts are 105 for C and 2 for R. Outside automation-only attempts, C accepts 205 against R's 163 under the strict rule. Under the argument rule, the remainders are 150 for C and 162 for R. So under the argument rule, C's TRAIN advantage over R does not remain outside automation-only attempts. These action analyses are exploratory, not registered metrics.

No held-out contrast yet. B, S, R and C each prove 0 of 8 DEV theorems. `dev_R` and `dev_C` each accept 0 of 32 attempts. C has no selection-set evaluation in these records. Both arms are at 0 on DEV, so the R/C contrast has no held-out difference to measure, and the TRAIN gains do not show transfer. Table 9 needs a familiar-panel result and a generalization-panel result. TRAIN acceptance is training accuracy, and the familiar-family panel uses held-out theorems instead. No row of that table applies yet.

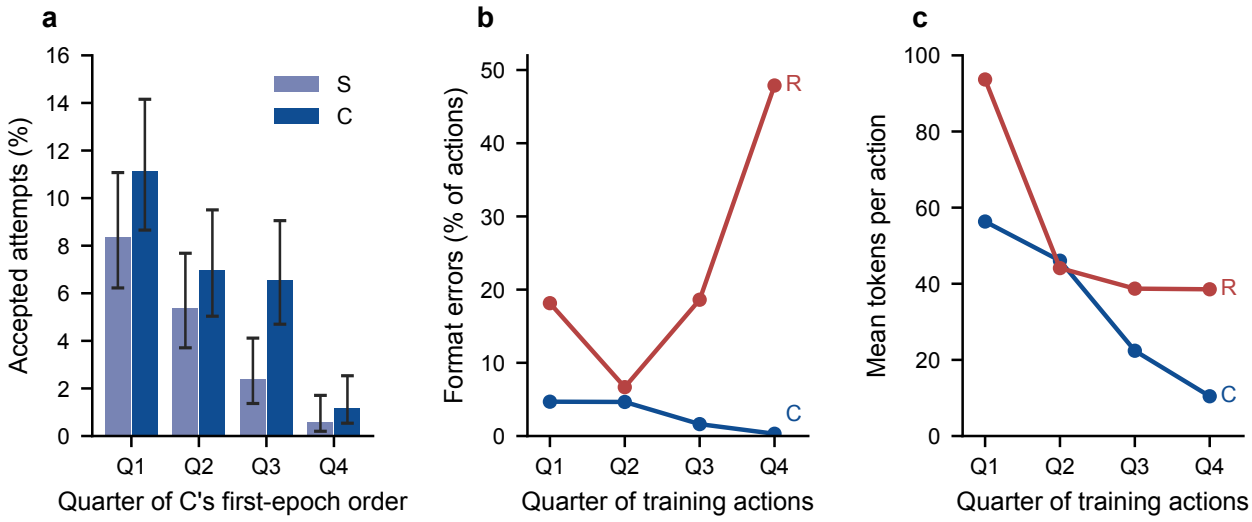


Figure 25: Curriculum order on the 506 TRAIN theorems, with R and C from the same S. (a) First-visit accepted attempts of 4 per theorem, by quarter of C's first-epoch order. S (`pass_rates`): 42/504, 27/504, 12/504 and 3/512. C: 56/504, 35/504, 33/504 and 6/512. Error bars: 95% Wilson score intervals, which treat correlated attempts as independent and so understate the uncertainty. (b) Format errors as a percentage of actions, by quarter of training actions, for C and R. (c) Mean generated tokens per action, by the same quarters. Each quarter holds 5,845 C actions or 5,935 R actions; the last of C's 23,381 actions falls outside the four quarters. Panels (b) and (c) have no error bars.

9.5 RL changes how the model fails

Figure 26 classifies every model action on the selection set for S (E0), R (E7) and `RL_filtered` (E6). Format errors are 373 of 1,953 actions for S. They are 1,288 of 1,963 for R and 18 of 1,948 for `RL_filtered`. Mean generated tokens per action are 95.8 for S, 34.6 for R and 30.8 for `RL_filtered`. In 1,636 of `RL_filtered`'s actions, Lean rejects the first tactic, against 1,411 for S and 468 for R. R mostly writes actions that fail the format check. `RL_filtered` mostly writes short tactics that Lean rejects. On DEV, inspected R format errors repeat one tactic beyond the 4-tactic limit, for example seven consecutive `simp` lines. `dev_R` makes 113 format errors in 192 actions, against 37 for S. `dev_C` makes 1 format error in 192 actions. It generates about 12 tokens per action, against 93.7 for S. This action analysis is exploratory, not a registered metric. The dated record names REINFORCE without a KL or entropy term as a likely contributor; no run tests it.

9.6 Why the TRAIN gain does not reach held-out theorems

RL raises acceptance on the theorems that it trains on (Sections 9.3 and 9.4). No arm shows a detected held-out gain (Section 9.2). On 20 October, Manas asked whether this comes from the RL update or from the reward. This subsection answers from the saved records of the runs above, with no new Lean or model run. We consider five candidate causes: the update, the reward, the starting policy S, the held-out measurement, and the change in the policy's actions. Every result here is exploratory and a postdiction, and none is a registered metric. The EXP-028 entry of 20 October in `docs/EXPERIMENTS.md` lists the scripts and their outputs.

Design. A *visit* is one group of 4 attempts on one TRAIN theorem. R and C visit each of the 506 TRAIN theorems twice, and `RL_filtered` visits each of its 62 theorems 16 or 17 times. The *S reference*

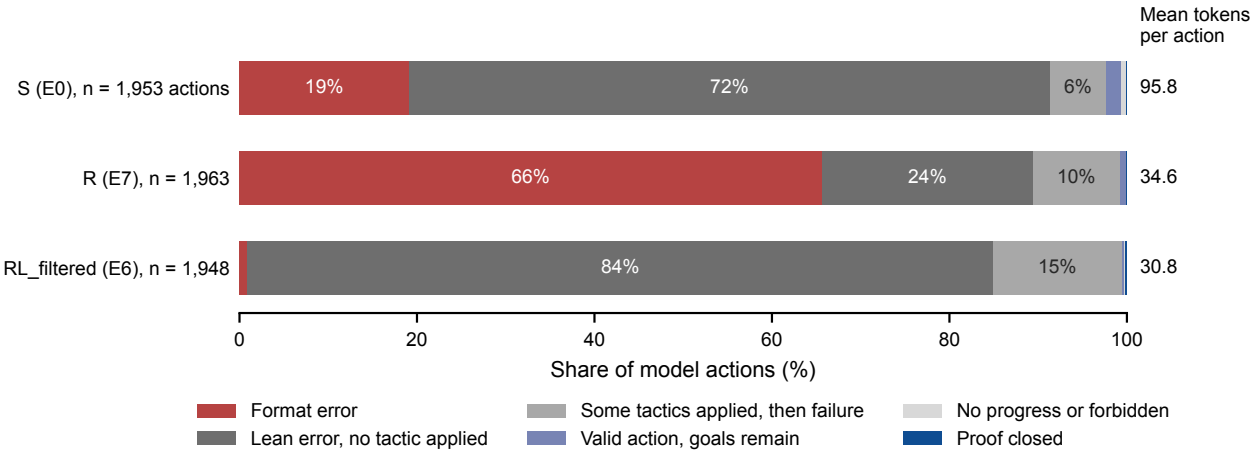


Figure 26: Status of every model action on the 164-theorem selection set, with 2 attempts per theorem: S (E0, 1,953 actions, 163 theorems assessed), R (E7, 1,963) and `RL_filtered` (E6, 1,948). Categories: format error; Lean error with no tactic applied (status `rejected`); some tactics applied, then a Lean error, unchanged goals or a repeated state (`partial`); a valid action with goals remaining (`changed`); no progress or a forbidden construct; proof closed. The right column gives mean generated tokens per action. Bars are 100% stacked shares, with no error bars. This analysis is exploratory, not a registered metric.

of a theorem is the number of S's 4 attempts in `pass_rates` that Lean accepts. For theorem τ and visit v, let $a_\tau^{(v)}$ be the arm's accepted attempts, s_τ the S reference, and $d_\tau = a_\tau^{(v)} - s_\tau$. The test is an exact two-sided paired sign-flip test on $\sum_\tau d_\tau$. We split the theorems by whether the arm's first visit has at least one accepted attempt. If the policy equalled S, the second visit and the S reference would be exchangeable given the theorem, so selecting on the first visit would not bias their difference. The split does not isolate the update from each theorem's own first visit, because the policy also changes through training on other theorems.

Acceptance rises where the reward is nonzero. Figure 27 shows the split. On the 72 theorems where C's first visit has a proof, C's second visit accepts 95 of 288 attempts, against 54 for S ($p = 0.011$; 0.087 after a Bonferroni factor of 8). On the 46 such theorems for R, R's second visit accepts 55 of 184 attempts, against 42 for S ($p = 0.199$). `RL_filtered` trains only on theorems that S proves in 1 to 3 of 4 attempts. Its mean acceptance per theorem rises from 0.306 at the first visit to 0.597 at the last (Section 9.3). At the last visit, 31 of its 62 theorems have 4 of 4 accepted attempts, and 23 still have 0. C's first visits already beat S over all 506 theorems (130 against 84 accepted attempts), so part of C's second-visit excess can come from training on earlier theorems. The rise of `RL_filtered` within its 62 theorems is the cleaner evidence.

The reward is rarely nonzero. S has no accepted attempt on 443 of the 506 TRAIN theorems. R's first visit has a proof on 46 of the 506 theorems, and C's has one on 72. On the other 460 theorems, R's second visit accepts 28 of 1,840 attempts, against 42 for S ($p = 0.136$). On the other 434 theorems, C's second visit accepts 30 of 1,736 attempts, the same count as S ($p = 1.0$). A group with no proof gives every attempt an advantage of 0 (Section 2.2), so the first visit to these theorems gives no update.

S applies fewer tactics on held-out theorems. S applies at least one tactic in 105 of 326 held-out attempts (E0). On TRAIN, it applies one in 1,199 of 2,024 attempts (Figure 28a). By action, the

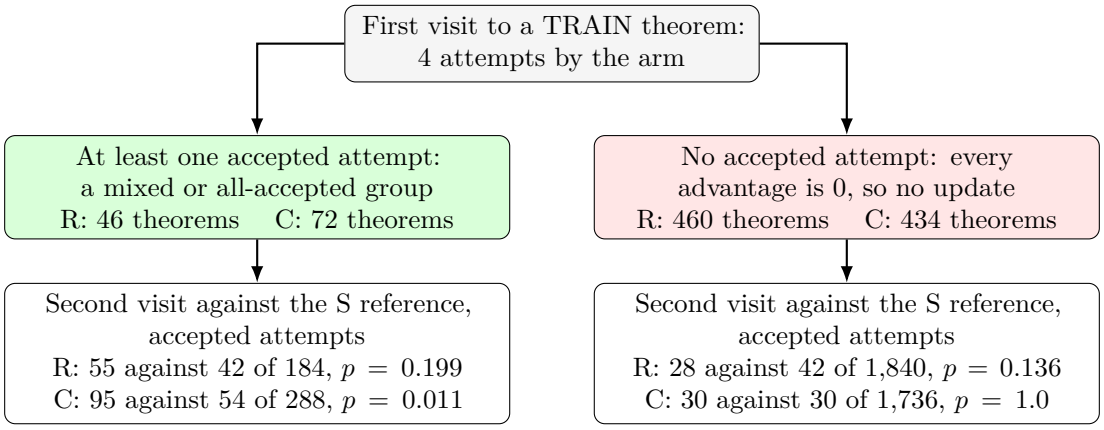


Figure 27: TRAIN theorems of R and C split by the outcome of the arm's first visit. The S reference is S's accepted attempts of 4 per theorem in `pass_rates`, a separate run on the same theorems. The p-values come from exact two-sided paired sign-flip tests on the per-theorem difference, with no correction for the 8 tests. The split puts all-accepted first visits with the mixed groups. This analysis is exploratory.

counts are 156 of 1,953 held-out actions and 2,106 of 11,890 TRAIN actions. The TRAIN theorems also carry SFT exposure. Of the 506 TRAIN theorems, 486 have at least one SFT example from their own reference proof, and so do 59 of the 62 `RL_filtered` theorems. No SFT example comes from a selection-set statement. On the selection set, 161 of S's 326 first actions start with `have` and have more than one line. Lean rejects 147 of them at the first tactic.

RL concentrates on one action template. Of `RL_filtered`'s 2,082 accepted training attempts, 1,627 close the proof with the first action (Figure 28c). S's 84 accepted TRAIN attempts close at every action from 1 to 6. Of the 1,627 closing first actions, 2 are a single fixed tactic without arguments. The 12 most frequent closing texts all start with `simp_all` and a lemma list, and they cover 1,076 of the 1,627. The most frequent text, `simp_all [Int.ModEq, Int.emod_eq_emod_iff_emod_sub_eq_zero]`, closes 451 attempts. In the last quarter of training, 40 theorems have a proof; 14 of them have one distinct proof text, and 10 have two. On the selection set, 326 of `RL_filtered`'s 328 first actions start with `simp_all` and a lemma list. Of these 326, Lean rejects 224 at the first tactic, 94 apply some tactics and then fail, 4 close the proof, and 4 are format errors. The learner has no KL, entropy or reference-model term that would keep the policy near S or keep its actions varied.

The action limits act on R and S. Figure 28b splits the selection-set format errors by cause. Of R's 1,288 format errors, 1,257 have more than 4 tactics, so the environment refuses them before Lean runs them. Of S's 373 format errors, 320 reach the 192-token limit. `RL_filtered` makes 18 format errors. On TRAIN, 893 of S's 1,416 format errors reach the token limit, and 465 more have more than 4 tactics.

The held-out test cannot detect the measured gain. We compute the power of the exact two-sided sign test at $\alpha = 0.05$ on theorems that one arm proves and the other does not. The computation uses two dependence assumptions: in the nested case, the arm proves every theorem that S proves; in the independent case, the two outcomes are independent. Negative dependence, where the arms prove disjoint theorems, is not computed, and it can give a lower power. At the observed rates (S 1 of 163, `RL_filtered` 2 of 164) and 164 theorems, the power is 0.001 (nested) to 0.006 (independent).

A grid search (every 10 theorems below 2,000, every 250 above) finds that the same gain needs 1,310 (nested) to 4,250 (independent) theorems for a power of 0.8. At the registered 5-point bar, 164 theorems give a power of 0.693 to 0.833. These values are for the sign test, not for the registered bootstrap rule.

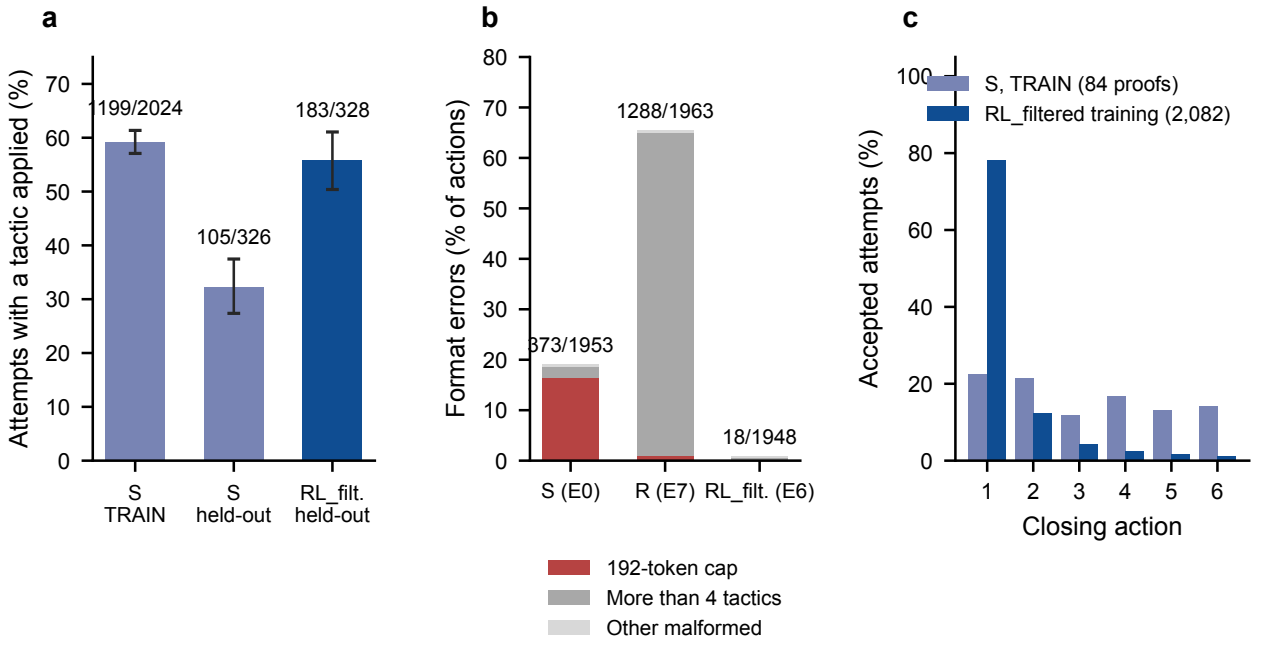


Figure 28: Three measurements behind Section 9.6. (a) Attempts in which at least one tactic applies: S on TRAIN (`pass_rates`) 1,199/2,024, S on the selection set (E0) 105/326, and `RL_filtered` on the selection set (E6) 183/328. Error bars: 95% Wilson score intervals, which treat correlated attempts as independent and so understate the uncertainty. (b) Selection-set format errors as a share of all model actions, split by cause in this order of checks: the 192-token limit, more than 4 tactics, and other malformed actions. S: 320, 44 and 9 of 1,953 actions. R: 20, 1,257 and 11 of 1,963. `RL_filtered`: 1, 10 and 7 of 1,948. (c) Accepted attempts by the action that closes the proof, as a share of each run's accepted attempts. S on TRAIN: 19, 18, 10, 14, 11 and 12 of 84. `RL_filtered` training: 1,627, 258, 90, 50, 34 and 23 of 2,082. Panels (b) and (c) have no error bars. This analysis is exploratory.

Interpretation. The records show no sign of a fault in the update. On the 62 `RL_filtered` theorems, which all have a nonzero S acceptance rate, acceptance rises, and the rise is not regression to the mean. The reward is Lean-verified, and it is rarely nonzero: S proves 63 of the 506 TRAIN theorems in 4 attempts. On held-out theorems, S applies a tactic in fewer attempts than on TRAIN. RL reinforces `simp_all` actions with a lemma list. On the selection set, 326 of `RL_filtered`'s 328 first actions start this way, and Lean rejects 224 of them at the first tactic. The missing held-out gain is therefore consistent with a rare reward under a weak starting policy and with narrow reinforcement, more than with a fault of the update. No run varies the reward, the starting policy, a KL or entropy term, or the action limits, so this attribution is not tested.

Alternative explanations, not controlled:

- Theorems with a first-visit proof are easier. On R's split, S accepts 42 of 184 attempts on them and 42 of 1,840 on the others. A general drift toward short actions could help easy theorems without a theorem-specific update.

- S's TRAIN advantage can come from SFT exposure, from the exclusion of theorems that one fixed tactic proves from the selection set, or from difficulty. No count separates them.
- The `simp_all` concentration changes how `RL_filtered` fails. S does not show it and also proves 1 of 163 held-out theorems, so it is not the only cause.

Limits: each arm is one training run. The visit comparisons use 8 sign-flip tests, 4 per arm. Under a Bonferroni factor of 8, C's first-visit gain over S stays below 0.05 (0.015), and C's split result does not (0.087).

9.7 What the results show

Measured contrasts:

- `RL_filtered` raises acceptance on its 62 trained theorems, from a mean of 0.306 at the first visit to 0.597 at the last ($p < 0.001$).
- C accepts more TRAIN attempts than R, 255 against 164 of 4,048 ($p = 0.0002$), in one training run per arm.
- C's first epoch accepts more attempts than S on the same theorems ($p = 0.0019$).
- R shows no detected rise on TRAIN.
- No arm shows a detected held-out gain: no paired interval on the selection set excludes 0.
- B, S, R and C each prove 0 of 8 DEV theorems.
- RL shortens the actions. R also makes more format errors, and C concentrates on three automation tactics. These action results are exploratory.
- Where an arm's first visit to a TRAIN theorem has no proof, the second visit shows no detected gain over S on that theorem, for R or for C. Where it has one, C's second visit gains over S ($p = 0.011$; 0.087 after a Bonferroni factor of 8). This split is exploratory (Section 9.6).
- `RL_filtered` starts 326 of its 328 held-out attempts with `simp_all` and a lemma list. This count is exploratory.

Hypotheses, not tested:

- RL sharpens behaviour that S already has, rather than adding new proofs. `RL_filtered` trains only on theorems that S proves at least once in 4 attempts, so its training gain cannot show new proofs. Yue et al. report that RL with verifiable rewards raises pass@k at small k but lowers it at large k, relative to the base model [15]. This study does not measure pass@k at large k.
- The training rise reflects recall of proofs seen in SFT. S's SFT examples come from reference proofs of TRAIN statements, including 59 of the 62 `RL_filtered` theorems, and no run separates recall from new skill.

Measurement floor. S proves 1 of 163 selection theorems and 0 of 8 DEV theorems. The promotion rule needs a paired gain of at least 5 points, so a smaller held-out gain cannot pass it. At the observed rates and 164 theorems, an exact sign test has a power of 0.001 to 0.006 (Section 9.6). No detected gain does not show that RL gives no transfer.

10 Post-hoc analyses of 20 and 21 October 2026

This section records every post-hoc analysis of the runs of 18 to 20 October that was done on 20 and 21 October 2026. No analysis used new Lean, model or GPU runs. Each one reads the saved run records

of Section 9. The section asks three questions: what RL changed, what it harmed, and what went well. It does not repeat the results of Sections 9 and 9.6; it cites them. The scripts and their outputs are in the private recount directory `.private/ops/report_recount_20261019_v1/`, in `scripts/` and `out/`. Every number below comes from a JSON file in `out/`.

10.1 How to read this section

- Every result is exploratory and a postdiction. None is a registered metric.
- Each arm is one training run with one seed. Section 10.5 shows that seed noise on the selection set is not small.
- A *valid step* is an action in which Lean applies at least one tactic. A *valid attempt* is an attempt with at least one valid step. A valid step is partial progress. It is not a proof.
- A *visit* is one group of 4 attempts on one TRAIN theorem (Section 9.6). A *decile* or *quarter* is one tenth or one quarter of a run's 1,012 groups, in training order.
- On the selection set, E0 is S, E5 is S with SFT seed 2, E6 is `RL_filtered` and E7 is R (Table 11).
- An *unknown-name error* is a Lean error that reports an unknown identifier or constant. Section 10.3 counts how many of these names are invented lemma names.
- The *template* is a first action that starts with `simp_all` and a lemma list (Section 9.6).

Section 10.5 states which tests survive a correction for multiple tests. It uses two families from `out/p8_findings.json`. The *headline family* holds 11 tests, one primary test per finding. The *278-test family* holds every test in that file. Results from the later scripts `p9_*`, `p10_*`, `p11_regressions` and `p12_positives` are outside both families.

10.2 What RL changed: the short version

RL raised acceptance on the TRAIN theorems it trained on (Sections 9.3 and 9.4). The rise came with one short action template. No arm shows a detected gain in proved held-out theorems (Section 9.2). Table 13 lists the measured changes that this section adds.

10.3 What went wrong

The learning signal is rare. A group gives a learning signal only if its 4 attempts get different returns. Of 1,012 groups, R has 81 mixed groups and 920 groups in which all 4 attempts fail. C has 103 mixed groups and 886 all-fail groups. `RL_filtered` has 207 mixed groups and 398 all-fail groups. Groups with a nonzero advantage are 91 for R, 122 for C and 319 for `RL_filtered` (`p6_signal`, `A.*.total`). In all three runs, the last decile has fewer such groups than the first: R 11 against 6, C 25 against 0, and `RL_filtered` 61 against 21 (Figure 29). The fall is not monotone, and it has no test. This rare signal is consistent with the missing held-out gain. No run changes the reward or the start model, so this cause is not tested.

The action limits cut off real proofs. Section 9.6 shows that the token and tactic limits cause most format errors. A greedy split of each proof into actions also shows how many known proofs fit the limits (Figure 30). Under the current limits, 314 of 516 TRAIN reference proofs fit. Under relaxed limits of 12 actions, 8 tactics and 384 tokens, 416 of 516 fit. For Kimina's accepted proofs, 118 of 180 fit the current limits and 166 of 180 fit the relaxed limits (`p6_limits`). The greedy split does not show that a model would find these proofs under relaxed limits.

Table 13: Scorecard of the post-hoc analyses. "Before" is S, or the start of training. "After" is the RL model, or the end of training. All entries are exploratory and come from one run per arm. Files are in `out/`.

Change	Measure	Before	After	n, test and file
Better	Selection theorems with a valid step, S against `RL_filtered`	84	116	163 paired; McNemar 55 gains, 23 losses, $p = 0.00038$; `p11_regressions`
Better	Next visit repeats the winning first action, `RL_filtered`	loser 0.023	winner 0.286	51 theorems; sign-flip $p = 1.0 \times 10^{-10}$; `p9_update_follow`
Better	TRAIN theorems proved where S had no accepted attempt; R and C had 8 attempts, S had 4	S 0	R 29; C 54	443 theorems; `p12_positives`
Worse	`RL_filtered` TRAIN theorems proved, early half to late half	50	40	62; McNemar 1 against 11, $p = 0.0063$; `p6_coverage`
Worse	R TRAIN theorems proved, S against R's first visit, 4 attempts each	63	46	506; McNemar 20 gains, 37 losses, $p = 0.033$; `p11_regressions`
Worse	Unknown-name first errors per 100 selection actions	S 23.91	`RL_filtered` 45.02	1,953 and 1,948 actions; `p10_remedies`
Worse	Identical repeat after an error, C, quarter 1 to quarter 4	72 of 4,557	1,872 of 4,190	error actions; `p7_feedback`
Worse	`RL_filtered` actions with a comment, quarter 1 to quarter 4	208 of 4,480	0 of 3,149	actions; `p7_feedback`
Worse	C distinct first actions, decile 1 to decile 10	403 of 408	84 of 404	first actions; `p5_diversity`
Worse	Jensen-Shannon distance of tactic heads to Kimina	S 0.477	R 0.814; `RL_filtered` 0.918	1,953; 1,963; 1,948 action texts; `p7_heldout`
Same	R TRAIN theorems proved, S with 4 attempts against R with 8	63	63	506; 29 gains, 29 losses, $p = 1.0$; `p11_regressions`
Same	Accepted TRAIN proofs equal to an SFT reference	S 1 of 84	`RL_filtered` 0 of 2,082	accepted attempts; `p5_recall`

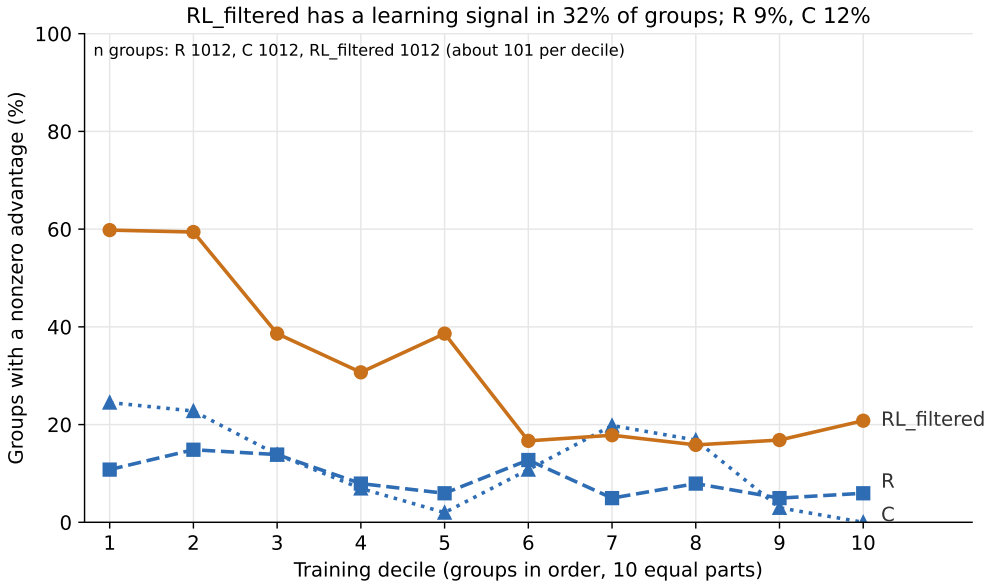


Figure 29: Groups with a nonzero advantage, as a percentage of the groups in each training decile. x-axis: training decile, groups in training order in 10 equal parts. y-axis: percentage of groups with a nonzero advantage. n: 1,012 groups per run, about 101 per decile. Totals: R 91, C 122, `RL_filtered` 319 of 1,012. No error bars. Source: `p6_signal`, `A.*.deciles`.

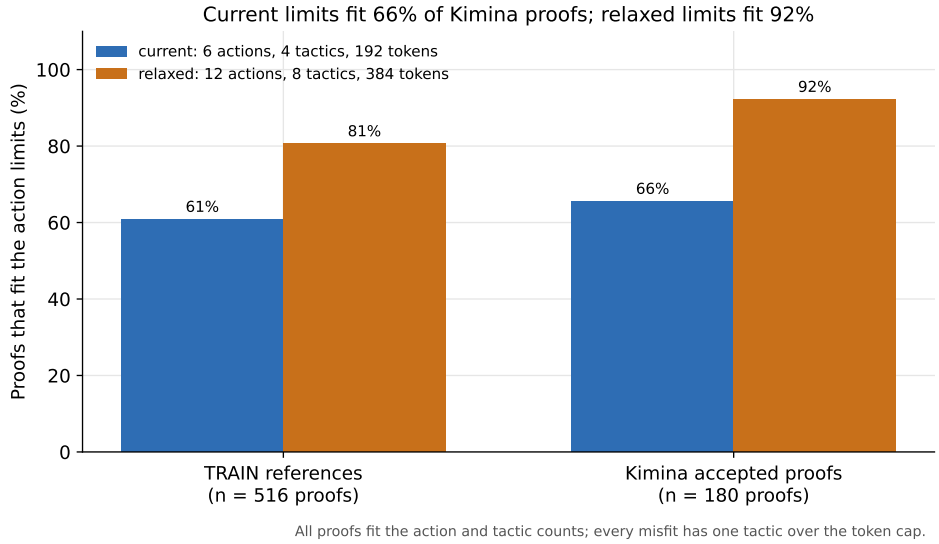


Figure 30: Proofs that fit the action limits after a greedy split into actions. x-axis: proof source, TRAIN SFT reference proofs ($n = 516$) and Kimina accepted proofs ($n = 180$). y-axis: percentage of proofs that fit. Bars: current limits (6 actions, 4 tactics, 192 tokens) and relaxed limits (12 actions, 8 tactics, 384 tokens). Counts: 314 and 416 of 516; 118 and 166 of 180. No test. Source: `p6_limits`.

The updates reinforced 1-action attempts. In quarter 1 of `RL_filtered`, the feature "1 action" gets an advantage mass of +46.96. The feature "6 actions" gets −100.84, and "0 tactics applied" gets −52.18 (`p7_reinforced`). The mean closing action of accepted attempts falls from 2.08 in quarter 1 to 1.14 in quarter 4, over 404 and 602 accepted attempts. After a mixed group, the next visit of `RL_filtered` repeats the winning first action in a mean share of 0.286 of its first actions, and the losing one in 0.023. Of 51 theorems, 37 are positive and 1 is negative (sign-flip $p = 1.0 \times 10^{-10}$; `p9_update_follow`). R (46 theorems, $p = 1.0$) and C (71 theorems, $p = 0.3125$) show no detected effect of this kind (Figure 31). Updates from other theorems happen between two visits, so this result is an association, not an isolated effect of one update.

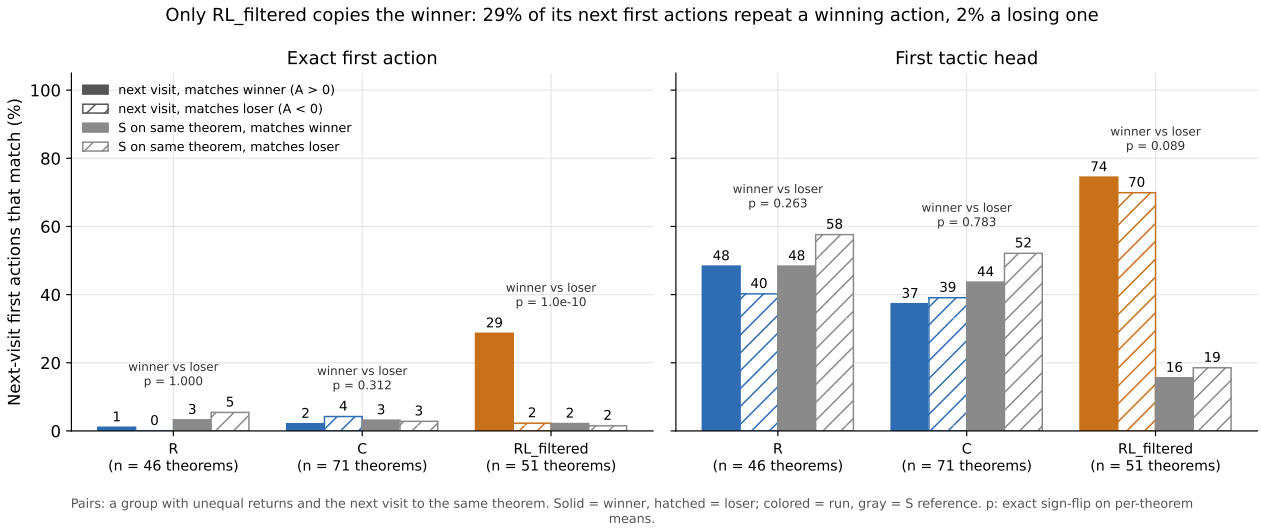


Figure 31: Does the next visit repeat the winner of a mixed group? Each pair is a group with unequal returns and the next visit to the same theorem. x-axis: run, with R ($n = 46$ theorems), C ($n = 71$) and `RL_filtered` ($n = 51$). y-axis: percentage of next-visit first actions that match the winning action (solid) or the losing action (hatched). Gray bars: S's samples on the same theorems. Left: exact first action. Right: first tactic head. p: exact sign-flip test on per-theorem means, winner against loser. No correction. Source: `p9_update_follow`.

One template took over. Section 9.6 reports the template on the selection set. During training, its share of `RL_filtered`'s first actions rises from 0.252 in decile 1 to 1.000 in decile 10 (Figure 32; `p5_diversity`). On a theorem, the template first becomes the majority of first actions after the first accepted visit on 33 theorems, and before it on 3 (binomial $p = 2.3 \times 10^{-7}$; `p7_spillover`). The test leaves out 15 theorems where both happen in the same visit and 11 theorems with no accepted attempt. Of 1,935 accepted attempts with a `simp_all` lemma list, 1,380 use a list that also occurs in an accepted attempt on another theorem, earlier or later in training. The most used list closes 814 attempts on 22 theorems (`p10_lemmas`). C changes too, on theorems that gave it no signal. On 423 such TRAIN theorems, the share of C's first actions that are one automation tactic rises from 0.031 at visit 1 to 0.517 at visit 2, against 0.0006 for S on the same theorems (sign-flip $p = 2.8 \times 10^{-91}$; Figure 33). The learner has no KL or entropy term (Table 12). No run adds one, so we cannot say that the missing term caused the collapse.

`RL_filtered` proves its usual theorems more often but proves fewer theorems. From the early half to the late half of training, the mean pass@1 per theorem rises from 0.446 to 0.583 (36

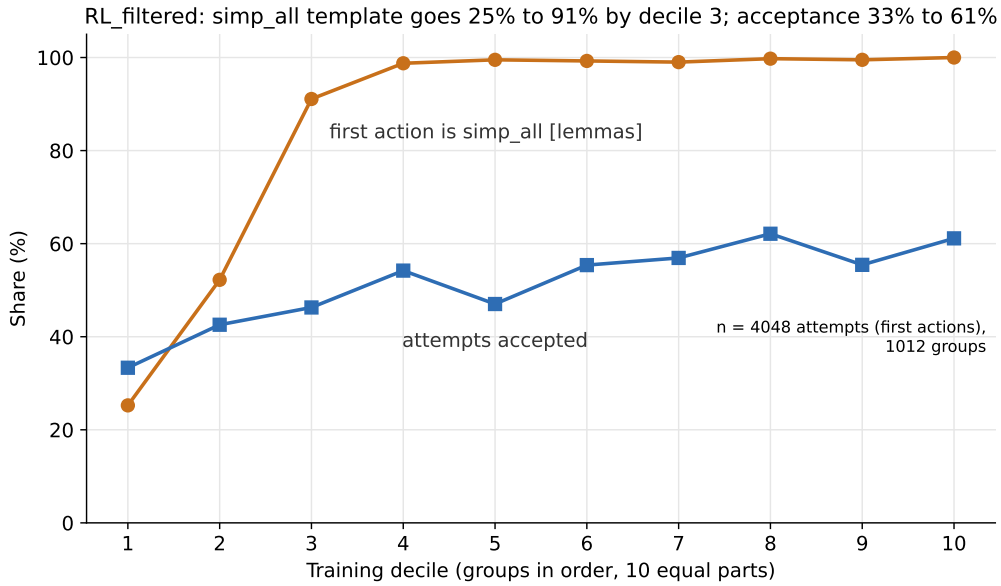


Figure 32: The template in `RL_filtered` training. x-axis: training decile, groups in order in 10 equal parts. y-axis: percentage of first actions. Orange: first actions that start with `simp_all` and a lemma list. Blue: accepted attempts. n: 4,048 attempts (one first action each) in 1,012 groups. No error bars. Source: `p5_diversity`, `runs.RL_filtered.deciles`.

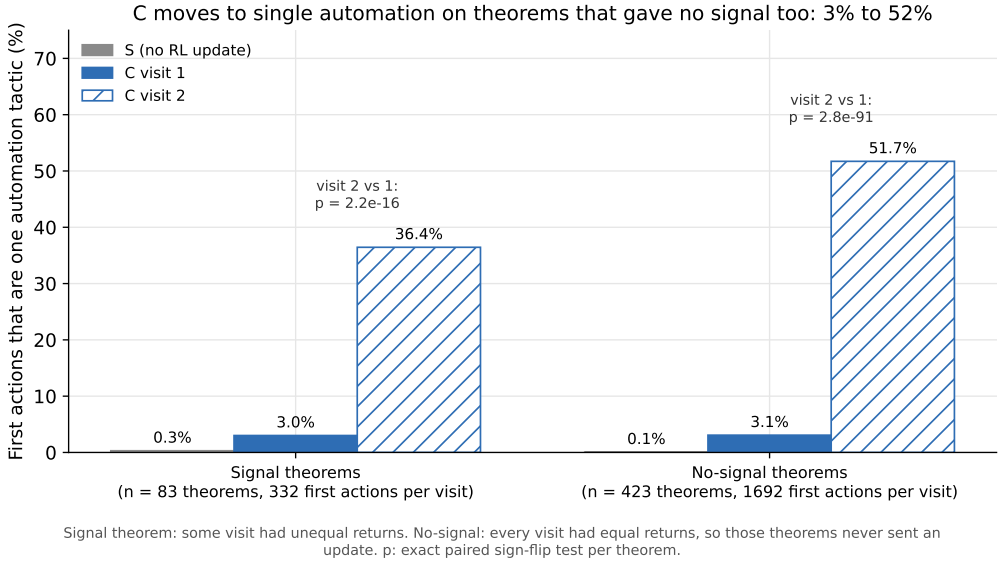


Figure 33: C moves to one automation tactic on theorems with and without a learning signal. A signal theorem had unequal returns in some visit ($n = 83$ theorems, 332 first actions per visit). A no-signal theorem never had unequal returns ($n = 423$, 1,692 first actions per visit). x-axis: theorem class. y-axis: percentage of first actions that are one automation tactic. Bars: S on the same theorems, C visit 1 and C visit 2. p: exact paired sign-flip test per theorem, visit 2 against visit 1. Source: `p7_spillover`, `part1.C`.

theorems higher, 14 lower, sign test $p = 0.0026$). Here pass@1 is the chance that one attempt is accepted. Theorems proved fall from 50 to 40 of 62 (McNemar 1 against 11, $p = 0.0063$; Figure 34). Each half mixes policies, so pass@k here is an approximation. A rule fixed before the results sorts the 62 theorems into 27 learned, 11 lost, 13 stable and 11 never proved (Figure 35; `p10_trajectories`). "Lost" means proved in visits 1 to 8 and never in visits 9 to 16. For the 11 lost theorems (`p11_regressions`, `B_rl_filtered_lost`):

- S's 14 accepted attempts on them include 0 that start with the template. 13 of the 14 use a tactic other than `simp_all` or `omega`, mostly `have` and `interval_cases`.
- The template's mean share of first actions on these theorems rises from 0.724 in visits 1 to 8 to 0.991 in visits 9 to 16.

This pattern is consistent with the template pushing out the `have` and `interval_cases` proofs, but no test isolates that.

R and C also lose TRAIN theorems that S proves (`p11_regressions`, `A_train`). With 4 attempts each, R's first visit loses 37 S-proved theorems and gains 20 ($p = 0.033$). C's first visit loses 29 and gains 38 ($p = 0.33$). 16 of the 63 S-proved theorems are proved by neither R nor C in 8 attempts. S proves each of these 16 in only 1 of its 4 attempts, so they can be sampling noise. No count separates noise from forgetting.

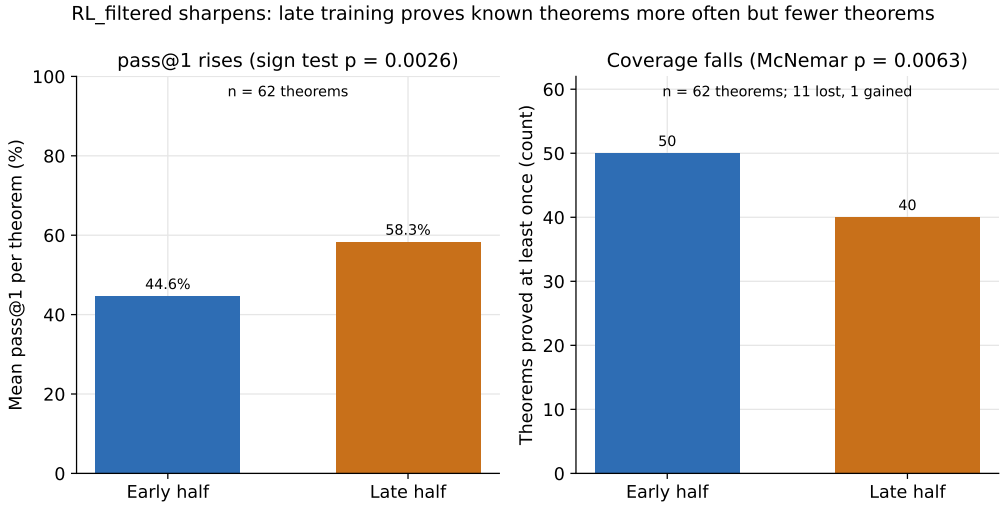


Figure 34: `RL_filtered` on its 62 TRAIN theorems, early half against late half of training. Left: y-axis is the mean pass@1 per theorem (%); 44.6% against 58.3%; sign test $p = 0.0026$. Right: y-axis is the number of theorems proved at least once; 50 against 40; McNemar $p = 0.0063$, 11 lost and 1 gained. $n = 62$ theorems. Each half mixes policies, so pass@1 is an approximation. Source: `p6_coverage`, `B`.

`RL_filtered` and C use Lean's error messages less. In `RL_filtered`, the next action applies a tactic after an error in 662 of 3,260 cases in quarter 1, and in 181 of 2,120 in quarter 4 (`p7_feedback`). In C, an identical repeat of the failed action follows an error in 72 of 4,557 cases in quarter 1, and in 1,872 of 4,190 in quarter 4. In `RL_filtered`, it rises from 122 of 3,260 to 420 of 2,120 (Figure 36). After a failed named tactic, `RL_filtered`'s next action starts with the same tactic in 4,663 of 5,953 pairs. After a valid action, the baseline is 213 of 885 (`p10_readerror`). After an unknown-name error, `RL_filtered` keeps the same name in 871 of 2,260 next actions, against a baseline of 103 of 808 (Figure 37). The Fisher tests in Figure 37 treat pairs inside one attempt as independent. They are not, so these p-values are optimistic.

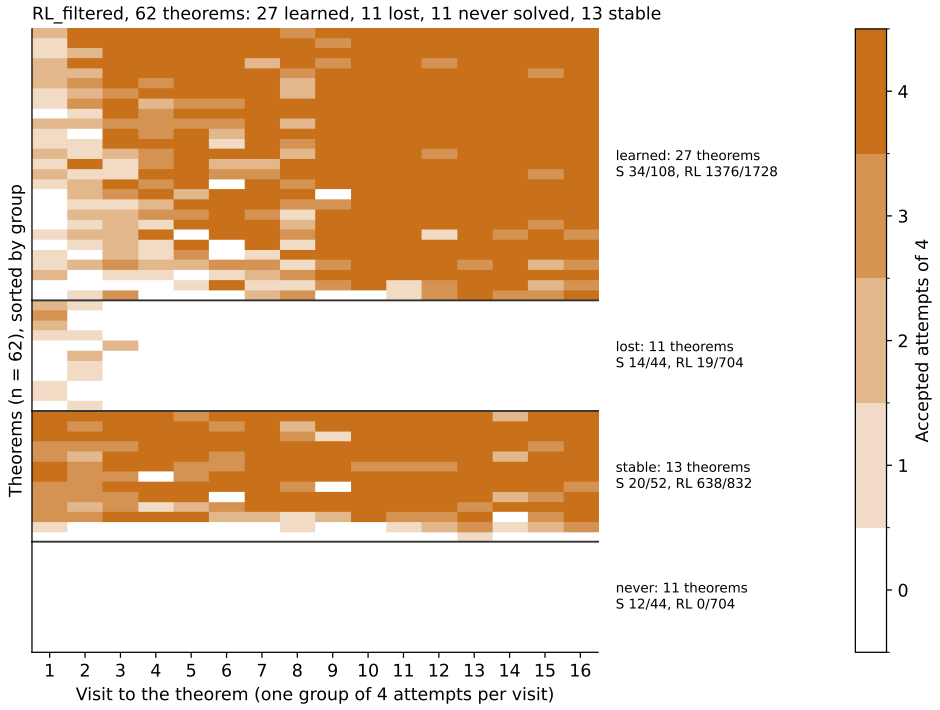


Figure 35: Accepted attempts of 4 at each visit, for each of the 62 `RL_filtered` theorems. x-axis: visit to the theorem (one group of 4 attempts), visits 1 to 16. y-axis: theorems sorted by group. Color: accepted attempts of 4. Groups follow a rule fixed before the results: learned (27), lost (11), stable (13) and never proved (11). The text beside each block gives S's and `RL_filtered`'s accepted attempts of all attempts on those theorems. Source: `p10_trajectories`, `A`.

Comments disappear. Actions with a comment fall in all three runs from quarter 1 to quarter 4. R falls from 728 of 5,905 to 30 of 5,955, C from 350 of 5,765 to 22 of 5,869, and `RL_filtered` from 208 of 4,480 to 0 of 3,149 (`p7_feedback`). Comments do not change what Lean checks. We did not test whether their loss changes proof success.

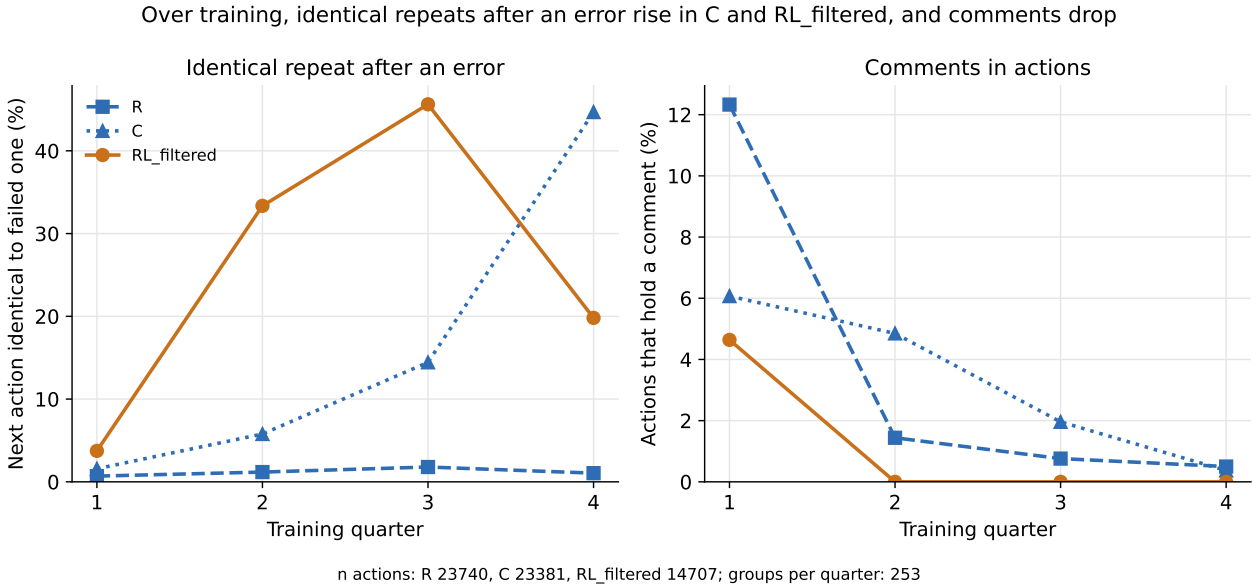


Figure 36: Feedback use and comments during training. x-axis: training quarter, 253 groups per quarter. Left: y-axis is the percentage of error actions whose next action in the same attempt is identical. Right: y-axis is the percentage of actions that hold a comment. n: R 23,740, C 23,381 and `RL_filtered` 14,707 actions. No error bars and no test. Source: `p7_feedback`, `runs.*.quarters`.

Invented lemma names on held-out theorems. On the selection set, `RL_filtered` makes an unknown-name first error in 45.02 of every 100 actions. S makes one in 23.91 of every 100 (`p10_remedies`). Of `RL_filtered`'s 1,919 failed selection actions, 877 are an unknown-name error and 948 are a failed named tactic, mostly `simp_all` (614) and `omega` (298). For S, 467 of 1,534 failed actions are an unknown-name error (Figure 38; `p5_errors`). Of `RL_filtered`'s distinct unknown names, 174 of 274 are wholly invented lemma names. For S, it is 88 of 178 (`p10_lemmas`). A name counts as wholly invented if its normalized edit distance to the nearest real name is above 0.2, for the full name and for its last component.

The RL models' tactics are further from Kimina's than S's tactics. Kimina proves 56 of 164 selection theorems with 16 samples each. S proves 1 of 163 (`p6_coverage`). The Jensen-Shannon distance of top-level tactic heads to Kimina's is 0.473 for the SFT examples and 0.477 for S. It is 0.814 for R and 0.918 for `RL_filtered` (Figure 39; `p7_heldout`). The distance is 0 for the same distribution and 1 for no overlap. Kimina proves theorems with shorter reference proofs: the median difference is -16.5 lines ($p = 7.0 \times 10^{-5}$). Kimina writes a whole proof in one output, with a larger token budget. The distance compares tactic names only.

C's late acceptance drop is consistent with theorem order. C accepts 39 of 408 attempts in decile 1 and 4 of 404 in decile 10 (`p5_diversity`). On about the same theorems, its first pass accepts 16 of 808 attempts and its second pass 9 of 808 (`p11_regressions`). These counts are consistent with

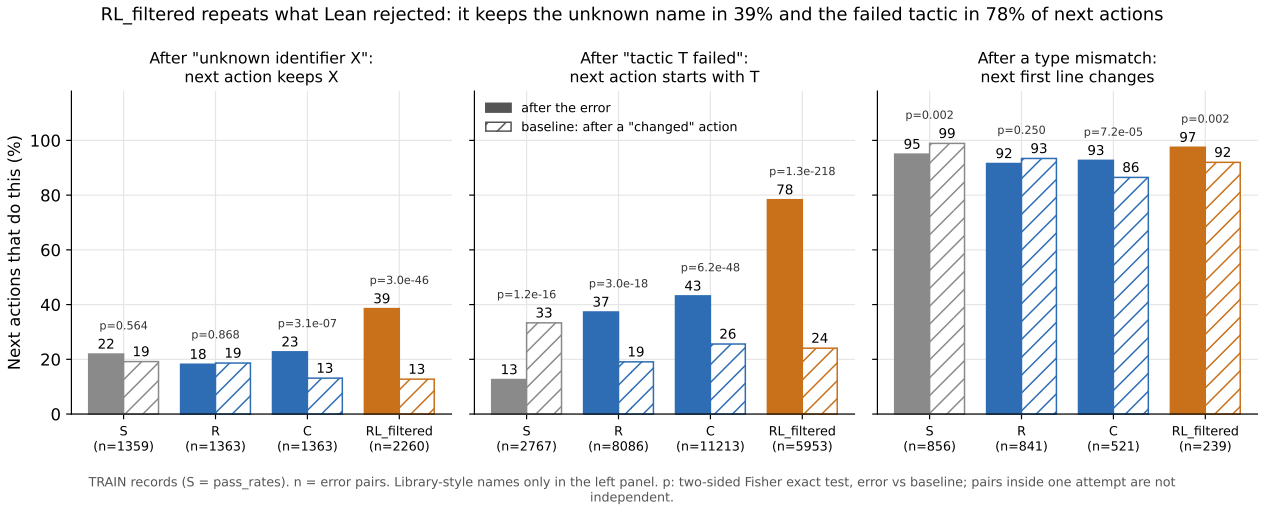


Figure 37: The next action after three Lean errors, on TRAIN records (S is `pass_rates`). y-axis: percentage of next actions that keep the unknown name (left), start with the failed tactic (middle), or change the first line after a type mismatch (right). Solid bars: after the error. Hatched bars: the baseline, after a valid (`changed`) action. n under each run: error pairs. p: two-sided Fisher exact test, error against baseline. Pairs inside one attempt are not independent, so the p-values are optimistic. Source: `p10_readerror`.

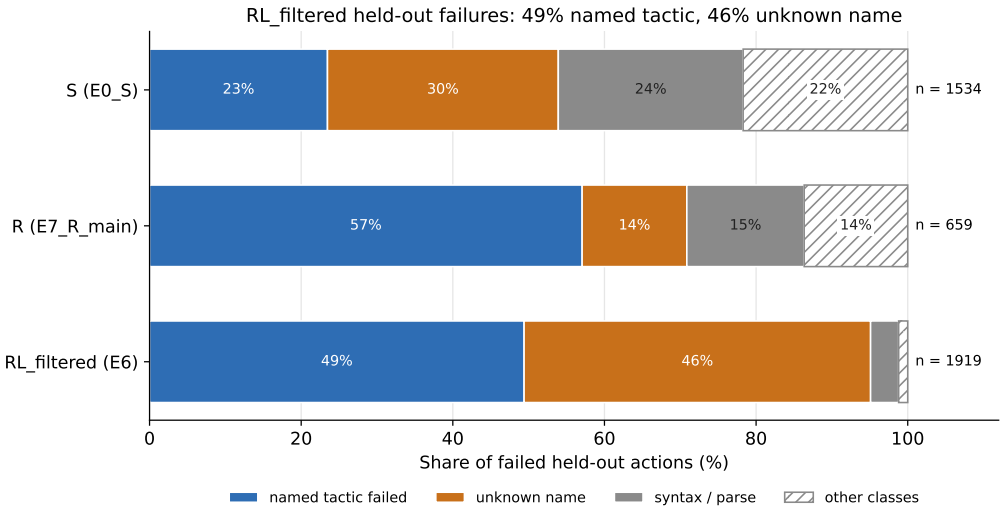


Figure 38: First-error classes of failed actions on the selection set. x-axis: share of failed held-out actions (%). y-axis: run. Classes: failed named tactic, unknown name, syntax or parse error, and other classes. n: failed actions, S (E0) 1,534, R (E7) 659 and `RL_filtered` (E6) 1,919. No test. Source: `p5_errors`, `runs.*.classes`.

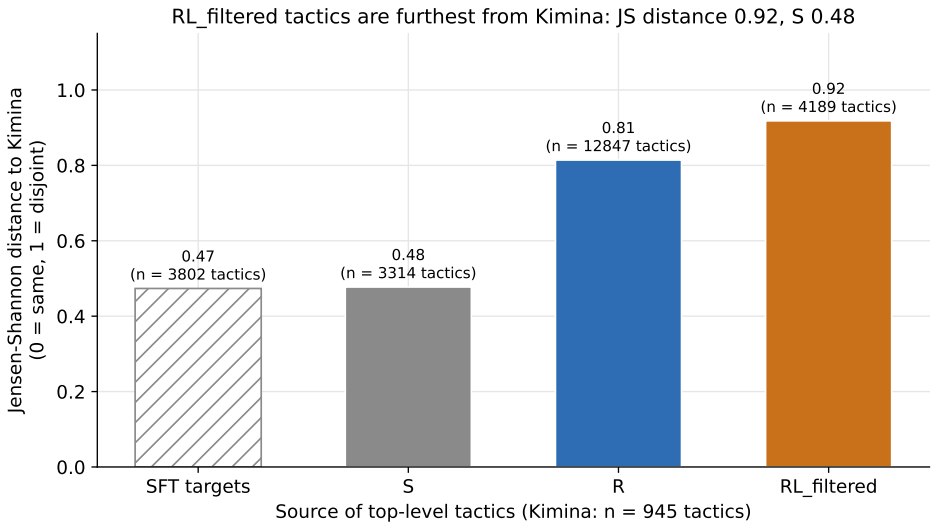


Figure 39: Jensen-Shannon distance between the top-level tactic heads of each source and those of Kimina's accepted proofs. x-axis: source. y-axis: distance, 0 for the same distribution and 1 for disjoint ones. n: SFT targets 3,802 tactics, S 3,314, R 12,847, `RL_filtered` 4,189 and Kimina 945. No interval. Source: `p7_heldout`, `vocabulary.top_level`.

theorem order as the main source of the decile-10 drop. The same-theorem drop is smaller. Neither comparison has a test.

The dense reward of EXP-030 gives penalized actions a positive weight. EXP-030 did not run (Section 11). A re-scoring of the recorded actions under its reward (Table 16) finds a design flaw. The baseline counts all actions of the other attempts, but the reward-to-go of a late action drops the early penalties of its own attempt. A late penalised action can therefore get a positive weight. Of R's 22,962 penalised actions, 19,750 get a positive weight. For `RL_filtered`, it is 8,306 of 11,656 (Figure 40; `p9_counterfactual`). The script reproduces the recorded advantages with a maximum absolute difference of 0.0. This is an off-policy re-scoring. It gives the direction of the first update only, not the result of a dense-reward run. The records do not show that this finding caused the stop of EXP-030.

10.4 What went well

Table 14 lists each positive with its control and its caveat. Positives that Section 9 already reports, for example the `RL_filtered` TRAIN rise and C against R, appear there with their tests.

Held-out partial progress beyond seed noise. Figure 41 shows valid steps on the selection set. The control is two S models: E0, and E5, a fresh SFT from B with SFT seed 2. Valid attempts are 105 for E0, 130 for E5, 183 for E6 and 146 for E7 (`p12_positives`). E6 beats both S models, and E5 against E0 measures the noise of S's seed and sampling. R (E7) against E5 gives 58 theorems up and 48 down ($p = 0.27$). Held-out theorems similar to the 27 learned theorems do not show more E6 progress. Only 6 of 163 are similar (token Jaccard at least 0.5). Their mean E6-minus-E0 difference is 0.0, against 0.4968 for the other 157 (exact permutation $p = 0.34$). With 6 similar theorems, this check cannot detect a small effect.

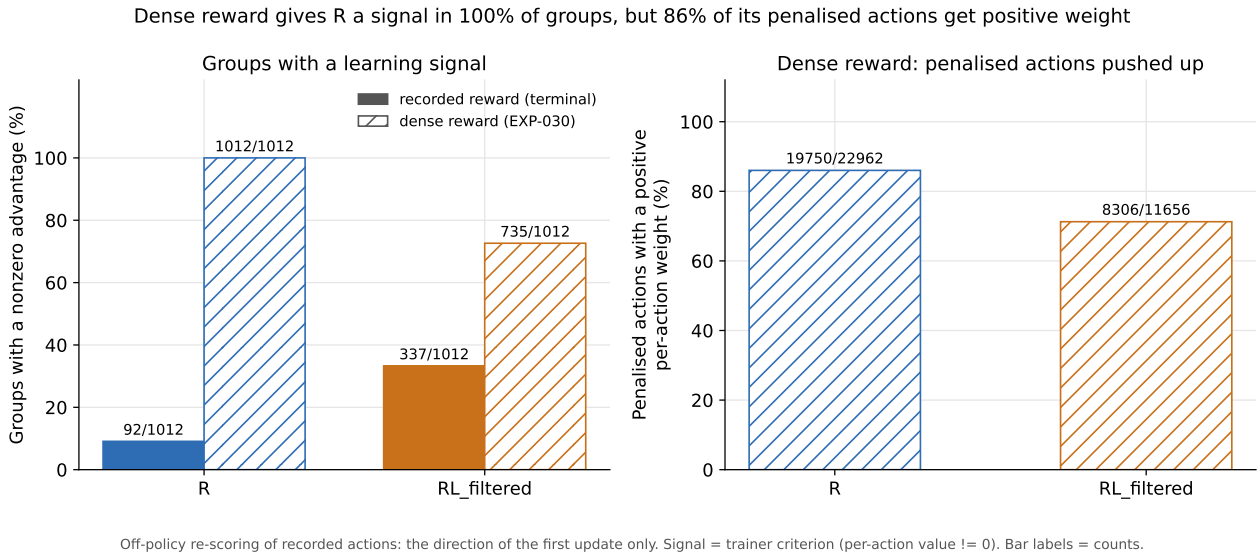


Figure 40: Off-policy re-scoring of recorded actions under the EXP-030 dense reward. Left: y-axis is the percentage of groups with a nonzero advantage, under the recorded terminal reward (solid) and the dense reward (hatched); counts of 1,012 groups. Right: y-axis is the percentage of penalised actions with a positive per-action weight; R 19,750 of 22,962 and `RL_filtered` 8,306 of 11,656. x-axis: run. The figure gives the direction of the first update only. Source: `p9_counterfactual`.

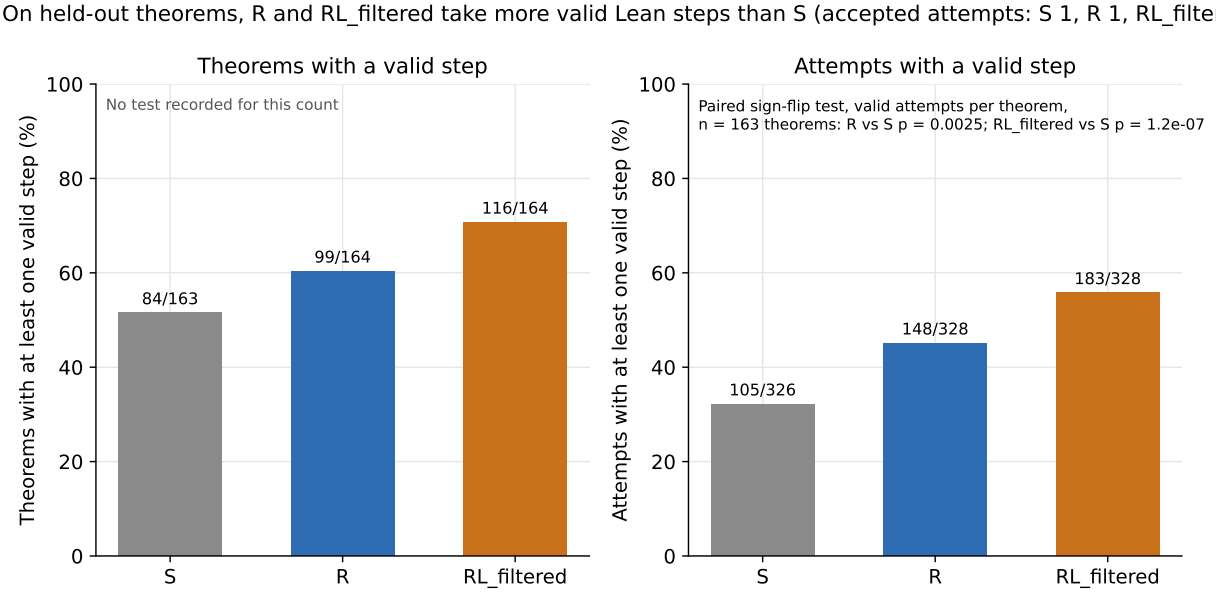


Figure 41: Valid steps on the selection set, 2 attempts per theorem. x-axis: model, S (E0), R (E7) and `RL_filtered` (E6). Left: y-axis is the percentage of theorems with at least one valid step; S 84/163, R 99/164, `RL_filtered` 116/164; no test. Right: y-axis is the percentage of attempts with a valid step; 105/326, 148/328 and 183/328; paired sign-flip test on valid attempts per theorem over 163 theorems, R against S $p = 0.0025$, `RL_filtered` against S $p = 1.2 \times 10^{-7}$. Source: `p7_heldout`, `selection`.

Table 14: Positives of the post-hoc analyses, each with its control and its caveat. All are exploratory and come from one run per arm. Files are in `out/`.

Positive	Result and test	Control	Caveat
`RL_filtered` held-out valid attempts beat two S models	Against E0: 82 theorems up, 31 down, $p = 1.2 \times 10^{-7}$. Against E5: 76 up, 40 down, $p = 0.00068$. 163 paired theorems; `p12_positives`	E0 and E5. E5 against E0: 50 up, 31 down, $p = 0.030$	Partial progress, not proofs. E6 proves 2 of 164
`RL_filtered` has the most valid attempts of E0, E5 and E6	63 of the 98 theorems with a unique maximum, of 163; chance gives 32.667; exact $p = 3.8 \times 10^{-10}$	E0 has the maximum on 12, E5 on 23	Many valid steps are `simp_all` steps that leave the goal open
The updates follow the winner	Share 0.286 against 0.023; 37 up, 1 down of 51 theorems; $p = 1.0 \times 10^{-10}$; `p9_update_follow`	The loser of the same group	Association. R and C show no detected effect
Learned proofs are new text	0 of 2,082 accepted attempts equal an SFT reference. On the 27 learned theorems, 69 of 69 late distinct texts differ from the references; `p5_recall`, `p12_positives`	S: 1 of 84 equal a reference	Distinct text is not a distinct method. Most texts are the template
New TRAIN proofs where S had none	Of 443 theorems, R proves 29 and C 54; C has 54 gains, 24 losses, McNemar $p = 0.0009$; `p12_positives`, `p11_regressions`	S with 4 attempts	R and C had 8 attempts, S had 4. R's gains equal its losses (29 and 29)
Task filtering gives more signal	Mixed groups: `RL_filtered` 207, R 81, C 103 of 1,012; `p6_signal`	R and C, same start model and group count	62 theorems, not 506. No held-out proof gain
Fewer format errors in `RL_filtered` training	142 of 4,480 actions in quarter 1, 10 of 3,149 in quarter 4; 48 theorems down, 2 up, sign test $p = 2.3 \times 10^{-12}$; `p12_positives`	The same theorems in quarter 1	TRAIN only. R's format errors rise
Fewer unknown-name errors in C training	Share of non-format actions with an unknown-name error: 0.1514 at visit 1, 0.0549 at visit 2. Sign test on the per-theorem rate over all actions: 374 down, 66 up, $p = 2.7 \times 10^{-53}$	The same theorems at visit 1	Comes with the move to automation, which cannot invent a name
Fewer tokens per accepted attempt	`RL_filtered`: 388.2 in quarter 1, 141.4 in quarter 4; `p12_positives`	The same 62 theorems	Shorter outputs are part of the template collapse
Step search raises partial progress	E2 231 against E0 105 valid attempts; 99 up, 6 down, $p = 1.0 \times 10^{-23}$; `p10_remedies`	E0 on the same 163 theorems	E2 proves 1 of 163
The selection set is provable	Kimina proves 56 of 164 with 16 samples each; `p6_coverage`	S proves 1 of 163	Not under our action limits
The analysis found the dense-reward flaw before a run	19,750 of 22,962 penalised R actions get a positive weight; `p9_counterfactual`	Recorded sparse advantages	Off-policy re-scoring

Other fixes during training. These fixes are on TRAIN only. In R, the share of non-format actions with an unknown-name error falls from 0.146 at visit 1 to 0.0861 at visit 2. A sign test on the per-theorem rate over all actions gives 296 theorems that fall and 115 that rise ($p = 1.7 \times 10^{-19}$). R's format errors rise in the same comparison, from 0.164 to 0.2928 of actions (311 theorems rise, 138 fall, $p = 2.0 \times 10^{-16}$). `RL_filtered`'s unknown-name rate is mixed. Per action, it rises from 0.1741 to 0.2328 (Fisher $p = 3.4 \times 10^{-10}$). Per theorem, it falls in 40 and rises in 22 ($p = 0.030$). On held-out theorems, `RL_filtered` makes more unknown-name errors than S (Section 10.3).

Records. The archived inputs match their recorded hashes: 19 of 19 data members, 2 of 2 schedule members and 3 of 3 selection-set members (`.private/ops/abhinav_data_handoff_20261020_v1/member_checks.txt`). The file `runtime_order_manifest.json` has no recorded hash. One record chunk has an S3 expiry date of 18 November 2026. Without a copy, the records stop being reproducible after that date.

10.5 How sure we are

Table 15 lists the corrected tests. In the 278-test family, 86 tests survive Holm at 0.05 (`p8_findings`). All 11 tests of the headline family survive Holm at 0.05.

Table 15: Holm-corrected p-values for the tests of `out/p8_findings.json`. "Headline" is the 11-test family. "278" is the family of every test in that file. H1 is not in the 278-test family.

H	Finding	Raw p	Headline Holm p	278 Holm p
Survives both families				
8	`RL_filtered` against S, held-out valid attempts	1.2×10^{-7}	1.2×10^{-6}	2.9×10^{-5}
10	C, no-signal theorems, single-automation share, visit 2 against visit 1	2.8×10^{-91}	3.1×10^{-90}	7.7×10^{-89}
11	`RL_filtered` template before against after the first acceptance	2.3×10^{-7}	2.0×10^{-6}	5.7×10^{-5}
6	Kimina proves theorems with shorter reference proofs	7.0×10^{-5}	4.9×10^{-4}	0.014
Survives the headline family only				
1	`RL_filtered` TRAIN acceptance, last visit against first	2.4×10^{-6}	1.9×10^{-5}	not in family
2	C first visit against S	0.0019	0.011	0.36
3	C second visit against S, where the first visit had a proof	0.011	0.019	1
4	`RL_filtered` coverage, early half against late half	0.0063	0.019	1
5	`RL_filtered` pass@1, late half against early half	0.0026	0.013	0.47
7	C transfer band (direction opposite to transfer)	0.0072	0.019	1
9	R against S, held-out valid attempts	0.0025	0.013	0.47

H7 contradicts transfer through similar theorems. C's theorems with an earlier accepted similar theorem gain less, not more: a mean of 0.040 against 0.209 (`p6_transfer`).

The following results are outside any correction:

- every result from `p9_*`, `p10_*`, `p11_regressions` and `p12_positives`, which includes the update-follow test, the error responses, the lemma reuse, the remedies, the dense-reward flaw, every per-theorem loss and gain, and the 48 tests of `p12_positives`;
- the breakdown cells of `p8_breakdown`;
- every DEV number, because DEV has 8 theorems.

Every arm is one training run with one seed. E5 against E0 gives 50 theorems up and 31 down ($p = 0.030$), so the seed noise of S is not small.

10.6 Replication in EXP-023

EXP-023 is the curriculum run of 13 October (Section 4.7), with a different start model and theorem set. Its records allow four of the harms to be checked (`p9_exp023`, first and last decile).

- The diversity loss replicates in a weaker form in arms B and C_seq, and not in arm A (Figure 42). Distinct first actions go from 94 of 120 to 97 of 112 in arm A, 107 of 120 to 93 of 112 in arm B, and 98 of 120 to 68 of 112 in arm C_seq. The EXP-023 arms lose at most 21 points, against 78 for C and 49 for `RL_filtered`.
- More identical repeats after an error replicate in two of three arms. Arm A goes from 20 of 345 to 31 of 301, and arm C_seq from 38 of 343 to 77 of 215. Arm B falls, from 20 of 452 to 11 of 164.
- The comment drop cannot be tested. EXP-023 has no comments in decile 1 (A 0 of 491, B 0 of 615, C_seq 0 of 503 actions).
- Acceptance does not fall late. Arm A goes from 69 of 120 to 63 of 112, arm B from 37 of 120 to 89 of 112, and arm C_seq from 65 of 120 to 70 of 112.

The EXP-023 deciles hold about 120 first actions, against about 404 in October. A smaller sample gives a higher distinct share, so only the trends are comparable.

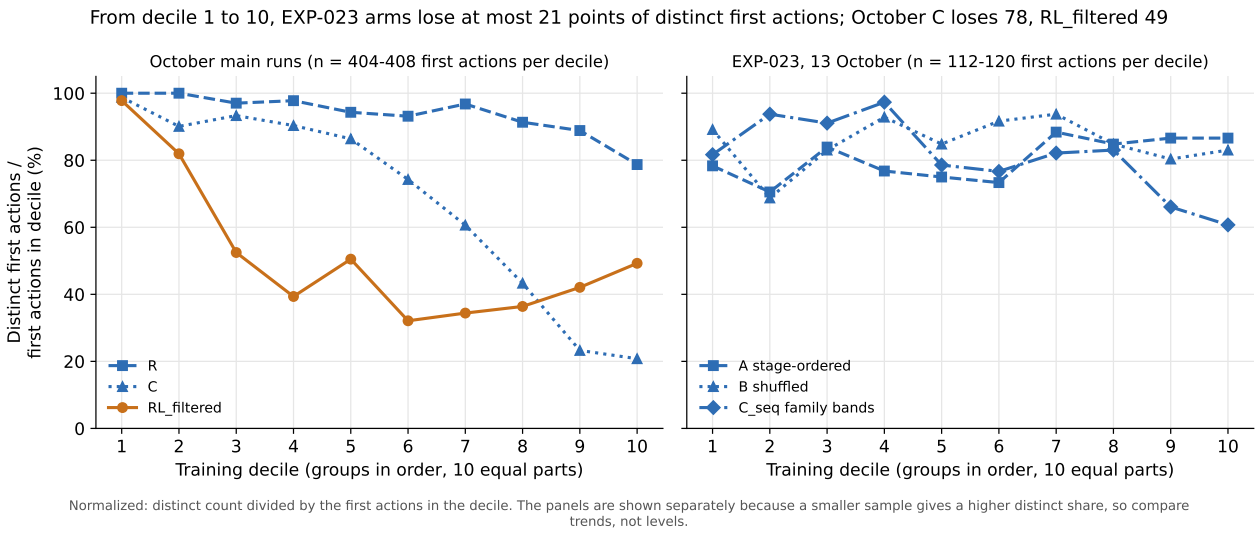


Figure 42: Distinct first actions as a percentage of the first actions in each training decile. x-axis: training decile, groups in order in 10 equal parts. Left: the October runs R, C and `RL_filtered`, 404 to 408 first actions per decile. Right: the EXP-023 arms of 13 October, 112 to 120 first actions per decile. The panels have different sample sizes, so compare trends, not levels. No error bars. Source: `p9_exp023` and `p5_diversity`.

10.7 Retracted and weakened claims

- C's late acceptance drop is consistent with theorem order, not with a loss on the same theorems (Section 10.3).
- Transfer through similar theorems is contradicted. H7 points in the opposite direction ($p = 0.0072$ raw; Table 15).
- Part of the held-out valid-step gain over E0 is seed noise. On the 163 paired theorems, E5, a second S, has a valid step on 94, against 84 for E0 (`p12_positives`).

- Lemma retrieval (E3) gives no detected change in unknown-name errors. E3 has 23.87 per 100 actions, against 23.32 for its control E5 ($p = 0.743$; `p10_remedies`).

10.8 Open questions and the test for each

No new run is planned. Each item names the test that would answer it.

1. Does a KL or entropy term stop the template collapse? Test: rerun `RL_filtered` from S with a KL term to S, with the same theorems, budget and seed. Compare the template share and the late-half coverage.
2. Is the rare reward the main cause of the missing held-out gain? Test: start RL from a model that proves more TRAIN theorems, or train on theorems that S proves at a higher rate. Measure proved held-out theorems on a larger set.
3. Would relaxed action limits change what RL reinforces? Test: rerun R under 12 actions, 8 tactics and 384 tokens. Compare format errors and the closing action.
4. Did the template push out the proofs of the 11 lost theorems? Test: in a KL run, check whether these 11 theorems keep their early proofs.
5. Are the 16 TRAIN theorems that only S proves forgetting or sampling noise? Test: sample S and R 32 times each on these 16 theorems and compare the per-theorem pass rates.
6. Would retrieval of real lemma names lower the unknown-name errors after RL? Test: run E3-style retrieval on `RL_filtered` and compare against `RL_filtered` without retrieval.
7. Is there a held-out gain too small to detect? Test: evaluate S and `RL_filtered` on a held-out set large enough for the observed rates (Section 9.6 gives the power).
8. Does a corrected dense reward help? Test: first make the baseline count the same actions as the reward-to-go. Then re-score the recorded actions as in `p9_counterfactual` before any run.

11 Follow-up designs that did not run

Two follow-up designs were registered on 19 and 20 October and prepared up to a dry run. On 20 October, Manas stopped all experiments before any launch. Neither design ran, so neither has a result or a GPU cost. This section records their motivation and setup, because they state the open questions. Their programs, configurations and dry-run quotes are in the EXP-029 and EXP-030 records of `docs/EXPERIMENTS.md`.

EXP-029: more attempts, screened strata and the action limits. Three observations motivate EXP-029:

- Two attempts per theorem put every arm at the measurement floor (Section 9.7).
- The data audit of 19 October found three unchecked risks in the selection set. At least 5 of the 22 statements with quantified hypotheses that were read by hand have contradictory premises; Lean has not checked this. No screen tried short automation chains. The near-duplicate screen compares text only, so it does not flag theorems close to TRAIN proofs.
- The selection set has already been used to choose arms, so any further result on it is exploratory.

EXP-029 evaluates S, C, R and `RL_filtered` with 8 attempts per theorem on the 164 selection theorems. Three screens split the theorems into four strata: vacuous hypotheses, automation chains, near TRAIN, and "other". The "other" stratum is the headline. EXP-029 also evaluates S and `RL_filtered` with 8

attempts on the 62 `RL_filtered` theorems. The paired comparisons, the bootstrap and the 5-point promotion rule are those of EXP-028.

The amendment of 20 October fits the program on one L4 GPU within about USD 50. Its dry-run quote is USD 49.51 for 28 hours, against USD 118.75 for the first program on one L40S. It adds an exploratory comparison on the 62 theorems: S with 12 actions, 384 action tokens and 8 tactics per action, against S at the registered limits. The limits recount of 20 October on TRAIN (`pass_rates`) motivates it:

- 1,935 of S's 2,024 attempts use all 6 actions without a proof.
- S's 84 proofs close at every action from 1 to 6 (Figure 28c).
- The token and tactic limits cause 1,358 of S's 1,416 format errors.
- At $\gamma = 0.95$, a proof closed by the last action earns 0.774 with 6 actions and 0.569 with 12.

The relaxed configuration also changes the instruction from "up to 4 tactics" to "up to 8 tactics". The comparison therefore measures the new limits and the new wording together.

EXP-030: a dense per-action reward. Most first visits in R have no proof, and a group with no proof gives no update (Section 9.6). Manas proposed a dense reward. EXP-030 asks whether per-action penalties for Lean errors make REINFORCE close more TRAIN proofs than the terminal proof reward alone. R_dense starts from the same S and uses R's seed and the first 506 groups of R's frozen task order, which cover 377 distinct TRAIN theorems. Its configuration differs from R's only in the reward mode, the run budgets and the run identifiers. Table 16 gives the reward. The control is R's existing record of the same 506 groups, in which R closes 87 of 2,024 attempts. The test is an exact two-sided sign-flip test on the per-group difference in accepted attempts, with groups paired by position. R_dense passes only if it closes more proofs than R with $p < 0.05$. A higher return or fewer errors without more proofs does not pass. A gaming audit compares the shares of valid actions and of no-progress actions between the arms. R trained on an L40S, and R_dense would train on an L4, so their sampled attempts would differ even with the same seed.

Table 16: The per-action reward of EXP-030 (`error_feedback_v1` in environment version 2). A partial action is scored by the failure that stopped it. An action whose next prompt exceeds the limit ends the attempt and keeps the reward of its outcome. The discount stays $\gamma = 0.95$.

Event after an action	Reward	Label
Lean accepts the complete proof	+1	Lean-verified
Format error, syntax error, forbidden code, unresolved name, or invalid backtrack	−0.2	Heuristic
Type or tactic error, unchanged goals, repeated state, unclassified Lean rejection, or rejected replay	−0.1	Heuristic
Valid action, timeout, or crash	0	None

The registration of `error_feedback_v1` on 11 October gave 0 to unchanged goals, repeated states and unclassified rejections. EXP-030 gives them −0.1. Errors do not end a version 2 attempt, so a reward of 0 would let a policy avoid every penalty by repeating an action that changes nothing. The penalties are heuristic: they score Lean errors, not proofs, so a lower error rate does not show more proof skill. `RL_filtered` makes 18 format errors in 1,948 held-out actions and proves 2 of 164 theorems (Section 9.6), so fewer format errors alone did not give held-out proofs. The dense reward also penalizes Lean rejections, which are 1,636 of `RL_filtered`'s 1,948 held-out actions. No record shows what that penalty would change; this is an inference, not a result of EXP-030.

12 Limitations

- The evaluation sets are small (3 DEV theorems in Experiment 19; 4 to 8 items in Experiment 11), and every run used one seed.
- Several sets were reused across experiments, so they now act as development sets.
- We do not know how much Lean or mathlib the model saw in pretraining.
- AI reviewers graded the natural-language answers in Experiments 9, 10, 11, and 13.
- The sample-size values come from simulations with assumed parameters.
- The numbers come from recorded run outputs; we did not repeat experiments for this report.
- The EXP-028 selection set and the 8 DEV theorems sit near the measurement floor: S proves 1 of 163 selection theorems and 0 of 8 DEV theorems. The promotion rule needs a paired gain of at least 5 points, so a smaller held-out gain cannot pass it.
- Section 9.6 is a postdiction from one training run per arm. No run tests its account of the missing held-out gain.

Sources. Each experiment has a record in `docs/EXPERIMENTS.md`. The earlier detailed report (Git commit `e3b7747`) contains every count of Experiments 1 to 13 and the full checker audit. The reward exploits are in `docs/REWARD_HACKING.md`. The script `scripts/report/make_report_figures.py` draws the data figures and names the record of each number.

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A Every evaluation theorem and its expected answer

This catalogue describes the encoded theorems from the evaluation review pack prepared on 17 October 2026. It preserves their source order, mathematical domains and task IDs. The source contains 34 DEV theorems and 82 TEST theorems. A description explains the statement; it does not repair or certify its intended mathematical meaning. Existing semantic concerns remain next to the affected theorem. Theorems without a warning still need the source-intent and exposure reviews recorded in `docs/EXPERIMENTS.md`.

Every expected answer is a complete Lean proof of the unchanged statement under the pinned verifier. Existence theorems require witnesses with proofs of their conditions. Equivalences require both directions; conjunctions require every conjunct. A negated universal claim requires its negation, rather than a proof of the positive claim. The proof need not match the private reference. No proof solutions, curriculum order, difficulty ranking or model outcomes are supplied here.

Notation. $\mathbb{N}$ includes zero, $\mathbb{Z}$ denotes integers, and $\mathbb{Q}$ denotes rationals. φ denotes Euler's totient function; J and the parenthesized symbol $\left(\frac{a}{n}\right)$ denote the Jacobi symbol. ord denotes Lean's multiplicative order, and $\mathbb{Z}/m\mathbb{Z}$ denotes the residue ring used by `ZMod m`. Divisor sums range over the finite natural divisor sets in the source. Natural subtraction truncates at zero; division domains are specified where they affect interpretation. The original statement and initial state remain in the remote review pack.

A.1 DEV theorems

DEV 1. **Odd powers and factorials modulo 16**

Prove all three claims: every odd integer y satisfies $y^8 \equiv 1 \pmod{16}$; every natural $x \geq 6$ satisfies $x! + 1 \equiv 1 \pmod{16}$ after casting $x!$ to the integers; and for natural $x \geq 6$ and integer y, $x! + 1 = y^8$ implies that y is odd.

Expected answer. A complete proof of all three universally quantified claims, including the parity conclusion from $x! + 1 = y^8$.

Task ID. `ntb_0849ff82c82325aedf67ae3d:whole:0`

DEV 2. **Decimal length after appending zeros**

For natural numbers a, n with $1 \leq a < 10$ and $0 < n$, prove that the decimal digit list of $a10^n$ has length $n + 1$.

Expected answer. A proof that $\text{length}(\text{digits}_{10}(a10^n)) = n + 1$ under both bounds on a and positivity of n.

Task ID. `ntb_0b44211e960985affabb0e96:whole:0`

DEV 3. **Prime divisor and final nine digits**

Let p be a natural prime with $10^{50} < p$. Assume $(10^{294} - 1) \bmod p = 0$, and assume $(10^n - 1) \bmod p \neq 0$ for every natural $n < 294$, including zero. Prove $p \bmod 10^9 = 572857143$.

Expected answer. A proof of the stated remainder 572857143 from all the encoded assumptions.

Review note. Existing review flags contradictory assumptions: the universal nonzero-remainder hypothesis includes exponent zero. Semantic clearance remains withheld; this description does not certify the other assumptions.

Task ID. `ntb_0c317e1120b03cf847d9597c:whole:0`

DEV 4. **Negation of a multiplicative-order conjunction**

Prove that the following conjunction is false. Its first part says that every natural prime p with equal multiplicative orders of 2 and 3 in $\mathbb{Z}/p\mathbb{Z}$, and with some natural $k > 1$ satisfying $p \mid 2^k - 3^k$ in the integers, satisfies $7 \leq p$. Its second part says that these same order-equality and existence conditions hold at $p = 7$.

Expected answer. A proof of the negation of the entire conjunction, not a proof that 7 is a least qualifying prime.

Task ID. `ntb_12c42551bcbe67163eb49a45:whole:0`

DEV 5. **Divisibility equivalence under square-sum premises**

Let $n \in \mathbb{N}$ satisfy $n \geq 4$. Assume $i^2 > 0$ for every natural $i \leq n$, and $\sum_{i=0}^{n} i^2 = 2n(n^2 - n + 2)$. Prove $n \mid n(5n - 3)$ if and only if $n = 4$, with all subtraction in $\mathbb{N}$ truncated at zero.

Expected answer. A proof of both directions of $n \mid n(5n - 3) \iff n = 4$ under every stated premise.

Review note. Existing review flags a contradictory premise: positive squares are required even at zero. A saved Lean check confirms that premise contradiction; this description does not provide semantic clearance.

Task ID. `ntb_24495954b801bd41cc09d7a9:whole:0`

DEV 6. **Factorial product divisibility**

For all natural numbers n, k with $k \leq n$, prove that $(n - k)!\,k!$ divides $n!$. Subtraction takes place in $\mathbb{N}$.

Expected answer. A universally quantified proof of $(n - k)!\,k! \mid n!$ whenever $k \leq n$.

Task ID. `ntb_26c13028f7860afcbd1b1330:whole:0`

DEV 7. **A rational sum is not one**

For a natural prime p, prove that $\sum_{k=0}^{p-1} pk \neq 1$, where each summand and the sum are rational numbers.

Expected answer. A proof that the indicated rational sum differs from 1 for every natural prime p.

Task ID. `ntb_28ce24a682cba3a80079ac05:whole:0`

DEV 8. **No square completion for odd inputs**

For every pair of odd integers x, y, prove that no integer z satisfies $x^2 + y^2 + z^2 = x^2y^2$.

Expected answer. A universally quantified nonexistence proof for integer z under both oddness assumptions.

Task ID. `ntb_2ac8f84bb1611299a7fdbc33:whole:0`

DEV 9. **Order characterizes powers equal to one**

Let $a \in \mathbb{Z}$ and $n \in \mathbb{N}$ satisfy $\gcd(a, n) = 1$ and $n > 0$. Let $d \in \mathbb{N}$ equal the multiplicative order of the residue of a in $\mathbb{Z}/n\mathbb{Z}$. For every natural k, prove $a^k = 1$ in that ring if and only if $d \mid k$.

Expected answer. A proof of both implications between the residue equation $a^k = 1$ and natural divisibility $d \mid k$.

Task ID. `ntb_2f522c6a2246200d9d42f7d4:whole:0`

DEV 10. **Integer divisibility implication**

For an integer x, assume $x \mid 39x - 24$ and $328x - 4 \mid x^3 - 4x^2 + 4x$. Prove $x \mid 4x^2 - 16x(x - 1)$, using integer arithmetic throughout.

Expected answer. A proof of the final divisibility statement under the two encoded divisibility premises.

Task ID. `ntb_34759c85c220863a9b934698:whole:0`

DEV 11. **No mutually adjacent prime divisors**

Prove that there are no natural primes p, q with $p > 2$, $q > 2$, $p \mid q + 1$, and $q \mid p + 1$.

Expected answer. A proof excluding every natural pair satisfying all six conditions.

Task ID. `ntb_42221e0e76e026e91ccae67b:whole:0`

DEV 12. **Prime powers modulo the predecessor**

Let $p, n \in \mathbb{N}$, with p prime and $n > 1$. Assume there exist natural a, b with $ab = p$ and $b \equiv 1 \pmod{p}$ in the integers. Prove $p^n \equiv 1 \pmod{p - 1}$, where the modulus is the integer cast of p minus 1.

Expected answer. A proof of the integer congruence $p^n \equiv 1 \pmod{p - 1}$ under the prime, exponent, and existential assumptions.

Task ID. `ntb_4678ae628e2cd00da60bb8e9:whole:0`

DEV 13. **Digit-sum congruence and a missing digit**

Prove both claims. For every positive natural N, its decimal digit sum is congruent to N modulo 9 in $\mathbb{N}$; and there exists a natural $x \leq 9$ such that $9 \mid 123456789 \cdot 10^{10} + x10^9 + 123456789$.

Expected answer. A proof of the universal digit-sum congruence together with a digit witness $x \leq 9$ and a proof of the specified divisibility.

Review note. Existing review records cross-split semantic overlap: the first conjunct overlaps the TEST decimal digit-sum congruence. This is not evidence of independent generalization.

Task ID. `ntb_540f08afe6aea5bca8e7f52f:whole:0`

DEV 14. **Integer-division sum bound**

Let $p \in \mathbb{N}$ be prime, and assume $p \equiv 1 \pmod{4}$ or $p \equiv 3 \pmod{4}$ in the integers. Prove $\sum_{i=0}^{p-1} (1 \,/_{\mathbb{Z}}\, i) \leq 2p^p - 1$, with p and each i cast to integers and each quotient evaluated using Lean integer division, including the term at $i = 0$.

Expected answer. A proof of the encoded integer sum inequality, not a rational harmonic-sum bound.

Review note. Existing review flags an intent mismatch: the summand uses integer division, not rational division. Semantic clearance remains withheld.

Task ID. `ntb_5623ea7db1926fecb82805a4:whole:0`

DEV 15. **An implicitly quantified exponent**

Let $m, p \in \mathbb{N}$, with p prime. Assume there exists a nonzero natural a such that $2a = p - 1$, where that subtraction is natural subtraction. Prove $p^m \equiv 1 \pmod{p-1}$ after casting p to the integers in the congruence and its modulus.

Expected answer. A proof of the congruence for arbitrary natural exponent m, which is an implicit parameter in the statement.

Task ID. `ntb_57b89c0669597337e6ce5691:whole:0`

DEV 16. **Negation of a factorial-divisibility count**

Prove that the following universal counting assertion is false: for every $n, m \in \mathbb{N}$ with $1 \le m \le n - 1$, the set $\{k \in \mathbb{N} : m \mid k!,\ 1 \le k \le n - 1\}$ has cardinality $n - m$. Both subtractions are natural subtraction, truncated at zero.

Expected answer. A proof of the negation of the universal cardinality formula; a counterexample with a checked failure of the formula is acceptable.

Task ID. `ntb_5ce565368939b4275182fc8c:whole:0`

DEV 17. **Prime-exponent premise and a nested power**

Let $p \in \mathbb{N}$ be prime with $p \neq 2$, and let $a \in \mathbb{N}$ satisfy $a^p = 2$. Prove $a^{(2^p)} \equiv 2 \pmod{p}$ after casting a, p to integers in the congruence; exponentiation is right-associative.

Expected answer. A proof of the stated nested-power congruence from all encoded assumptions, including $a^p = 2$.

Review note. Existing review flags contradictory assumptions: no natural base has a natural prime exponent different from two and power equal to two. This is a review finding, not fresh Lean verification.

Task ID. `ntb_60328a061f4b90dcae8c1b6b:whole:0`

DEV 18. **Absolute value of a square minus one**

Let $n \in \mathbb{N}$ satisfy $2 \le n$ and $3 < n$. Prove that if the integer cast of n is at least 2 and n is not prime, then the natural absolute value of the integer $n^2 - 1$ is not 1.

Expected answer. A proof of the conditional inequality $\mathrm{natAbs}((n : \mathbb{Z})^2 - 1) \neq 1$ under all the encoded bounds and the nonprimality premise.

Task ID. `ntb_66128a7673636dccc946e79c:whole:0`

DEV 19. **No odd multiple as a fourth-power difference**

For every odd integer k, prove that there are no integers x, y with $x^4 - y^4 = 1223334444k$.

Expected answer. A universally quantified nonexistence proof for integer solutions x, y whenever k is odd.

Task ID. `ntb_6795aa26f8089aa7c1f49e08:whole:0`

DEV 20. **Power of an element with known order**

Let $p \in \mathbb{N}$ be prime, let $a, b \in \mathbb{Z}/p\mathbb{Z}$, and let $m, k \in \mathbb{N}$. Assume the multiplicative order of a is m and $a^k = b$. Prove $b^m = 1$ in $\mathbb{Z}/p\mathbb{Z}$.

Expected answer. A proof of $b^m = 1$ under the stated order and power relation, without adding a nonzero assumption on a.

Task ID. `ntb_6bf87afbdba942d3d665e9d7:whole:0`

DEV 21. **Digit-sum exclusion on an interval**

Prove that no natural number x in the inclusive interval $200 \leq x \leq 288$ has decimal digit sum divisible by 19.

Expected answer. A proof of nonexistence of such an x, including both interval bounds and the digit-sum divisibility condition.

Task ID. `ntb_8cadc503b92a3294a3770016:whole:0`

DEV 22. **Negation of a decimal-remainder set equality**

Let S be the set of natural remainders $(100000A + 10000B + 1000C + 100D + 48) \bmod 72$, where $A, B, C, D \in \mathbb{N}$ are each less than 10. Prove $S \neq \{12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68\}$. Leading digit $A = 0$ is permitted.

Expected answer. A proof that the two sets differ, not a proof that the displayed set lists all possible remainders.

Task ID. `ntb_8fb14729a7d6626bd3782104:whole:0`

DEV 23. **Remainder after division by multiplicative order**

Let $a, m, n, k, q, r \in \mathbb{N}$ satisfy $m > 1$, $\gcd(a, m) = 1$, and $k = \mathrm{ord}_m(a)$. Assume $a^n = 1$ in $\mathbb{Z}/m\mathbb{Z}$, $n = qk + r$, and $r < k$. Prove $r = 0$.

Expected answer. A proof that the specified remainder r is zero under the order, residue-power, decomposition, and bound assumptions.

Task ID. `ntb_92a8e6497c8f664aa1696ccf:whole:0`

DEV 24. **Zero residue and zero remainder**

Let $n \in \mathbb{N}$ satisfy $6 < n$, $10 < n$, and $\gcd(n, 3) = 1$. Prove that its residue in $\mathbb{Z}/3\mathbb{Z}$ is zero if and only if its natural remainder $n \bmod 3$ is zero.

Expected answer. A proof of both directions of $(n : \mathbb{Z}/3\mathbb{Z}) = 0 \iff n \bmod 3 = 0$ under every given assumption.

Task ID. `ntb_9c107931350d9e67fa24c1f8:whole:0`

DEV 25. **Decimal digit sum modulo nine**

For every natural n, including zero, prove that n is congruent modulo 9 to its decimal digit sum, with both values cast to integers.

Expected answer. A proof of the integer congruence $n \equiv \sum \mathrm{digits}_{10}(n) \pmod 9$ for all natural n.

Review note. Existing review records cross-split semantic overlap with the TEST decimal digit-sum congruence, despite different positivity assumptions and notation.

Task ID. `ntb_9f2790ee0b18fb114caa8b87:whole:0`

DEV 26. **Order divides a positive exponent**

Let $q \in \mathbb{N}$ be prime and $a \in \mathbb{Z}$ with $q \nmid a$ in the integers. Let $k \in \mathbb{N}$ be the multiplicative order of the residue of a in $\mathbb{Z}/q\mathbb{Z}$. If natural $m > 0$ satisfies $a^m = 1$ in that ring, prove $k \mid m$.

Expected answer. A proof that the natural order k divides the positive natural exponent m under the nondivisibility and residue-power assumptions.

Task ID. `ntb_be9c16ca3754ba8a9ab01583:whole:0`

DEV 27. **A decimal quotient with an implicit parameter**

For an arbitrary fixed natural parameter N, prove that no natural n satisfies $k \leq N \,/_{\mathbb{N}}\, 10^{\mathrm{length}(\mathrm{digits}_{10}(N))}$ for every natural k with $1 \leq k \leq 9$. The bound variable n does not occur in the quotient.

Expected answer. A proof of the stated nonexistence claim for each natural N, using natural division and the inclusive range from 1 through 9.

Review note. New possible intent issue: the initial state declares a natural parameter whose identifier is n!, so the repeated n! tokens denote that parameter rather than a factorial of the bound n. The description renames this parameter N and preserves the unused existential variable; no fresh Lean check or source repair is claimed.

Task ID. `ntb_bea1d7078c9528575e9bf318:whole:0`

DEV 28. **A larger prime does not divide a factorial**

Let $m, p \in \mathbb{N}$ satisfy $m > 0$, p prime, and $p > m$. Prove $p \nmid m!$.

Expected answer. A proof that the prime p does not divide the factorial of m under both positivity and size assumptions.

Task ID. `ntb_cb242a8876b8745c69ac06ff:whole:0`

DEV 29. **A greatest-common-divisor bound attained**

Prove that every positive natural n satisfies $\gcd(9n + 4, 5n + 2) \leq 2$. Also prove that some positive natural n attains equality $\gcd(9n + 4, 5n + 2) = 2$.

Expected answer. A proof of the universal upper bound and a positive natural witness attaining the bound, with its equality proof.

Task ID. `ntb_d62af373edae6d51dcf617f2:whole:0`

DEV 30. **Nonempty decimal digits and reconstruction bound**

Let $b, n \in \mathbb{N}$ satisfy $1 < b$ and $0 < n$. Prove that the decimal digit list of nb is nonempty and that interpreting that list in base 10 gives a natural number at most $2nb^2$.

Expected answer. A proof of both nonemptiness and $\mathrm{ofDigits}_{10}(\mathrm{digits}_{10}(nb)) \leq 2nb^2$.

Task ID. `ntb_d6d6dd394cac5461186aa8d4:whole:0`

DEV 31. **A factorial plus a residue is not a square**

Let $c \in \mathbb{Z}$ satisfy $c \equiv 3 \pmod 4$. Prove that for every natural $m \geq 4$, no integer k satisfies $k^2 = m! + c$, with the factorial cast to $\mathbb{Z}$.

Expected answer. A proof excluding all integer square roots of $m! + c$ for every natural $m \geq 4$ under the congruence assumption on c.

Task ID. `ntb_e244bf908dd21056c803610d:whole:0`

DEV 32. **No solution with an even integer input**

For every even integer x, prove that there is no integer y satisfying $14x^2 + y^2 = 727$.

Expected answer. A universally quantified nonexistence proof for integer y whenever integer x is even.

Task ID. `ntb_ed3f3886e6597e95184acb2d:whole:0`

DEV 33. **Contradiction below multiplicative order**

Let $p \in \mathbb{N}$ be prime, and let $b \neq 0$ in $\mathbb{Z}/p\mathbb{Z}$. Let natural d be the multiplicative order of b. Assume natural r satisfies $0 < r < d$ and $b^r = 1$. Derive a contradiction.

Expected answer. A proof of False from the encoded assumptions about p, b, d, r, rather than an independently chosen numerical answer.

Task ID. `ntb_ee0e013b74d5f43c8a8e241a:whole:0`

DEV 34. **Four cubes realize every residue modulo nine**

For every natural $r < 9$, prove that there exist natural a, b, c, d such that $(a^3 + b^3 + c^3 + d^3) \bmod 9 = r$.

Expected answer. For each permitted residue r, provide four natural witnesses and prove that their cube sum has remainder r modulo 9.

Task ID. `ntb_f995e1335705564e3d7db684:whole:0`

A.2 TEST theorems

TEST 1. **Negated least-prime Jacobi claim**

Prove that the following combined claim is false: 43 is prime; the Jacobi symbols $J(2, 43)$, $J(3, 43)$ and $J(7, 43)$ all equal -1; and every natural prime $q < 43$ fails to have all three symbols equal to -1. Here $J(a, n)$ denotes the Jacobi symbol.

Expected answer. A complete Lean proof of the negation of the entire conjunction, including its least-prime condition.

Review note. The target negates the combined claim; it does not separately negate each conjunct.

Task ID. `ntb_05fad5360cfbe30eebe17b91:whole:0`

TEST 2. **Natural-power inequality**

Let $n \in \mathbb{N}$ satisfy $1 < n$ and suppose n is not prime. Prove $n^n \neq n^2 + (1 \operatorname{div} 2)$, where the division is natural-number division.

Expected answer. A complete Lean proof of the stated inequality under both assumptions.

Review note. Recorded intent warning: natural division makes $1/2$ zero. Preserve the formal target; do not replace it with a rational half.

Task ID. `ntb_07b7569a02e930619d9f4d6b:whole:0`

TEST 3. **Parity of two variables**

Let $x, y, n \in \mathbb{N}$ with $x > 0$, $y > 0$, n odd and $n \geq 1$. If $3x^2 + 5y^2 = 2^n$, prove that x, y are both even or both odd.

Expected answer. A complete Lean proof of $(\mathrm{Even}(x) \wedge \mathrm{Even}(y)) \vee (\mathrm{Odd}(x) \wedge \mathrm{Odd}(y))$.

Task ID. `ntb_08d43e44ee8db479c3f3713e:whole:0`

TEST 4. **Factors of a prime power**

Let $p, a, b, k \in \mathbb{N}$. Assume p is prime, $a > 0$, $b > 0$, $ab = p^k$ and $\gcd(a, b) = 1$. Prove that a and b are each powers of p with natural exponents.

Expected answer. A complete Lean proof supplying $i, j \in \mathbb{N}$ with $a = p^i$ and $b = p^j$. No relation between i, j beyond these equalities is requested.

Task ID. `ntb_1413c3ebe87c42aaf582105d:whole:0`

TEST 5. **Order modulo three**

In $\mathbb{Z}/3\mathbb{Z}$, prove that 10 has multiplicative order 1. Also prove that some positive natural k satisfies $10^k = 1$ and no natural j with $0 < j < k$ satisfies $10^j = 1$.

Expected answer. A complete Lean proof of the order equality and an existential witness k with positivity, power equality and minimality.

Review note. All powers and equalities involving 10 occur in ZMod 3.

Task ID. `ntb_16aa8a2ecd721698a83e6389:whole:0`

TEST 6. **Divisor totient sum bound**

Let $n \in \mathbb{N}$ with $1 < n$. Prove $\sum_{d|n} \varphi(d) \leq n^2 - 1$, summing over the natural divisors of n. The subtraction is natural-number subtraction.

Expected answer. A complete Lean proof of the divisor-sum inequality under $1 < n$.

Task ID. `ntb_16d0b14eb4f46af62451c672:whole:0`

TEST 7. **Prime divisibility and a square**

Let $n \in \mathbb{Z}$ and let $p \in \mathbb{N}$ be prime. Prove both $p \mid n \iff n^2 \bmod p = 0$ and $p \mid ((p+1)^2 - 1)$, interpreting p as an integer in both claims.

Expected answer. A complete Lean proof of both the divisibility equivalence and the second divisibility statement.

Review note. The remainder and subtraction occur in the integers.

Task ID. `ntb_17574657fab0aeeab228be04:whole:0`

TEST 8. **Negated Fibonacci congruence**

Prove that it is false that every natural prime p satisfies $\sum_{i=1}^{p} F_i \equiv J(p, 5) \pmod{p}$. Here F_i is the natural Fibonacci number, cast to an integer in the sum, and J is the Jacobi symbol.

Expected answer. A complete Lean proof negating the universal prime congruence, rather than proving that congruence for all primes.

Task ID. `ntb_1a7891d49276f2fdb7643bdf:whole:0`

TEST 9. **Decimal digit-sum remainder**

For every positive $n \in \mathbb{N}$, prove that the sum of its base-ten digits has the same remainder modulo 9 as n.

Expected answer. A complete Lean proof of $(\sum \mathrm{digits}_{10}(n)) \bmod 9 = n \bmod 9$ for every $n > 0$.

Review note. Recorded semantic overlap: DEV theorem 25 states the digit-sum congruence, and the first conjunct of DEV theorem 13 includes it. This is not a certified independent family.

Task ID. `ntb_1aa39c678607b996200182fc:whole:0`

TEST 10. **Order and prime exponent equivalence**

Let $n \in \mathbb{N}$ be prime and let $a \in \mathbb{Z}/n\mathbb{Z}$ be nonzero. Prove $(a^{n-1} \neq 0 \wedge a^n = 1) \iff (\mathrm{ord}(a) = 1 \vee \mathrm{ord}(a) = n)$, where ord is multiplicative order.

Expected answer. A complete Lean proof of the equivalence, preserving the conjunction on its left and the disjunction on its right.

Task ID. `ntb_1af4833b840d0b0fbec15e8c:whole:0`

TEST 11. **Cancel a coprime congruence factor**

Let $a, b, m, x, y \in \mathbb{Z}$ satisfy $ax \equiv b \pmod{m}$, $ay \equiv b \pmod{m}$ and $\gcd(a, m) = 1$. Prove $x \equiv y \pmod{m}$.

Expected answer. A complete Lean proof of the final congruence under all three assumptions.

Review note. No positivity or nonzero assumption on the integer modulus is stated.

Task ID. `ntb_1d4a55a1396005c641f975de:whole:0`

TEST 12. **Odd square divisibility**

Let $n \in \mathbb{N}$ be odd. Assuming $3 \neq 0$ in $\mathbb{N}$ and $2 < n$, prove both $8 \mid (n^2 - 1)$ and $((n^2)^2 \bmod 8) \leq 9$. The subtraction is natural-number subtraction.

Expected answer. A complete Lean proof of the implication from the stated conjunction to both target claims.

Review note. The second target is the stated remainder bound; do not replace it with a sharper congruence.

Task ID. `ntb_20e25d7da730561b7776bb3e:whole:0`

TEST 13. **Divisibility equivalence from a product**

Let $a, b, c \in \mathbb{Z}$ and assume $ab \mid c$. Prove $a \mid c \iff b \mid c$.

Expected answer. A complete Lean proof of both directions of the divisibility equivalence under the product-divisibility assumption.

Task ID. `ntb_21c9319bec71cef5f8c5a55d:whole:0`

TEST 14. **Square root of a field quotient**

Let $p \in \mathbb{N}$ be an odd prime and let $a, b, k, l \in \mathbb{Z}/p\mathbb{Z}$. Assume $a \neq 0$, $b \neq 0$, $ka^2 + lb^2 = 0$, l is a square and k is not a square. Prove that a nonzero $x \in \mathbb{Z}/p\mathbb{Z}$ satisfies $x^2 = -(l/k)$.

Expected answer. A complete Lean proof supplying x and proving both $x \neq 0$ and $x^2 = -(l/k)$.

Review note. Division is field division in ZMod p, not integer or natural division.

Task ID. `ntb_2b2b05cfd427f7eccfae7a28:whole:0`

TEST 15. **Odd integer positive powers**

Let $a \in \mathbb{Z}$ satisfy $a \equiv 1 \pmod 2$. For every positive natural exponent k, prove $a^k \equiv 1 \pmod 2$.

Expected answer. A complete Lean proof of the congruence for every exponent in the positive naturals.

Task ID. `ntb_2c9caaaff14dc84be05601b1:whole:0`

TEST 16. **Natural quotient lower bound**

Let $a, b, k \in \mathbb{N}$ satisfy $b \geq 3$, b odd, $a > 0$, $k > 0$ and $2kb(b-1) \geq a(b-1)^2$. Prove $k \operatorname{div} a \geq (b-1) \operatorname{div} (2b)$. Both quotients and all subtractions are natural-number operations.

Expected answer. A complete Lean proof of the inequality between the natural quotients under every stated assumption.

Review note. Recorded intent warning: for natural b at least three, the right quotient is zero, so the ratio bound degenerates. This is not a rational-ratio statement.

Task ID. `ntb_317a357e031c74d97e1ba8f4:whole:0`

TEST 17. **Negated Jacobi residue formula**

Prove that it is false that every odd natural prime p satisfies $J(-3, p) = 1$ when $p \bmod 6 = 1$ and $J(-3, p) = -1$ otherwise. Here J is the Jacobi symbol with integer numerator.

Expected answer. A complete Lean proof negating the universally quantified conditional formula for odd primes.

Review note. The outer negation covers the whole universal statement; the otherwise branch includes every residue other than one.

Task ID. `ntb_3260c7ed4ee56866de06e982:whole:0`

TEST 18. **No coprime square representation**

Prove that there are no positive natural numbers a, b satisfying $\gcd(a, b) = 1$ and $b^2 = 5a$.

Expected answer. A complete Lean proof of the nonexistence of a, b satisfying all four conditions.

Task ID. `ntb_36c456b2166347d83b0ed3f3:whole:0`

TEST 19. **Prime divisors and nonsquare two**

Prove three claims: every natural prime p has exactly two natural divisors; 2 has exactly two natural divisors; and no natural number k satisfies $k^2 = 2$.

Expected answer. A complete Lean proof of the full conjunction, including the universal prime claim, the concrete divisor count and the nonexistence claim.

Task ID. `ntb_38d4e0e8d6c4a9e630133c64:whole:0`

TEST 20. **Existence of prime product witnesses**

Prove that natural numbers p, q, r, n exist such that p, q, r are prime, $pqr = n$ and $(p+1)(q+1)r = n + 138$.

Expected answer. A complete Lean proof supplying all four witnesses and proving all three primality claims and both equalities.

Review note. The primes are not required to be distinct.

Task ID. `ntb_398249d36acc8ef9a5e1ca40:whole:0`

TEST 21. **Empty set of nonsquare conditions**

Prove that no $n \in \mathbb{N}$ has this property: for every natural prime p, both $n^2 + 4n + 1 - p$ and $2n^2 + n + 2 - p$ are nonsquares in $\mathbb{N}$. Both subtractions truncate at zero.

Expected answer. A complete Lean proof that the set of natural n satisfying the stated universal conjunction is empty.

Review note. Natural truncated subtraction is part of the statement; do not read these as integer polynomial differences.

Task ID. `ntb_3ca1d1b12b5dfddc44f1290e:whole:0`

TEST 22. **Remainder of prime-square minus one**

Let $p \in \mathbb{N}$ be prime, and let $a \in \mathbb{N}$ be arbitrary. Prove $(p^2 - 1) \bmod (p^2 + 1) = p^2 - 1$, with natural-number subtraction and remainder.

Expected answer. A complete Lean proof of the remainder identity in the stated context.

Review note. The arbitrary parameter a appears in the statement but not in the assumptions or target.

Task ID. `ntb_3fd7520aa33f27729d8a7a07:whole:0`

TEST 23. **Negated quadratic representation theorem**

Prove that the following universal claim is false. Every natural prime $p > 5$ for which -5 is a square in $\mathbb{Z}/p\mathbb{Z}$ has positive natural m, n, k with $p^2 = m^2 + 5n^2 + 6k^2$; every such positive representation has $m + n + k \geq 6$; and some such representation has $m + n + k = 6$.

Expected answer. A complete Lean proof negating the universal conjunction of representation existence, the lower bound for every positive representation, and attainment of that bound.

Review note. The target is the negation of the entire universal claim, not a request to construct those representations for every qualifying prime.

Task ID. `ntb_43bce97fd20bfa368ddf2b30:whole:0`

TEST 24. **Nonprime exponential with an odd factor**

Let $N \in \mathbb{N}$ be positive. Suppose some odd natural prime divides N. Prove that $3^N + 1$ is not prime.

Expected answer. A complete Lean proof of nonprimality under the positivity and odd-prime-divisor assumptions.

Task ID. `ntb_44ecc8d8d9a04797f8633215:whole:0`

TEST 25. **Totient of a prime power**

Let $p \in \mathbb{N}$ be prime and let $k \in \mathbb{N}$ be positive. Prove $\varphi(p^k) = p^k - p^{k-1}$, where φ is Euler totient and the subtractions occur in $\mathbb{N}$.

Expected answer. A complete Lean proof of the totient identity under primality and positive exponent assumptions.

Review note. Recorded semantic overlap: TEST theorem 40 states the same prime-power totient identity.

Task ID. `ntb_45847c8fcf7e4ce252bf9497:whole:0`

TEST 26. **Least prime square residue modulo seven**

Prove that 11 is prime, $11 > 7$ and its residue in $\mathbb{Z}/7\mathbb{Z}$ is a square. Also prove $11 \leq q$ for every natural prime $q > 7$ whose residue modulo 7 is a square.

Expected answer. A complete Lean proof of all three properties of 11 and the universal minimality statement.

Task ID. `ntb_4613017e2c6d8b3546380196:whole:0`

TEST 27. **Reciprocal-sum disjunction**

Prove that positive natural numbers a, b, c exist for which at least one branch holds: they are pairwise coprime and $1/a + 1/b + 1/c = 9/(a+b+c)$; or $1/a + 1/b + 1/c = 10/(a+b+c)$; or $1/a + 1/b + 1/c = 11/(a+b+c)$. All fractions are rational numbers.

Expected answer. A complete Lean proof supplying positive witnesses and proving a branch of the exact disjunction. Pairwise coprimality is required only for the branch with numerator 9.

Review note. Recorded intent warning: pairwise coprimality constrains only the first branch, not the branches with numerators 10 or 11.

Task ID. `ntb_49e6cc8d82bba05f96425e21:whole:0`

TEST 28. **Sum of prime proper divisors**

Let $p \in \mathbb{N}$ be prime. Prove that the sum of its natural divisors strictly less than p equals 1.

Expected answer. A complete Lean proof of $\sum_{d|p,\ d<p} d = 1$.

Task ID. `ntb_4b422de08469fc142a8bd3f2:whole:0`

TEST 29. **Even-term sum bound by a prime power**

Let $p \in \mathbb{N}$ be prime with $p \neq 2$ and assume $p^p \geq p + 1$. Prove $\sum_{k=0}^{p-1}(2k+2) \leq p^p$.

Expected answer. A complete Lean proof of the finite-sum bound under primality, exclusion of two and the stated power bound.

Task ID. `ntb_4c9e096311c691a10fc18f4c:whole:0`

TEST 30. **Prime exceeding a square-root threshold**

Let $n \in \mathbb{N}$ satisfy $1 < n$. Prove that some natural prime $p \neq n$ satisfies $\lfloor \sqrt{n} \rfloor < p^2$, where the square root is the natural integer square root.

Expected answer. A complete Lean proof supplying p with primality, $p \neq n$ and the square-root inequality.

Review note. The conclusion does not require p to divide n or to be less than n.

Task ID. `ntb_507de6b94de228dd2f022572:whole:0`

TEST 31. **Divisor sum of a prime square**

Let $p \in \mathbb{N}$ be prime, and let S be the natural sum of the divisors of p^2. Prove both $S = 1 + p + p^2$ and, after casting to $\mathbb{Q}$, $S/(p^2 + p) = 1 + 1/(p(p + 1))$.

Expected answer. A complete Lean proof of the natural divisor-sum equality and the rational quotient equality.

Review note. The second equality uses rational division after casts; it is not a natural quotient.

Task ID. `ntb_549962e77307dd430ec3b45c:whole:0`

TEST 32. **Negated weighted Jacobi-sum identity**

Prove that it is false that every natural prime p with $p \bmod 4 = 3$ satisfies $\sum_{k=1}^{p-1} k^3 J(k, p) - p^2 \sum_{k=1}^{p-1} kJ(k, p) = 0$. Both sums and the subtraction are integer-valued, and J denotes the Jacobi symbol.

Expected answer. A complete Lean proof negating the universal weighted-sum identity under the stated prime and residue conditions.

Task ID. `ntb_554b2dc790fab84a168853ef:whole:0`

TEST 33. **Integer function determined at primes**

Let $x : \mathbb{N} \to \mathbb{Z}$ satisfy $n = \prod_{d|n} x(d)$ for every natural $n \geq 1$, with n cast to an integer. Prove both $x(1) = 1$ and $x(p) = p$ for every natural prime p, again casting p to an integer.

Expected answer. A complete Lean proof of the value at one and the universal prime-value claim under the divisor-product hypothesis.

Task ID. `ntb_58cf341d56782a157b4f37da:whole:0`

TEST 34. **Totient ratio at a prime**

Let $p \in \mathbb{N}$ be prime. After casting p and $\varphi(p)$ to $\mathbb{Q}$, prove $\varphi(p)/p = 1 - 1/p$.

Expected answer. A complete Lean proof of the rational identity under the prime assumption.

Task ID. `ntb_5a32ab19491895603fc1af22:whole:0`

TEST 35. **Even modulus fails a totient divisibility**

Let $n \in \mathbb{N}$ be even with $n > 2$. Prove $n \nmid (2^{\varphi(n)} - 1)$, where the subtraction is natural-number subtraction.

Expected answer. A complete Lean proof that the stated natural divisibility does not hold.

Task ID. `ntb_6be76276083ed05501bfa715:whole:0`

TEST 36. **Prime powers differ from a truncated difference**

Let $p, m, n \in \mathbb{N}$ with p prime and assume $m \geq 0 \wedge n \geq 0$. Prove $p^{m+n} \neq p^m - p^n$, where subtraction truncates at zero.

Expected answer. A complete Lean proof of the inequality under all stated assumptions.

Review note. The nonnegativity assumptions are explicit but already follow from the natural-number domains.

Task ID. `ntb_73b60c27b218a377c1d1c37e:whole:0`

TEST 37. **Congruent exponents modulo a totient**

Let $a, b \in \mathbb{N}$ satisfy $a \equiv b \pmod{\varphi(125)}$. Prove $2^a \equiv 2^b \pmod{125}$ using natural-number congruence.

Expected answer. A complete Lean proof of the power congruence under the exponent congruence assumption.

Task ID. `ntb_767e128f330effc40b628557:whole:0`

TEST 38. **Nondivisibility of a sum of squares**

Let $p \in \mathbb{N}$ be prime with $p \bmod 4 = 3$, and let $a, b \in \mathbb{Z}$. Assume $p \mid a$ and $p \nmid b$, casting p to an integer. Prove $p \nmid (a^2 + b^2)$.

Expected answer. A complete Lean proof of nondivisibility under all four assumptions.

Task ID. `ntb_771a85cbc730dded6e9f1fe8:whole:0`

TEST 39. **Residue class with two nonsquares**

For arbitrary $a, b \in \mathbb{Z}$, prove that $m \in \mathbb{N}$ and $c \in \mathbb{Z}$ exist with $m > 0$ such that every natural prime $p \equiv c \pmod{m}$ has both $-a$ and $-b$ nonsquare in $\mathbb{Z}/p\mathbb{Z}$.

Expected answer. A complete Lean proof supplying m, c, proving $m > 0$, and proving the universal pair of nonsquare claims for primes in the class.

Review note. Recorded intent warning: the residue class need not contain a prime, so the universal claim can be vacuous. No coprimality of c and m or prime existence is required.

Task ID. `ntb_78234cf56468fef0944a0a2a:whole:0`

TEST 40. **Repeated prime-power totient identity**

Let $p, k \in \mathbb{N}$ with p prime and $k > 0$. Prove $\varphi(p^k) = p^k - p^{k-1}$, where φ is Euler totient and the subtractions occur in $\mathbb{N}$.

Expected answer. A complete Lean proof of the totient identity under both assumptions.

Review note. Recorded semantic overlap: TEST theorem 25 states the same prime-power totient identity.

Task ID. `ntb_797a92fae369277d867f92aa:whole:0`

TEST 41. **Rational inequality for an even sequence sum**

Let $p \in \mathbb{N}$ be prime with $p \neq 2$. Let $a : \mathbb{N} \to \mathbb{N}$ satisfy $a(i) = 2i$ for every natural i. If $S = \sum_{k=1}^{100} a(k)$, prove $S \neq 2S/p$ after casting the sums and p to $\mathbb{Q}$.

Expected answer. A complete Lean proof of the rational inequality under the prime, exclusion and sequence assumptions.

Review note. The division and comparison are rational after casts; the sum has natural terms.

Task ID. `ntb_79908f21d3a04c4fdaa934ac:whole:0`

TEST 42. **Prime divisor of adjacent factorial sum**

Prove two claims about natural numbers. First, there exist $n > 0$ and a prime p such that $p \nmid (5n)!$, $p \mid (5n+1)! + (5n+2)!$, and $p \bmod 5 = 1$. Second, every positive n and prime p satisfying those two divisibility conditions has $p \bmod 5 \geq 1$.

Expected answer. Supply witnesses for the existence claim and prove the universal remainder bound. Both conjuncts require complete Lean proofs.

Task ID. `ntb_7a12343ef5f6b7715d2b8dd7:whole:0`

TEST 43. **Unequal powers for a nonprime exponent**

Let $n \in \mathbb{N}$ satisfy $1 < n$ and assume n is not prime. Prove $2^{n^2-n} \neq 3^n$, with the exponent subtraction performed in natural numbers.

Expected answer. Prove the stated inequality under both hypotheses.

Task ID. `ntb_7a8b6877ed0dbbf567a789ba:whole:0`

TEST 44. **Maximum digit product under divisibility constraints**

For natural-number digits $A, B < 10$ with $A \geq 1$, impose divisibility of $100000A + 20210 + B$ by both 4 and 9. Prove that an admissible pair has $AB = 42$, and that every admissible pair has $AB \leq 42$.

Expected answer. Provide admissible digit witnesses attaining 42 and prove the upper bound for every pair satisfying all digit and remainder conditions.

Task ID. `ntb_7b9304b912534907d37cabf6:whole:0`

TEST 45. **Number of divisors of a prime**

Let $n \in \mathbb{N}$ be prime. Prove that its finite set of positive divisors has cardinality 2.

Expected answer. Prove $|\text{divisors}(n)| = 2$.

Task ID. `ntb_7ed9af586fa024354709e2f9:whole:0`

TEST 46. **A power differs from its natural quotient**

Let $n \in \mathbb{N}$ satisfy $1 < n$ and assume n is even. Prove $2^n \neq \lfloor 2^n/4 \rfloor$, where the quotient is Lean natural-number division.

Expected answer. Prove the inequality of these two natural numbers.

Review note. The division is natural-number division, not rational division.

Task ID. `ntb_819c891a3cb42e3ed8f8c31c:whole:0`

TEST 47. **Different prime-base powers after subtraction**

Let $p, n \in \mathbb{N}$. Assume p is prime, $4 < p$, $1 < n$, and n is not prime. Prove $p^n - 1 \neq 2^n - 1$, with both subtractions performed in natural numbers.

Expected answer. Prove the inequality while retaining all four assumptions.

Task ID. `ntb_81a6446c304a5842e1b56484:whole:0`

TEST 48. **Inverse of an assumed quadratic equivalence**

Let $n \in \mathbb{N}$ be positive and not prime. Assume $\sigma : \mathbb{N} \to \mathbb{N}$ is a bijection satisfying $\sigma(x) = x^2+2$ for every natural x. Prove $\sigma^{-1}(2^3 n) \neq n$.

Expected answer. Prove the inverse-value inequality from the exact stated assumptions.

Review note. Existing review flags contradictory assumptions: a bijection must attain zero, whereas $x^2 + 2$ cannot. This is a source warning from manual review, not fresh Lean verification.

Task ID. `ntb_8a04bce03beb48fdc3862d35:whole:0`

TEST 49. **Negation of a Jacobi sum identity**

Prove that the following identity does not hold for every natural prime p with $p \bmod 4 = 3$: $\sum_{k=1}^{p-1} k^4 \left(\frac{k}{p}\right) - p \sum_{k=1}^{p-1} k^2 \left(\frac{k}{p}\right) = 0$. Here $\left(\frac{k}{p}\right)$ is the Jacobi symbol, and both sums are integer-valued.

Expected answer. Refute the universally quantified identity. The target is its negation; it does not request a proof that every eligible prime violates it.

Review note. Preserve the outer negation and the scope of the universal quantifier.

Task ID. `ntb_8da859b76deb65034cb08063:whole:0`

TEST 50. **A prime divisor-square sum is not fourteen**

Let $n \in \mathbb{N}$ be prime and assume $n \neq 2$. Prove $\sum_{k \mid n,\ k>0} k^2 \neq 14$, where the sum ranges over the finite set of positive divisors of n.

Expected answer. Prove the inequality for the natural-number divisor sum.

Task ID. `ntb_8fa705e94721f2d74586f12b:whole:0`

TEST 51. **Bound on a sum of squared natural quotients**

Let $n \in \mathbb{N}$ satisfy $1 < n \leq 100$ and $3 \mid n$. Prove $\sum_{k=1}^{n} (2\lfloor n/k \rfloor)^2 \leq 2n(n^2 - 1)$. The quotients and subtraction are natural-number operations.

Expected answer. Prove the sum bound for every n satisfying all three hypotheses.

Review note. The summands contain natural-number division, not rational fractions.

Task ID. `ntb_947d302024239a0c66ed6c03:whole:0`

TEST 52. **Equivalence of a remainder and divisibility claim**

Let $a \in \mathbb{N}$ be prime with $2 < a$. Prove $a^{a-1} \bmod a \neq 0 \iff a \mid a^{a-1} + 1$, with a natural-number exponent and remainder.

Expected answer. Prove both directions of the exact biconditional.

Review note. An equivalence does not assert that either side is true.

Task ID. `ntb_966e922f641f91f7ef67f209:whole:0`

TEST 53. **Prime square divisibility equivalence**

Let $a \in \mathbb{N}$ be prime with $2 < a$. Prove $a \mid a^2 + 1 \iff a = a^2 + 1$.

Expected answer. Prove both implications under the prime and size hypotheses.

Review note. An equivalence does not assert that either side is true.

Task ID. `ntb_a0396886f1b22e0a21fd7e72:whole:0`

TEST 54. **Least totient witness with two prime factors**

Prove that there exists a positive $n \in \mathbb{N}$ of the form $n = 2^x 3^y$ for positive natural x, y, with $\varphi(n) = \lfloor n/3 \rfloor$. Require n to be no larger than every other positive m of that form satisfying the same totient equation. Also require the sum of the distinct prime factors of n to equal 5, where φ is Euler's totient function.

Expected answer. Supply n and its positive exponent witnesses; prove the totient equality, the universal minimality condition, and the distinct-prime-factor sum.

Review note. The factor list is converted to a finite set before summation, so multiplicities are removed. Division is natural-number division.

Task ID. `ntb_a53c82259ef05c3220f305af:whole:0`

TEST 55. **One is a least positive totient divisibility exponent**

Prove $1 > 0$ and $2 \cdot 1 \mid \varphi(2^1 + 3^1)$. Also prove that every positive natural x satisfying $2x \mid \varphi(2^x + 3^x)$ obeys $1 \leq x$, where φ is Euler's totient function.

Expected answer. Prove the two concrete claims and the universally quantified lower bound.

Task ID. `ntb_a77aff91cab6f59d7fe10c04:whole:0`

TEST 56. **Nonzero residue power equivalence**

Let $n, a \in \mathbb{N}$. Assume $1 < n$, $2 < n$, and $a \neq 0$. Prove $(a \bmod n)^{n-1} \neq 0 \iff a \bmod n \neq 0$, with natural-number subtraction in the exponent.

Expected answer. Prove both directions while retaining both stated bounds on n and the assumption on a.

Task ID. `ntb_a8321a47269f6f807338a3be:whole:0`

TEST 57. **Zero solutions for a nonsquare coefficient**

Let $p \in \mathbb{N}$ be prime and $a \in \mathbb{Z}$. Assume the integer p does not divide a, and the residue of a is not a square in $\mathbb{Z}/p\mathbb{Z}$. Prove that every $x, y \in \mathbb{Z}/p\mathbb{Z}$ satisfying $ax^2 = y^2$ has $x = 0$ and $y = 0$, with a cast into that residue ring.

Expected answer. Prove the universal implication and both zero conclusions in $\mathbb{Z}/p\mathbb{Z}$.

Task ID. `ntb_a980417194ef95918cb0bf7d:whole:0`

TEST 58. **Identify a multiple from its divisor sum**

Let $x \in \mathbb{N}$. Assume the sum of its positive divisors is 24 and $15 \mid x$. Prove $x = 15$.

Expected answer. Prove the exact value of x from both hypotheses.

Task ID. `ntb_add27f3592c556b6b88b8c09:whole:0`

TEST 59. **Congruence equivalence under a negative-square premise**

Let $a, b, p \in \mathbb{N}$, with p prime. Assume there exists an integer x with $x^2 = -p$, where p is cast into $\mathbb{Z}$. Prove $(p \mid a \vee p \mid b) \iff (a \equiv 0 \pmod p \vee b \equiv 0 \pmod p \vee a \equiv b \pmod p)$, where the congruences are integer congruences after casting a, b, p.

Expected answer. Prove both directions of the full biconditional, preserving all three alternatives on the right.

Review note. Existing review flags contradictory assumptions: an integer square cannot equal the negative of a natural prime. The initial state reveals that implicit a, b are natural numbers. The contradiction warning is manual, not fresh Lean verification.

Task ID. `ntb_b23977f4ad063101726eb32a:whole:0`

TEST 60. **Square-sum inequality for a constrained integer sequence**

Let $n \in \mathbb{N}$ satisfy $1 < n$ and $1 < n$, and assume n is not prime. Let $a : \mathbb{N} \to \mathbb{Z}$ satisfy $a(k) = n\,a(k)$ for every k. Writing $S = \sum_{k=0}^{n-1} a(k)^2$, prove $Sn \geq (nS)/(n-1)$, where n is cast to an integer and the division and subtraction on the right are integer operations.

Expected answer. Prove the integer inequality under the duplicated size hypothesis and the universal sequence constraint.

Review note. Existing intent review notes that the sequence constraint forces every $a(k)$ to zero when $1 < n$. Division is integer division. The source warning is manual, not fresh Lean verification.

Task ID. `ntb_b6e5df954bbe811b2595e721:whole:0`

TEST 61. **Negation of a Jacobi residue characterization**

Prove it is not true for every natural prime p that $\left(\frac{-75}{p}\right) = 1 \iff p \bmod 12 = 1$, where the fraction notation denotes the Jacobi symbol.

Expected answer. Refute the universal biconditional. The goal is not to prove the negated biconditional for every prime.

Review note. Preserve the outer negation and its scope over all primes.

Task ID. `ntb_b967b0cf42b2c29b96f24210:whole:0`

TEST 62. **Sum bound from least common multiple and gcd**

Let $a, b \in \mathbb{N}$ be positive. Assume $\operatorname{lcm}(a, b) = 140$ and $\gcd(a, b) = 5$. Prove $a + b \leq 145$.

Expected answer. Prove the upper bound using the positivity, least-common-multiple, and greatest-common-divisor hypotheses.

Task ID. `ntb_ba0da67564e4aee3b28f8bc4:whole:0`

TEST 63. **A nonsquare product has a nonsquare factor**

Let $p \in \mathbb{N}$ be prime and odd. Assume 39 is not a square in $\mathbb{Z}/p\mathbb{Z}$. Prove that 3 is not a square in that ring or 13 is not a square in that ring.

Expected answer. Prove the disjunction of the two nonsquare claims in the same residue ring.

Task ID. `ntb_bb24d44ab7f1b559a280beda:whole:0`

TEST 64. **Finite-set sum under an interval equivalence**

Let S be a finite set of natural numbers. Assume that for every natural p, the conjunction 'p is prime and $p \in S$' is equivalent to $1 < p \leq 10$. Prove $\sum_{k \in S} k = 17$.

Expected answer. Prove the finite-set sum equality from the exact universal biconditional.

Review note. Existing review flags contradictory assumptions: substituting $p = 4$ forces 4 to be prime. This does not encode the usual set of primes at most 10. The warning is manual, not fresh Lean verification.

Task ID. `ntb_bb8109462d7df8e0ab8a17eb:whole:0`

TEST 65. **Least square satisfying an iterated divisor sum**

For positive natural u, write $\sigma(u)$ for the sum of its positive divisors. Prove that there exist natural n, k with $n > 0$, an odd natural m satisfying $n = 2^k m$, a natural s satisfying $s^2 = n$, and $\sigma(\sigma(n)) = 2n$. Require that no positive $n' < n$ simultaneously admits an odd-part representation $n' = 2^{k'} m'$, is a square, and satisfies $\sigma(\sigma(n')) = 2n'$.

Expected answer. Supply the witnesses and prove every existence condition, the iterated divisor-sum equality, and the exclusion of all smaller positive candidates.

Review note. No positivity condition is imposed on k or k'. The oddness condition is the natural remainder equation $m \bmod 2 = 1$.

Task ID. `ntb_beedbd080221783debb0b0ff:whole:0`

TEST 66. **Totient congruence and positivity for a coprime base**

Let $p, a \in \mathbb{N}$, with p prime and $\gcd(a, p) = 1$. Prove $a^{\varphi(p)} \equiv 1 \pmod{p}$, $a^p \neq 0$, and $a > 0$, where φ is Euler's totient function. The congruence is over integers after casting a and p; the other two conclusions are over naturals.

Expected answer. Prove all three conjuncts, including nonvanishing and positivity.

Task ID. `ntb_bf724e76836b4276f40bbe4d:whole:0`

TEST 67. **Divisor sum bound for a perfect nonprime**

Let $n \in \mathbb{N}$ satisfy $1 < n$ and assume n is not prime. Assume its positive proper divisors sum to n. Prove that the sum of all its positive divisors is greater than $\lfloor n^2/(n+1) \rfloor$.

Expected answer. Prove the strict inequality of natural numbers under all three hypotheses.

Review note. The quotient on the right is natural-number division.

Task ID. `ntb_c0570ab949614d32f731fda0:whole:0`

TEST 68. **Power-product congruence modulo four**

Let $a, b, k \in \mathbb{N}$. Assume $a \bmod 2 = 1$ or $a \bmod 2 = 0$, assume the corresponding disjunction for b, and assume $(ab) \bmod 4 = 0$ or $(ab) \bmod 4 = 1$. Prove $a^k b^k \equiv (ab)^k \pmod{4}$ using natural-number congruence.

Expected answer. Prove the congruence while preserving all three disjunctive hypotheses.

Task ID. `ntb_c150b80beff0c40596c26203:whole:0`

TEST 69. **Base-seven digit length of a power minus one**

Let $k \in \mathbb{N}$ satisfy $k \geq 1$. Prove that the base-7 digit list of $7^k - 1$ has length k, where subtraction is over natural numbers.

Expected answer. Prove the exact digit-list length equality.

Task ID. `ntb_c40ed915e8bcab761ac71f09:whole:0`

TEST 70. **Negation of a squarefree counting identity**

Prove that the following equality does not hold for every positive natural n. Its left side counts natural k with $0 \le k \le n$, $k > 0$, k squarefree, and the natural quotient $\lfloor n/k \rfloor$ odd. Its right side is $\sum_{d \mid n,\ d>0} \mu(d)((n : \mathbb{Z})/(d : \mathbb{Z}))$, with integer division and the integer-valued Moebius function μ; the count is cast to an integer.

Expected answer. Refute the universally quantified equality with these exact counting and arithmetic conventions.

Review note. The left-hand filter does not require $k \mid n$. Its quotient is natural-number division, whereas the right-hand quotient is integer division. Preserve the outer negation.

Task ID. `ntb_c881044f36da98a364d84d31:whole:0`

TEST 71. **Bound for a totient sum over divisors of a prime minus one**

Let $p \in \mathbb{N}$ be prime, and let $n \in \mathbb{N}$ satisfy $n \neq 0$. Prove $\sum_{k \mid p-1,\ k>0} \varphi(k) \le p^{p-1} + p - 1$, where φ is Euler's totient function and subtraction is over natural numbers.

Expected answer. Prove the natural-number upper bound while retaining the stated nonzero parameter n.

Review note. The parameter n and its hypothesis do not occur in the target expression.

Task ID. `ntb_cb9923d415db6eaf0b10d6bd:whole:0`

TEST 72. **Odd totient of a power of two**

For every $k \in \mathbb{N}$, prove that $\varphi(2^k)$ is odd if and only if $k \le 1$, where φ is Euler's totient function.

Expected answer. Prove both directions, including the case $k = 0$.

Task ID. `ntb_d703b92567dd399885123b13:whole:0`

TEST 73. **Prime conclusion under a universal quadratic bound**

Let $p \in \mathbb{N}$ be prime with $p \equiv 1 \pmod 2$ and $4 < p$. Assume $2x^2 \le p^2$ for every natural x. Prove $p^4 = 1$ or $p = 2$.

Expected answer. Prove the exact disjunction under the prime, congruence, size, and universal-bound hypotheses.

Review note. Existing review flags contradictory assumptions: the universal bound at $x = p$ conflicts with $4 < p$. The warning is manual, not fresh Lean verification.

Task ID. `ntb_e17448422e4f2cbf449a13be:whole:0`

TEST 74. **Bound for a square with a square decimal prefix**

For every $N \in \mathbb{N}$, assume N is a square, $N \bmod 10 \neq 0$, and $\lfloor N/100 \rfloor$ is also a square. Prove $N \le 1681$.

Expected answer. Prove the universal upper bound with all three premises.

Review note. The quotient $N/100$ is natural-number division, so it removes the final two decimal digits.

Task ID. `ntb_e5cc8ba57ce19eaf086ba3dd:whole:0`

TEST 75. **Divisor counts for a prime and its square**

For every natural prime p, prove that p has exactly two positive divisors and p^2 has exactly three positive divisors.

Expected answer. Prove both cardinality equalities for the corresponding finite divisor sets.

Task ID. `ntb_e5f06a50a73ffcbedc223452:whole:0`

TEST 76. **Integer witnesses for reduced rational fractions**

Find integers a, b, c, d such that $a, b, c \geq 0$, $\gcd(a, b) = 1$, $\gcd(c, d) = 1$, $b \neq 0$, $d \neq 0$, and $a/b + c/d = 1$ in $\mathbb{Q}$. Also require $a + b = c$ or $a + c = b$ or $b + c = a$, together with $a + c = 2b$ and $a + b + c = 6$.

Expected answer. Supply all four integer witnesses and prove every sign, gcd, nonzero-denominator, rational-equality, disjunction, and linear-equality condition.

Review note. The source does not require $d \geq 0$. Division is rational after casting the integer numerators and denominators.

Task ID. `ntb_f098a3f400e5643c0856230f:whole:0`

TEST 77. **Transfer of nonsquare residues through two congruences**

Let $p, q \in \mathbb{N}$ be distinct odd primes, and let $a, b, c \in \mathbb{Z}$. Assume $a \equiv b \pmod{p}$ and $a \equiv c \pmod{q}$, with b nonsquare in $\mathbb{Z}/p\mathbb{Z}$ and c nonsquare in $\mathbb{Z}/q\mathbb{Z}$. Prove that a is nonsquare in both respective residue rings.

Expected answer. Prove the two nonsquare conclusions, one modulo p and one modulo q.

Review note. The displayed initial state suppresses the two residue moduli; the statement distinguishes them.

Task ID. `ntb_f0ed5c021d48de136cceb5c7:whole:0`

TEST 78. **Power inequality from square divisibility**

Let $a, b \in \mathbb{Z}$ satisfy $1 < a$ and $1 < b$, and let $n \in \mathbb{N}$. Assume $a^2 \mid b^2$ over integers. Prove $a^{2n} \leq (b^n)^4$.

Expected answer. Prove the integer inequality for every natural exponent n, including zero.

Task ID. `ntb_f4cbc48f0a66be7e4c5910e7:whole:0`

TEST 79. **Least positive multiple with digit sum twelve**

Prove that there exists $s \in \mathbb{N}$ with $s = 48$, $s > 0$, decimal digit sum 12, and $s \bmod 12 = 0$. Also require $s \leq t$ for every positive natural t whose decimal digit sum is 12 and whose remainder modulo 12 is zero.

Expected answer. Supply s and prove its fixed value, positivity, digit sum, divisibility, and universal minimality.

Task ID. `ntb_f571b2ea2148e532749d00e3:whole:0`

TEST 80. **Prime divisors and Jacobi symbol at one hundred sixty-three**

Prove two claims. For every natural p, if p is prime, $p \mid 163$, and 163 is prime, then $p = 163$. Also prove $\left(\frac{-163}{163}\right) = 0$, where the notation denotes the Jacobi symbol.

Expected answer. Prove the universally quantified implication and the concrete Jacobi-symbol equality.

Review note. Primality of 163 is an antecedent in the first conjunct; the statement does not separately request its proof.

Task ID. `ntb_f61adf6ee5fe0552363aedb5:whole:0`

TEST 81. **Integer gcd with a prime**

Let $p \in \mathbb{N}$ be prime and $k \in \mathbb{Z}$. Let g be the nonnegative natural-valued gcd of the integers k and p. Prove $g = 1$ or $g = p$; the second source comparison casts g and p to integers.

Expected answer. Prove one of the two stated gcd equalities with the source casts respected.

Task ID. `ntb_fc84491591d39fa8112b24ae:whole:0`

TEST 82. **Fixed square-divisor biconditional**

Prove that there exists a natural k with $2024 = k^2$ if and only if every natural square d dividing 2024^3 satisfies $d \leq 2024^2$. The square premise on d means that some natural m satisfies $d = m^2$.

Expected answer. Prove both directions of the exact biconditional at the fixed number 2024.

Review note. The goal does not assert either side separately and is not a universally quantified characterization for arbitrary base numbers.

Task ID. `ntb_fdb9aa58f002498f2660c655:whole:0`